



Universidade de Brasília

Score-Based Density Modeling Framework through Stochastic Differential Equations

Jose Daniel Arismendy Tibaduiza

Advisors: Dr. Alberto Masahoshi Faria Ohashi
Dr. Francesco Russo

Department of Mathematics
Universidade de Brasília

Dissertation submitted in partial fulfillment of the requirements for the
degree of
Magister in Mathematics

Brasília, january 26, 2026

Abstract

Score-based generative modeling through stochastic differential equations (SDEs) provides a unified and mathematically principled framework for learning and sampling from complex probability distributions. This dissertation offers a self-contained exposition of that framework, making explicit the calculations and theoretical choices that the original literature presents in condensed form, from score matching and denoising objectives through to time-reversal theory and practical discretization schemes for reverse SDE and probability flow ODE samplers.

Building on this exposition, we propose the Linear Variance (LV) SDE,

$$d\mathbf{X}_t = -\theta\mathbf{X}_t dt + \sqrt{1 + 2\theta t} d\mathbf{W}_t, \quad \theta > 0,$$

a new member of the affine diffusion family engineered so that $\text{Var}(\mathbf{X}_t \mid \mathbf{X}_0) = tI$ independently of θ . This structural regularity yields a score field that is smooth across the entire temporal trajectory, with uniformly low score magnitude and stable reverse dynamics. A controlled study on MNIST evaluates the LV SDE alongside the Variance Exploding, Variance Preserving, and sub-VP families of Song et al. [19] under identical training conditions. The results confirm that a principled, minimal modification within the general SDE ansatz suffices to produce a model with a distinctive and potentially advantageous profile.

Contents

List of Figures	ix
List of Tables	xi
Symbols	xiii
Introduction	1
1 Noise and Distributions	5
1.1 Parametric Model	6
1.1.1 Normalization Constant Problem and Score Matching Solution	8
1.1.2 Training Objectives and the Fisher Divergence	10
1.1.3 Practical Limitations of Score Matching	11
1.2 Denoising Score Matching	12
1.2.1 Setup and Conditional Distribution	13
1.2.2 Equivalence between ESM and DSM	16
1.2.3 Noise-Conditioned Score Networks (NCSNs)	18
1.3 Empirical Validation of Denoising Score Matching	19
2 Score-Based Modeling via Stochastic Differential Equations	23
2.1 Stochastic Differential Equations as Continuous Perturbation Processes	23
2.1.1 From Discrete Noise Scales to Continuous-Time SDEs	25
2.1.2 Time Reversal of Diffusion Processes	26
2.2 Continuous-Time Score Matching	29
2.2.1 The Affine, State-Independent Case	30
2.2.2 The Continuous-Time Denoising Score Matching Objective . . .	36
2.3 The Role of Noise and Transition Kernels	37
2.3.1 Necessary Conditions for a Well-Behaved Noise Process	37
2.3.2 Why the Gaussian Distribution Is Inevitable	38

3	Families of Stochastic Differential Equations	41
3.1	Variance Exploding (VE) SDE	42
3.2	Variance Preserving (VP) SDE	43
3.3	The sub-VP SDE	44
3.4	Linear Variance (LV) SDE	45
3.4.1	Derivation from the Linear Variance Hypothesis	46
3.4.2	Transition Kernel	47
3.4.3	Properties and Motivation	48
3.5	Visualizing the Forward Perturbation Process	48
4	Sampling	53
4.1	Discretisation and Reverse-Time Sampling	53
4.1.1	General Discretisation Framework	54
4.1.2	Reverse-Time SDE Discretisation	55
4.1.3	Model-Specific Discretisations	56
4.1.4	Convergence, Implementation, and Practical Considerations	59
4.2	Probability Flow ODE	61
4.2.1	Derivation from the Fokker–Planck Equation	61
4.2.2	Likelihood Computation	62
4.2.3	Sampling with the Probability Flow ODE	63
4.2.4	Connection to Reverse SDE	64
5	Experimental Implementation and Results	65
5.1	Score Network Architecture	65
5.1.1	U-Net Encoder-Decoder Structure	68
5.1.2	Learned Representations after Training	73
5.2	Training Procedure	74
5.2.1	Dataset and Preprocessing	74
5.2.2	Denoising Score Matching Objective	74
5.2.3	Optimisation Details	75
5.3	Sampling Implementation	76
5.4	Results	77
5.4.1	Cross-Family and Cross-Sampler Comparison	85
	Bibliography	87

Appendix A Proof of the Time Reversal Theorem	89
A.1 Problem Setting and Regularity Conditions	89
A.2 Proof of the Theorem	91
Appendix B The Fokker-Planck-Kolmogorov Equation	95
B.1 Preliminaries and Notation	95
B.2 The Infinitesimal Generator	96
B.3 Derivation of the Fokker-Planck-Kolmogorov Equation	96
B.3.1 Operator Formulation	99
B.4 Transition Densities and Markov Properties	99
B.4.1 Backward Kolmogorov Equation	100
B.4.2 Chapman-Kolmogorov Equation	101
B.5 Application to the SDE Families	101
B.5.1 Variance Exploding (VE) SDE	101
B.5.2 Variance Preserving (VP) SDE	102
B.5.3 Linear Variance (LV) SDE	104
B.6 Equations for Moments and Covariances	105
B.6.1 General Derivation	106
B.6.2 Linear (Affine) Case	106
B.6.3 Application to the LV SDE	107

List of Figures

1.1	Conceptual framework	5
1.2	transformation sequence	9
1.3	Denoising score matching	12
1.4	Training loss curve for the NCSN model.	19
1.5	Comparison between true score and approximate	21
1.6	Scatter plot of 10,000	21
1.7	Marginal histograms	22
2.1	Diffusion with Gaussian vs. Cauchy noise	39
3.1	Forward Process	51
5.1	Overall U-Net architecture	66
5.2	tensor size	68
5.3	Activations of the stem	69
5.4	Encoder level 1	69
5.5	first downsampling	70
5.6	Encoder level 2	70
5.7	second downsampling	71
5.8	Bottleneck activations	71
5.9	first transposed convolution	72
5.10	Decoder level 2	72
5.11	Decoder level 1	72
5.12	Output score estimate	73
5.13	Activation maps of selected layers after training	74
5.14	VE SDE comparison	77
5.15	VP SDE comparison	78
5.16	sub-VP SDE comparison	79
5.17	LV SDE comparison	80

5.18	Score magnitude and variability across noise levels	82
5.19	U-Net feature maps at $t = 0.5$	83
5.20	Denoising trajectories from $t = 0.9$	84

List of Tables

4.1	Discrete coefficients SDE	59
4.2	Reverse-time	59
4.3	Discrete coefficients ODE	64
5.1	Forward kernels	75
5.2	Shared training hyperparameters	76

Symbols

$(\Omega, \mathcal{F}, \{\mathcal{F}_t\}, \mathbb{P})$	Filtered probability space	20
$\alpha(t)$	Auxiliary function $1 - e^{-2\bar{\beta}(t)}$ in the sub-VP SDE	38
$\bar{\beta}(t)$	Integrated β schedule, $\int_0^t \beta(s)ds$	38
$\bar{W}_t$	Standard Wiener process in backward-time SDE (Song et al.)	24
$\beta(t)$	Noise schedule for the VP and sub-VP SDEs	37
ϵ	Standard Gaussian noise vector	15
θ	Parameter vector, $\theta \in \Theta \subset \mathbb{R}^m$	2
$\lambda(\sigma)$	Weighting function for noise scales	14
$\lambda(t)$	Weighting function in continuous-time DSM objective	31
$\mathbb{E}_{\mathbf{x} \sim p}$	Expectation with respect to the density p	3
$\mathbf{A}(t)$	Time-dependent diffusion matrix in affine case	26
$\mathbf{D}(\mathbf{x}, t)$	Diffusion matrix $G(\mathbf{x}, t)G(\mathbf{x}, t)^\top$	24
$\mathbf{f}(\mathbf{x}, t)$	Drift coefficient of the forward SDE	20
$\mathbf{m}(t)$	Mean function in characteristic function ansatz	27
$\mathbf{m}_t(\mathbf{x}_0)$	Conditional mean of $\mathbf{X}_t$ given $\mathbf{X}_0 = \mathbf{x}_0$	30
$\mathbf{s}_\theta$	Score of the parametric model (score network)	5
$\mathbf{W}_t$	Standard r -dimensional Wiener process	20
$\mathbf{X}_t$	State of the forward diffusion at time t	20

$\mathbf{Y}_t$	Time-reversed process $\mathbf{Y}_t = \mathbf{X}_{T-t}$	23
$\mathcal{B}(\mathbb{R}^d)$	Borel σ -algebra on $\mathbb{R}^d$	9
$\mathcal{L}_{\text{DSM}}$	Continuous-time denoising score matching loss	31
$\mathcal{L}_{\text{NCSN}}$	Noise-Conditioned Score Network training loss	15
$\mathcal{N}(\mathbf{x}, \sigma^2 I_d)$	Gaussian distribution with mean $\mathbf{x}$ and covariance $\sigma^2 I_d$	11
$\nabla_{\mathbf{x}} \cdot$	Divergence operator with respect to $\mathbf{x}$	6
$\nabla_{\mathbf{x}} \log p_{\text{data}}$	Score function of the data distribution	2
$\Phi(t, s)$	Fundamental matrix associated with $\mathbf{F}(t)$	28
ϕ_{θ}	Neural network approximating $\log p_{\text{data}}$	3
σ	Noise standard deviation (scale)	8
$\Sigma(t)$	Covariance matrix in characteristic function ansatz	27
$\sigma(t)$	Noise schedule (standard deviation) in the VE SDE	36
Σ_t	Conditional covariance of $\mathbf{X}_t$ given $\mathbf{X}_0$ (affine case)	30
σ_t	Isotropic standard deviation at time t	31
θ	Drift parameter in the LV SDE	40
$\tilde{\mathbf{f}}(\mathbf{x}, t)$	Drift of the reverse-time SDE	23
$\tilde{\mathbf{W}}_t$	Wiener process with respect to the reverse filtration	23
$\tilde{\mathbf{x}}$	Corrupted/noisy observation	8
$\tilde{\mathcal{F}}_t$	Reverse filtration	23
$D(\boldsymbol{\theta}; q_{\sigma})$	Denoising score matching objective for perturbed data	12
$d\mathbf{W}_t$	Itô differential of $\mathbf{W}_t$	20
D_F	Fisher divergence	6
D_{KL}	Kullback–Leibler divergence	3
$E(\boldsymbol{\theta}; p)$	Explicit score matching objective w.r.t. density p	6

$f(t)$	Scalar drift coefficient in affine SDE	25
$G(\mathbf{x}, t)$	Diffusion coefficient matrix of the forward SDE	20
$g(t)$	Scalar diffusion coefficient in affine SDE	25
$I(\boldsymbol{\theta}; p)$	Implicit score matching objective w.r.t. density p	6
p_T	Prior distribution at time T	20
p_t	Marginal density of $\mathbf{X}_t$ at time t	20
$p_{0,t}$	Transition kernel (conditional density of $\mathbf{X}_t$ given $\mathbf{X}_0 = \mathbf{x}_0$)	25
$p_{\boldsymbol{\theta}}$	Parametric model density	2
$p_{\mathbf{z}}$	Noise density	9
p_{data}	True data distribution on $\mathbb{R}^d$	2
q_{σ}	Smoothed/perturbed data distribution with noise level σ	8
$q_{\boldsymbol{\theta}}$	Unnormalized density (energy model)	2
T	Final time of the forward process	20

Introduction

The problem of approximating an unknown probability distribution from a finite set of samples is fundamental to statistics and machine learning, with implications that extend well beyond the image generation systems dominating today’s literature. Given i.i.d. samples $\{\mathbf{x}_1, \dots, \mathbf{x}_N\} \subset \mathbb{R}^d$ from an unknown density p_{data} , the goal is to construct a parametric model p_θ that both approximates p_{data} and permits efficient sampling. The technology underlying the most successful current approaches to this problem is score-based generative modeling through stochastic differential equations (SDEs), a framework consolidated by Song et al. [19] in 2021.

This dissertation undertakes a careful mathematical exposition of that framework. The aim is not to replicate known results, but to reconstruct the intellectual edifice from its conceptual origins, making explicit the calculations, theoretical motivations, and methodological choices that the original article presents in compressed form. A deeper understanding of the framework’s internal logic reveals natural directions for exploration. As a concrete illustration, we introduce and analyze a new parametrization of the forward diffusion process, the Linear Variance (LV) SDE, which emerges organically from a systematic interrogation of the general theory.

The chain of ideas begins with the intractability of the normalization constant. A model $p_\theta(\mathbf{x}) = q_\theta(\mathbf{x})/Z(\theta)$ cannot be trained via maximum likelihood because $Z(\theta) = \int q_\theta$ is computationally prohibitive in high dimensions. Hyvärinen [9] bypassed this obstacle by targeting the score function $\nabla_{\mathbf{x}} \log p(\mathbf{x})$, which is independent of $Z(\theta)$. The resulting implicit score matching compares scores via the Fisher divergence and eliminates the unknown true score through integration by parts.

Classical score matching nonetheless suffers from two practical drawbacks: the cost of evaluating the divergence of the score model, and the absence of a reliable learning signal in low-density regions. Denoising Score Matching (DSM) [21] addressed both

by corrupting clean data $\mathbf{x}$ with Gaussian noise of scale σ , making the conditional score

$$\nabla_{\tilde{\mathbf{x}}} \log p(\tilde{\mathbf{x}} | \mathbf{x}) = -\frac{\tilde{\mathbf{x}} - \mathbf{x}}{\sigma^2}$$

available in closed form. Song and Ermon [17] then extended DSM to multiple noise scales via Noise-Conditioned Score Networks (NCSN), injecting informative gradients throughout the data space.

The decisive unification came with Song et al. [19], who took the limit of infinitely many noise scales. The discrete perturbation becomes a continuous diffusion governed by an Itô SDE,

$$d\mathbf{X}_t = \mathbf{f}(\mathbf{X}_t, t) dt + G(t) d\mathbf{W}_t, \quad \mathbf{X}_0 \sim p_{\text{data}},$$

whose marginals p_t evolve from the data distribution toward a tractable prior p_T , typically a standard Gaussian. A classical result of Anderson [1], later made rigorous by Haussmann and Pardoux [5], shows that the time-reversed process is again a diffusion whose drift depends on the score $\nabla_{\mathbf{x}} \log p_t(\mathbf{x})$. Learning this score therefore enables generation by simulating the reverse dynamics from p_T . Well-posedness of the reverse process is further guaranteed by Izydorczyk et al. [10], who established uniqueness for the associated Fokker–Planck equation and confirmed that the forward map from data distribution to prior is injective, legitimizing a learned score network as a surrogate for the true score.

A defining feature of the framework is its flexibility: the drift $\mathbf{f}$ and diffusion coefficient G can be chosen freely, subject to mild regularity, each choice yielding a distinct model family with an analytic transition kernel, a closed-form conditional score, and specific reverse dynamics. Song et al. [19] introduced three canonical families: the Variance Exploding (VE) SDE, the Variance Preserving (VP) SDE (the continuous limit of DDPM [6]), and the sub-VP SDE. Once the algebra of the affine, state-independent case is made explicit, the framework presents itself as an invitation to systematic exploration.

This dissertation takes up that invitation through a twofold contribution. First, it provides a self-contained, rigorous exposition of the score-based SDE framework, unpacking mathematical details often compressed in the original literature. Chapters 1 through 4 cover the historical path from score matching to NCSN, the continuous-time formalism and time-reversal theory, the four SDE families and their properties, and unified discretization schemes for both the reverse SDE and the probability flow ODE samplers.

Second, it proposes and validates the Linear Variance (LV) SDE,

$$d\mathbf{X}_t = -\theta\mathbf{X}_t dt + \sqrt{1 + 2\theta t} d\mathbf{W}_t, \quad \theta > 0,$$

which belongs to the affine family and is engineered so that the conditional variance grows linearly in time,

$$\text{Var}(\mathbf{X}_t \mid \mathbf{X}_0) = t I,$$

independently of θ . This regularity produces a score field that is unusually smooth across the entire temporal trajectory, leading to stable reverse sampling and uniformly low score magnitude. Chapter 5 reports a controlled study on MNIST where all four families are trained under identical conditions. The LV SDE is not offered as a replacement for existing approaches, but as evidence that a principled, minimal modification within the general SDE ansatz can yield a model with a distinctive and potentially advantageous profile.

Chapter 1

Noise and Distributions

This first chapter is conceived as a chain of heuristic reasoning together with a conceptual mathematical foundation, whose purpose is to arrive at an abstraction grounded in the cited literature. It serves as a conceptual introduction to the algorithm developed later in the dissertation, providing the essential ideas without yet entering into the full technical machinery.



Figure 1.1 Conceptual framework of score-based generative modeling.

Let $p_{\text{data}}(\mathbf{x})$ denote an unknown probability distribution on $\mathbb{R}^d$ from which we observe a finite collection of independent samples $\{\mathbf{x}_1, \dots, \mathbf{x}_N\}$. The central problem in generative modeling is to construct a parametric model $p_{\theta}(\mathbf{x})$ that approximates p_{data} and enables efficient sample generation.

This dissertation presents a rigorous mathematical investigation of **score-based generative modeling via Stochastic Differential Equations (SDEs)**. Instead of directly modeling the density $p_{\text{data}}(\mathbf{x})$, this paradigm learns the *score function* $\nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x})$ and uses it to invert a diffusion process that maps data to noise. This approach circumvents the intractability of the normalization constant and provides a unified framework with strong theoretical guarantees.

1.1 Parametric Model

Let $x_1, \dots, x_N \in \mathbb{R}^n$ be independent and identically distributed samples from a probability distribution with density $p_{\text{data}} : \mathbb{R}^n \rightarrow [0, \infty)$ and let $K \subset \mathbb{R}^d$ be a compact set containing the main mass of the distribution p_{data} . We assume that $p_{\text{data}} \in \mathcal{L}^1(K)$, $p_{\text{data}} \in C^2(K)$, and that $p_{\text{data}}(\mathbf{x}) \rightarrow 0$ sufficiently fast as $\|\mathbf{x}\| \rightarrow \infty$.

The goal is to approximate the unknown distribution $p_{\text{data}}(\mathbf{x})$ by means of a parametric model inferred from the data $p_{\theta}(\mathbf{x})$, where:

$$q_{\theta}(\mathbf{x}) \geq 0, \quad p_{\theta}(\mathbf{x}) = \frac{q_{\theta}(\mathbf{x})}{Z(\theta)}, \quad Z(\theta) = \int_{\mathbb{R}^n} q_{\theta}(\xi) d\xi,$$

with $\theta \in \Theta \subset \mathbb{R}^m$.

A natural measure of discrepancy between two probability densities is the *Kullback–Leibler divergence*, introduced in the context of information theory. Intuitively, $D_{\text{KL}}(p\|q)$ quantifies the expected amount of information (measured in nats) lost when the distribution q is used to approximate the true distribution p . It arises naturally from the principle of maximum likelihood: given independent samples from p_{data} , the parameter θ that minimizes $D_{\text{KL}}(p_{\text{data}}\|p_{\theta})$ is exactly the one that maximizes the log-likelihood of the data under the model p_{θ} .

Definition 1.1.1. [14] Let p and q be two probability densities on $\mathbb{R}^d$ with $p \ll q$ (i.e., p is absolutely continuous with respect to q). The Kullback–Leibler divergence from q to p is defined as

$$D_{\text{KL}}(p\|q) = \mathbb{E}_{\mathbf{x} \sim p} [\log p(\mathbf{x}) - \log q(\mathbf{x})].$$

It is non-negative and vanishes if and only if $p = q$ almost everywhere.

Theorem 1.1.1 (Universal Approximation for Densities). *For any continuous density $p_{\text{data}}(\mathbf{x})$ on $\mathbb{R}^d$ and for any $\epsilon > 0$, there exists a neural-network-based parametric model $p_{\theta}(\mathbf{x})$ such that*

$$D_{\text{KL}}(p_{\text{data}} \| p_{\theta}) < \epsilon,$$

where D_{KL} denotes the Kullback-Leibler divergence.

Proof. The proof relies on the Universal Approximation Theorem of [4] and [7], which states that neural networks with a single hidden layer and sigmoidal activation function can uniformly approximate any continuous function on compact sets.

Let $K \subset \mathbb{R}^d$ be a compact set containing the main mass of p_{data} . For the continuous function $f(\mathbf{x}) = \log p_{\text{data}}(\mathbf{x})$ and for any $\epsilon_1 > 0$, there exists a neural network $\phi_{\theta}(\mathbf{x})$ such that

$$|\log p_{\text{data}}(\mathbf{x}) - \phi_{\theta}(\mathbf{x})| < \epsilon_1 \quad \forall \mathbf{x} \in K.$$

Construct the parametric model as

$$p_{\theta}(\mathbf{x}) = \frac{\exp(\phi_{\theta}(\mathbf{x}))}{Z(\theta)}, \quad Z(\theta) = \int_K \exp(\phi_{\theta}(\mathbf{x})) d\mathbf{x}.$$

The KL divergence between the distributions is

$$\begin{aligned} D_{\text{KL}}(p_{\text{data}} \| p_{\theta}) &= \mathbb{E}_{p_{\text{data}}} [\log p_{\text{data}}(\mathbf{x}) - \log p_{\theta}(\mathbf{x})] \\ &= \mathbb{E}_{p_{\text{data}}} [\log p_{\text{data}}(\mathbf{x}) - \phi_{\theta}(\mathbf{x}) + \log Z(\theta)] \\ &= \mathbb{E}_{p_{\text{data}}} [\log p_{\text{data}}(\mathbf{x}) - \phi_{\theta}(\mathbf{x})] + \log Z(\theta). \end{aligned} \quad (1.1)$$

For the first term, using the uniform approximation and the fact that p_{data} is concentrated on K :

$$|\mathbb{E}_{p_{\text{data}}} [\log p_{\text{data}}(\mathbf{x}) - \phi_{\theta}(\mathbf{x})]| \leq \mathbb{E}_{p_{\text{data}}} [|\log p_{\text{data}}(\mathbf{x}) - \phi_{\theta}(\mathbf{x})|] < \epsilon_1 \int_K p_{\text{data}}(\mathbf{x}) d\mathbf{x} \approx \epsilon_1.$$

For the second term, write

$$\log Z(\theta) = \log \int_K \exp(\phi_{\theta}(\mathbf{x})) d\mathbf{x} = \log \int_K p_{\text{data}}(\mathbf{x}) \exp(\phi_{\theta}(\mathbf{x}) - \log p_{\text{data}}(\mathbf{x})) d\mathbf{x}.$$

From the uniform approximation $|\phi_{\theta}(\mathbf{x}) - \log p_{\text{data}}(\mathbf{x})| < \epsilon_1$, we have

$$e^{-\epsilon_1} < \exp(\phi_{\theta}(\mathbf{x}) - \log p_{\text{data}}(\mathbf{x})) < e^{\epsilon_1},$$

and integrating against p_{data} gives

$$e^{-\epsilon_1} \int_K p_{\text{data}}(\mathbf{x}) d\mathbf{x} < \int_K p_{\text{data}}(\mathbf{x}) \exp(\phi_{\theta}(\mathbf{x}) - \log p_{\text{data}}(\mathbf{x})) d\mathbf{x} < e^{\epsilon_1} \int_K p_{\text{data}}(\mathbf{x}) d\mathbf{x}.$$

Since $\int_K p_{\text{data}}(\mathbf{x}) d\mathbf{x} \approx 1$, we obtain $e^{-\epsilon_1} < Z(\theta) < e^{\epsilon_1}$ and therefore $|\log Z(\theta)| < \epsilon_1$.

Combining the two estimates,

$$D_{\text{KL}}(p_{\text{data}} \| p_{\theta}) < \epsilon_1 + \epsilon_1 = 2\epsilon_1.$$

Choosing $\epsilon = 2\epsilon_1$ completes the proof. $\square$

This theorem theoretically justifies the use of parametric models for density estimation.

1.1.1 Normalization Constant Problem and Score Matching Solution

Despite the guaranteed existence of parametric models that approximate $p_{\text{data}}(\mathbf{x})$, practical implementation faces a fundamental obstacle: the intractability of the normalization constant $Z(\theta)$. Consider the model

$$p_{\theta}(\mathbf{x}) = \frac{1}{Z(\theta)} q_{\theta}(\mathbf{x}),$$

where $q_{\theta}(\mathbf{x})$ is a known and computationally tractable function (energy model), but

$$Z(\theta) = \int_{\mathbb{R}^n} q_{\theta}(\xi) d\xi$$

is analytically intractable for most interesting models. In high dimensions ($n > 2$), numerical calculation of this integral becomes practically impossible due to the curse of dimensionality.

To circumvent this problem, [9] proposed the *score matching* method. This article introduces implicit score matching: minimizing the expected squared distance between score functions to avoid computing $Z(\theta)$. The cornerstone of the method is the *score function*, defined as the gradient of the logarithm of the probability density.

Heuristically, $\log p$ turns multiplicative structure into additive form, and its gradient $\nabla_{\mathbf{x}} \log p$ yields a vector field pointing toward regions of higher probability, independent of any global scaling.

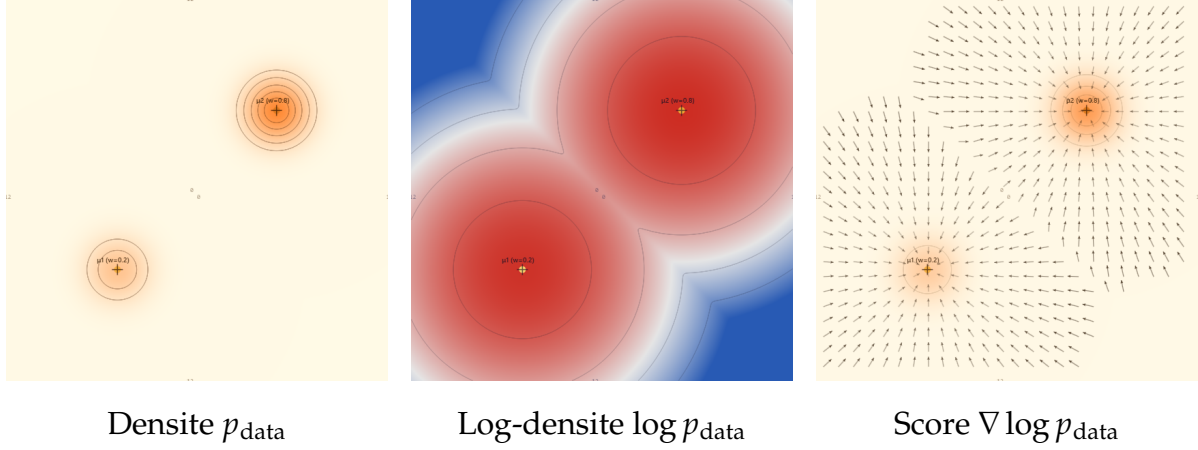


Figure 1.2 transformation sequence

For the parametric model $p_{\theta}(\mathbf{x})$, the score function is

$$\mathbf{s}_{\theta}(\mathbf{x}) = \nabla_{\mathbf{x}} \log p_{\theta}(\mathbf{x}).$$

The crucial observation is that the score function is independent of the normalization constant $Z(\theta)$:

$$\mathbf{s}_{\theta}(\mathbf{x}) = \nabla_{\mathbf{x}} \log q_{\theta}(\mathbf{x}) - \nabla_{\mathbf{x}} \log Z(\theta) = \nabla_{\mathbf{x}} \log q_{\theta}(\mathbf{x}),$$

because $\log Z(\theta)$ does not depend on $\mathbf{x}$. Hence we can compute $\mathbf{s}_{\theta}(\mathbf{x})$ using only the unnormalized function $q_{\theta}(\mathbf{x})$, without ever needing $Z(\theta)$. Analogously, the score of the true data distribution is $\nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x})$.

Geometrically, the score $\nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x})$ defines a vector field over $\mathbb{R}^d$: at each point, the vector points in the direction of greatest local increase of the log-density, i.e., toward regions of higher probability concentration. Importantly, when the data distribution is supported on a lower-dimensional submanifold $\mathcal{M} \subset \mathbb{R}^d$ a common assumption in high-dimensional applications, the density p_{data} is degenerate off $\mathcal{M}$, and the score is either undefined or exhibits explosive growth near the manifold. This geometric degeneracy is precisely why score-based methods incorporate noise perturbations: smoothing the distribution expands its support to the whole space, yielding a well-behaved score field everywhere.

1.1.2 Training Objectives and the Fisher Divergence

Since the Kullback–Leibler divergence is intractable for unnormalized models, we require an alternative discrepancy measure that does not involve the normalization constant. The *Fisher divergence* achieves this by comparing the *scores* $\nabla \log p$ rather than the densities themselves. Because the score of a model is invariant to multiplicative constants, minimizing the Fisher divergence yields a computationally feasible surrogate for maximum likelihood.

Definition 1.1.2. [9][Fisher Divergence] For two sufficiently smooth densities p and q on $\mathbb{R}^d$, the Fisher divergence is defined as

$$D_F(p\|q) = \frac{1}{2} \int_{\mathbb{R}^d} p(\mathbf{x}) \|\nabla_{\mathbf{x}} \log p(\mathbf{x}) - \nabla_{\mathbf{x}} \log q(\mathbf{x})\|_2^2 d\mathbf{x}.$$

For a given (possibly unnormalized) target density p , we introduce the **explicit score matching objective**

$$E(\boldsymbol{\theta}; p) \triangleq \frac{1}{2} \mathbb{E}_{\mathbf{x} \sim p} \left[\|\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}) - \nabla_{\mathbf{x}} \log p(\mathbf{x})\|_2^2 \right]. \quad (1.2)$$

When $p = p_{\text{data}}$, this is exactly the Fisher divergence $D_F(p_{\text{data}}\|p_{\boldsymbol{\theta}})$. However, $E(\boldsymbol{\theta}; p_{\text{data}})$ involves the unknown true score $\nabla_{\mathbf{x}} \log p_{\text{data}}$. The crucial observation of [9] is that integration by parts eliminates the unknown score, yielding an equivalent objective that depends only on the model score and its divergence.

Proposition 1.1.1 (Implicit Score Matching). Assume p is a smooth density on $\mathbb{R}^d$ such that $p(\mathbf{x}) \rightarrow 0$ and $\|\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x})\| \rightarrow 0$ as $\|\mathbf{x}\| \rightarrow \infty$. Then

$$E(\boldsymbol{\theta}; p) = I(\boldsymbol{\theta}; p) + C,$$

where C is a constant independent of $\boldsymbol{\theta}$ and

$$I(\boldsymbol{\theta}; p) = \mathbb{E}_{\mathbf{x} \sim p} \left[\nabla_{\mathbf{x}} \cdot \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}) + \frac{1}{2} \|\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x})\|_2^2 \right]. \quad (1.3)$$

Proof. Expand the squared norm in (1.2) to obtain

$$E(\boldsymbol{\theta}; p) = \frac{1}{2} \mathbb{E}_p [\|\mathbf{s}_\theta\|_2^2] - \mathbb{E}_p [\mathbf{s}_\theta \cdot \nabla_{\mathbf{x}} \log p] + \frac{1}{2} \mathbb{E}_p [\|\nabla_{\mathbf{x}} \log p\|_2^2].$$

The last term is independent of $\boldsymbol{\theta}$; call it C . For the cross term, write

$$\mathbb{E}_p [\mathbf{s}_\theta \cdot \nabla_{\mathbf{x}} \log p] = \int_{\mathbb{R}^d} \mathbf{s}_\theta(\mathbf{x}) \cdot \nabla_{\mathbf{x}} p(\mathbf{x}) d\mathbf{x}.$$

Integrating by parts and using the decay hypotheses gives

$$\int_{\mathbb{R}^d} \mathbf{s}_\theta \cdot \nabla_{\mathbf{x}} p d\mathbf{x} = - \int_{\mathbb{R}^d} p(\mathbf{x}) (\nabla_{\mathbf{x}} \cdot \mathbf{s}_\theta(\mathbf{x})) d\mathbf{x} = -\mathbb{E}_p [\nabla_{\mathbf{x}} \cdot \mathbf{s}_\theta].$$

Substituting back yields

$$E(\boldsymbol{\theta}; p) = \mathbb{E}_p \left[\nabla_{\mathbf{x}} \cdot \mathbf{s}_\theta + \frac{1}{2} \|\mathbf{s}_\theta\|_2^2 \right] + C = I(\boldsymbol{\theta}; p) + C. \quad \square$$

The two terms of $I(\boldsymbol{\theta}; p)$ have complementary roles: the energy term $\frac{1}{2} \mathbb{E}_p [\|\mathbf{s}_\theta\|_2^2]$ penalizes artificially steep gradients, while the curvature term $\mathbb{E}_p [\nabla_{\mathbf{x}} \cdot \mathbf{s}_\theta]$ forces the divergence of the estimated field to match the local geometry of the log-density.

1.1.3 Practical Limitations of Score Matching

Despite its theoretical foundation, the direct application of score matching encounters two main practical obstacles.

First Problem: Computational Cost of the Divergence

The evaluation of $I(\boldsymbol{\theta}; p)$ requires calculating $\nabla_{\mathbf{x}} \cdot \mathbf{s}_\theta(\mathbf{x})$, whose cost is $O(d)$ for data of dimension d . Although methods like *Sliced Score Matching* [18] circumvent this issue with Hessian-vector products, a more fundamental limitation persists.

Second Problem: Lack of Signal in Low-Density Regions

The expectation in $I(\boldsymbol{\theta}; p_{\text{data}})$ is taken with respect to p_{data} . Consequently, regions of low probability contribute negligibly to the loss, regardless of the model error $\mathbf{s}_\theta$ there. Thus the model does not receive informative gradients where data are sparse, leading to inaccurate or undefined score estimates in those regions. This is a critical

impediment for generative sampling, as processes like Langevin dynamics require a reliable estimate of the score throughout the entire space. Figure 1.3 illustrates this phenomenon: the estimate is reliable only near the modes of the distribution.

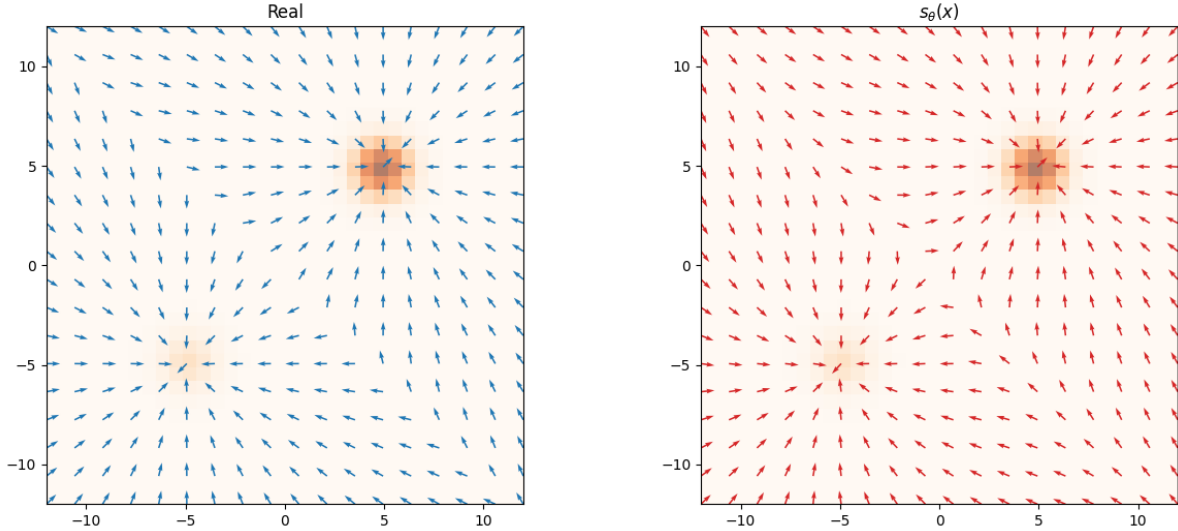


Figure 1.3 Comparison between the true score (left) and the one estimated score matching (right). The estimate deteriorates away from the distribution's modes.

1.2 Denoising Score Matching

The fundamental idea of *Denoising Score Matching* (DSM) is to replace the intractable objective on the sparse data distribution with a tractable one defined on a *perturbed* version of the data. By corrupting $\mathbf{x}$ with additive noise we effectively spread the empirical distribution across all of $\mathbb{R}^d$, ensuring informative gradients everywhere. We denote the corrupted observation by $\tilde{\mathbf{x}} = \mathbf{x} + \mathbf{z}$, where $\mathbf{z}$ is an independent noise variable.

More precisely, DSM overcomes the two limitations of classical score matching (Subsection 1.1.3). First, it replaces the intractable marginal score $\nabla \log q_\sigma$ with the conditional score $\nabla \log q_\sigma(\cdot \mid \mathbf{x})$, which for many noise distributions has a closed form depending only on $(\mathbf{x}, \tilde{\mathbf{x}})$; this avoids computing $\nabla \cdot \mathbf{s}_\theta$. Second, the perturbed distribution q_σ has full support, so the DSM objective receives non-negligible signal everywhere, and as $\sigma \rightarrow 0$ its score converges to the true score, even in low-density regions.

1.2.1 Setup and Conditional Distribution

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a complete probability space. We consider the following random variables:

- $\mathbf{x} : \Omega \rightarrow \mathbb{R}^d$, the clean data, with density p_{data} with respect to the Lebesgue measure λ^d .
- $\mathbf{z} : \Omega \rightarrow \mathbb{R}^d$, the additive noise, with density $p_{\mathbf{z}}$ with respect to λ^d .
- **Independence hypothesis:** $\mathbf{x} \perp\!\!\!\perp \mathbf{z}$, i.e. for all Borel sets $A, B \in \mathcal{B}(\mathbb{R}^d)$, $\mathbb{P}(\mathbf{x} \in A, \mathbf{z} \in B) = \mathbb{P}(\mathbf{x} \in A) \cdot \mathbb{P}(\mathbf{z} \in B)$.

The noisy observation is defined as $\tilde{\mathbf{x}} = \mathbf{x} + \mathbf{z}$. Since $\mathbf{x}$ and $\mathbf{z}$ are independent, the joint density of $(\mathbf{x}, \tilde{\mathbf{x}})$ is simply the product of the density of $\mathbf{x}$ and the density of $\tilde{\mathbf{x}} - \mathbf{x}$ (which equals the density of $\mathbf{z}$ evaluated at $\tilde{\mathbf{x}} - \mathbf{x}$).

lemma 1.2.1 (Conditional Distribution under Independent Additive Noise). *Under the hypotheses above, the conditional density of $\tilde{\mathbf{x}}$ given $\mathbf{x}$ is*

$$q(\tilde{\mathbf{x}} \mid \mathbf{x}) = p_{\mathbf{z}}(\tilde{\mathbf{x}} - \mathbf{x})$$

for λ^{2d} -almost every pair $(\mathbf{x}, \tilde{\mathbf{x}}) \in \mathbb{R}^{2d}$.

Proof. Because $\mathbf{x}$ and $\mathbf{z}$ are independent, the joint density factorizes as the product of the marginal densities:

$$f_{(\mathbf{x}, \mathbf{z})}(\mathbf{x}_0, \mathbf{z}_0) = p_{\text{data}}(\mathbf{x}_0) p_{\mathbf{z}}(\mathbf{z}_0), \quad \lambda^{2d}\text{-almost every } (\mathbf{x}_0, \mathbf{z}_0) \in \mathbb{R}^{2d}.$$

Define the transformation $T : \mathbb{R}^{2d} \rightarrow \mathbb{R}^{2d}$ by

$$T(\mathbf{x}_0, \mathbf{z}_0) = (\mathbf{x}_0, \mathbf{x}_0 + \mathbf{z}_0) = (\mathbf{x}_0, \tilde{\mathbf{x}}_0).$$

This map is bijective with inverse

$$T^{-1}(\mathbf{x}_0, \tilde{\mathbf{x}}_0) = (\mathbf{x}_0, \tilde{\mathbf{x}}_0 - \mathbf{x}_0).$$

Both T and T^{-1} are affine functions, hence Borel-measurable.

The Jacobian matrix of T^{-1} with respect to $(\mathbf{x}_0, \tilde{\mathbf{x}}_0)$ is the $2d \times 2d$ lower-triangular block matrix

$$J_{T^{-1}} = \begin{pmatrix} I_d & 0 \\ -I_d & I_d \end{pmatrix},$$

where I_d denotes the d -dimensional identity matrix and 0 the zero matrix. The determinant of this matrix is $\det(J_{T^{-1}}) = \det(I_d)^2 = 1$. Consequently, the transformation T^{-1} preserves volume.

Applying the change-of-variables theorem for densities,

$$\begin{aligned} f_{(\mathbf{x}, \tilde{\mathbf{x}})}(\mathbf{x}_0, \tilde{\mathbf{x}}_0) &= f_{(\mathbf{x}, \mathbf{z})}(T^{-1}(\mathbf{x}_0, \tilde{\mathbf{x}}_0)) |\det J_{T^{-1}}| \\ &= p_{\text{data}}(\mathbf{x}_0) p_{\mathbf{z}}(\tilde{\mathbf{x}}_0 - \mathbf{x}_0) \cdot 1, \end{aligned}$$

which holds for λ^{2d} -almost every $(\mathbf{x}_0, \tilde{\mathbf{x}}_0) \in \mathbb{R}^{2d}$.

For any $\mathbf{x}_0$ such that $p_{\text{data}}(\mathbf{x}_0) > 0$, the conditional density of $\tilde{\mathbf{x}}$ given $\mathbf{x} = \mathbf{x}_0$ is well defined as the ratio of the joint density to the marginal density of $\mathbf{x}$:

$$q(\tilde{\mathbf{x}} \mid \mathbf{x} = \mathbf{x}_0) = \frac{f_{(\mathbf{x}, \tilde{\mathbf{x}})}(\mathbf{x}_0, \tilde{\mathbf{x}}_0)}{p_{\text{data}}(\mathbf{x}_0)}.$$

Substituting the expression for the joint density obtained above,

$$q(\tilde{\mathbf{x}} \mid \mathbf{x} = \mathbf{x}_0) = \frac{p_{\text{data}}(\mathbf{x}_0) p_{\mathbf{z}}(\tilde{\mathbf{x}}_0 - \mathbf{x}_0)}{p_{\text{data}}(\mathbf{x}_0)} = p_{\mathbf{z}}(\tilde{\mathbf{x}}_0 - \mathbf{x}_0).$$

The equality holds for λ^d -almost every $\mathbf{x}_0$ (with respect to the measure $p_{\text{data}} d\mathbf{x}$) and for every $\tilde{\mathbf{x}}_0 \in \mathbb{R}^d$. This completes the proof. $\square$

The Gaussian Case and the Conditional Score

Theorem 1.2.1 is fully general. For the choice $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \sigma^2 I_d)$ the conditional distribution takes an explicit Gaussian form, and its score admits a closed-form expression that is the analytical engine of DSM.

Corollary 1.2.1 (Gaussian Noise: Conditional Distribution and Score). *If $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \sigma^2 I_d)$, then*

$$q_{\sigma}(\tilde{\mathbf{x}} \mid \mathbf{x}) = \mathcal{N}(\tilde{\mathbf{x}}; \mathbf{x}, \sigma^2 I_d) = \frac{1}{(2\pi\sigma^2)^{d/2}} \exp\left(-\frac{\|\tilde{\mathbf{x}} - \mathbf{x}\|^2}{2\sigma^2}\right),$$

and the conditional score satisfies

$$\nabla_{\tilde{\mathbf{x}}} \log q_{\sigma}(\tilde{\mathbf{x}} \mid \mathbf{x}) = -\frac{\tilde{\mathbf{x}} - \mathbf{x}}{\sigma^2}. \quad (1.4)$$

Proof. By Theorem 1.2.1, the conditional density is

$$q_\sigma(\tilde{\mathbf{x}} \mid \mathbf{x}) = p_{\mathbf{z}}(\tilde{\mathbf{x}} - \mathbf{x}),$$

where $p_{\mathbf{z}}$ is the density of the noise distribution. For Gaussian noise with variance σ^2 , the noise density is

$$p_{\mathbf{z}}(\mathbf{z}) = (2\pi\sigma^2)^{-d/2} \exp\left(-\frac{\|\mathbf{z}\|^2}{2\sigma^2}\right).$$

Substituting $\mathbf{z} = \tilde{\mathbf{x}} - \mathbf{x}$ yields

$$q_\sigma(\tilde{\mathbf{x}} \mid \mathbf{x}) = (2\pi\sigma^2)^{-d/2} \exp\left(-\frac{\|\tilde{\mathbf{x}} - \mathbf{x}\|^2}{2\sigma^2}\right),$$

which is precisely the density of $\mathcal{N}(\mathbf{x}, \sigma^2 I_d)$.

To obtain the conditional score, take the logarithm:

$$\log q_\sigma(\tilde{\mathbf{x}} \mid \mathbf{x}) = -\frac{d}{2} \log(2\pi\sigma^2) - \frac{\|\tilde{\mathbf{x}} - \mathbf{x}\|^2}{2\sigma^2}.$$

The first term is constant with respect to $\tilde{\mathbf{x}}$, so its gradient vanishes. For the second term, differentiate component-wise. For each coordinate i ,

$$\frac{\partial}{\partial \tilde{x}_i} \left(-\frac{\|\tilde{\mathbf{x}} - \mathbf{x}\|^2}{2\sigma^2} \right) = -\frac{1}{2\sigma^2} \cdot 2(\tilde{x}_i - x_i) = -\frac{\tilde{x}_i - x_i}{\sigma^2}.$$

Collecting all components gives

$$\nabla_{\tilde{\mathbf{x}}} \log q_\sigma(\tilde{\mathbf{x}} \mid \mathbf{x}) = -\frac{\tilde{\mathbf{x}} - \mathbf{x}}{\sigma^2}. \quad \square$$

The analytical tractability of (1.4) is the cornerstone of DSM: it provides a closed-form target score that does not require knowledge of p_{data} , transforming a density estimation problem into a supervised denoising regression task.

For a general noise distribution, the conditional score is simply the score of the noise density evaluated at the residual:

$$\nabla_{\tilde{\mathbf{x}}} \log q(\tilde{\mathbf{x}} \mid \mathbf{x}) = \nabla_{\mathbf{u}} \log p_{\mathbf{z}}(\mathbf{u}) \Big|_{\mathbf{u}=\tilde{\mathbf{x}}-\mathbf{x}}.$$

1.2.2 Equivalence between ESM and DSM

We now establish the core theoretical result of this section: minimizing a tractable objective on perturbed samples is equivalent to matching the score of the smoothed distribution q_σ . For the perturbed distribution we define the explicit and denoising objectives by

$$E(\boldsymbol{\theta}; q_\sigma) \triangleq \frac{1}{2} \mathbb{E}_{\tilde{\mathbf{x}} \sim q_\sigma} \left[\left\| \mathbf{s}_\theta(\tilde{\mathbf{x}}) - \nabla_{\tilde{\mathbf{x}}} \log q_\sigma(\tilde{\mathbf{x}}) \right\|_2^2 \right], \quad (1.5)$$

$$D(\boldsymbol{\theta}; q_\sigma) \triangleq \frac{1}{2} \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}} \mathbb{E}_{\tilde{\mathbf{x}} \sim q_\sigma(\cdot | \mathbf{x})} \left[\left\| \mathbf{s}_\theta(\tilde{\mathbf{x}}) - \nabla_{\tilde{\mathbf{x}}} \log q_\sigma(\tilde{\mathbf{x}} | \mathbf{x}) \right\|_2^2 \right]. \quad (1.6)$$

The ESM objective (1.5) targets the score of the marginal q_σ and requires evaluating $\nabla \log q_\sigma$, which is intractable. The DSM objective (1.6) replaces it with the conditional score, which by Corollary 1.2.1 is the closed-form quantity $-(\tilde{\mathbf{x}} - \mathbf{x})/\sigma^2$ for Gaussian noise.

Theorem 1.2.1 (Equivalence ESM-DSM on q_σ , [21]). *For any $\sigma > 0$ and any measurable $\mathbf{s}_\theta$ with $\mathbb{E}_{q_\sigma} [\|\mathbf{s}_\theta(\tilde{\mathbf{x}})\|_2^2] < \infty$, there exists a constant C independent of $\boldsymbol{\theta}$ such that*

$$E(\boldsymbol{\theta}; q_\sigma) = D(\boldsymbol{\theta}; q_\sigma) + C.$$

Proof. We start from the explicit objective $E(\boldsymbol{\theta}; q_\sigma)$. By Proposition 1.1.1 applied to the density $p = q_\sigma$, we have the decomposition

$$E(\boldsymbol{\theta}; q_\sigma) = I(\boldsymbol{\theta}; q_\sigma) + C_1,$$

where C_1 does not depend on $\boldsymbol{\theta}$ and

$$I(\boldsymbol{\theta}; q_\sigma) = \mathbb{E}_{q_\sigma} \left[\nabla_{\tilde{\mathbf{x}}} \cdot \mathbf{s}_\theta(\tilde{\mathbf{x}}) + \frac{1}{2} \|\mathbf{s}_\theta(\tilde{\mathbf{x}})\|_2^2 \right].$$

Expanding $I(\boldsymbol{\theta}; q_\sigma)$ gives

$$I(\boldsymbol{\theta}; q_\sigma) = \frac{1}{2} \mathbb{E}_{q_\sigma} [\|\mathbf{s}_\theta(\tilde{\mathbf{x}})\|_2^2] + \mathbb{E}_{q_\sigma} [\nabla_{\tilde{\mathbf{x}}} \cdot \mathbf{s}_\theta(\tilde{\mathbf{x}})]. \quad (1.7)$$

On the other hand, expanding the denoising objective (1.6) yields

$$\begin{aligned} D(\boldsymbol{\theta}; q_\sigma) &= \frac{1}{2} \mathbb{E}_{p_{\text{data}}} \mathbb{E}_{q_\sigma(\tilde{\mathbf{x}} | \mathbf{x})} [\|\mathbf{s}_\theta(\tilde{\mathbf{x}})\|_2^2] \\ &\quad - \mathbb{E}_{p_{\text{data}}} \mathbb{E}_{q_\sigma(\tilde{\mathbf{x}} | \mathbf{x})} [\mathbf{s}_\theta(\tilde{\mathbf{x}}) \cdot \nabla_{\tilde{\mathbf{x}}} \log q_\sigma(\tilde{\mathbf{x}} | \mathbf{x})] + C_2, \end{aligned} \quad (1.8)$$

where C_2 is a constant independent of θ .

By the law of total expectation, the first terms of (1.7) and (1.8) coincide:

$$\mathbb{E}_{p_{\text{data}}} \mathbb{E}_{q_{\sigma}(\tilde{\mathbf{x}}|\mathbf{x})} [\|\mathbf{s}_{\theta}(\tilde{\mathbf{x}})\|_2^2] = \mathbb{E}_{q_{\sigma}} [\|\mathbf{s}_{\theta}(\tilde{\mathbf{x}})\|_2^2].$$

It remains to prove the equality of the cross terms. Using q_{σ} as a marginal, we compute

$$\mathbb{E}_{q_{\sigma}} [\mathbf{s}_{\theta}(\tilde{\mathbf{x}}) \cdot \nabla_{\tilde{\mathbf{x}}} \log q_{\sigma}(\tilde{\mathbf{x}})] = \int_{\mathbb{R}^d} \mathbf{s}_{\theta}(\tilde{\mathbf{x}}) \cdot \nabla_{\tilde{\mathbf{x}}} q_{\sigma}(\tilde{\mathbf{x}}) d\tilde{\mathbf{x}}.$$

Substituting the integral representation of q_{σ} and exchanging the gradient with the integral (justified by the integrability condition) gives

$$\int_{\mathbb{R}^d} \mathbf{s}_{\theta}(\tilde{\mathbf{x}}) \cdot \nabla_{\tilde{\mathbf{x}}} \left(\int_{\mathbb{R}^d} p_{\text{data}}(\mathbf{x}) q_{\sigma}(\tilde{\mathbf{x}} | \mathbf{x}) d\mathbf{x} \right) d\tilde{\mathbf{x}} = \int_{\mathbb{R}^d} \mathbf{s}_{\theta}(\tilde{\mathbf{x}}) \cdot \left(\int_{\mathbb{R}^d} p_{\text{data}}(\mathbf{x}) \nabla_{\tilde{\mathbf{x}}} q_{\sigma}(\tilde{\mathbf{x}} | \mathbf{x}) d\mathbf{x} \right) d\tilde{\mathbf{x}}.$$

Writing $\nabla_{\tilde{\mathbf{x}}} q_{\sigma}(\tilde{\mathbf{x}} | \mathbf{x}) = q_{\sigma}(\tilde{\mathbf{x}} | \mathbf{x}) \nabla_{\tilde{\mathbf{x}}} \log q_{\sigma}(\tilde{\mathbf{x}} | \mathbf{x})$ and applying Fubini, we obtain

$$\int_{\mathbb{R}^d} \int_{\mathbb{R}^d} \mathbf{s}_{\theta}(\tilde{\mathbf{x}}) \cdot \nabla_{\tilde{\mathbf{x}}} \log q_{\sigma}(\tilde{\mathbf{x}} | \mathbf{x}) p_{\text{data}}(\mathbf{x}) q_{\sigma}(\tilde{\mathbf{x}} | \mathbf{x}) d\mathbf{x} d\tilde{\mathbf{x}} = \mathbb{E}_{p_{\text{data}}} \mathbb{E}_{q_{\sigma}(\tilde{\mathbf{x}}|\mathbf{x})} [\mathbf{s}_{\theta}(\tilde{\mathbf{x}}) \cdot \nabla_{\tilde{\mathbf{x}}} \log q_{\sigma}(\tilde{\mathbf{x}} | \mathbf{x})].$$

Integrating by parts the term $\mathbb{E}_{q_{\sigma}} [\nabla_{\tilde{\mathbf{x}}} \cdot \mathbf{s}_{\theta}(\tilde{\mathbf{x}})]$ in (1.7) gives exactly $-\mathbb{E}_{q_{\sigma}} [\mathbf{s}_{\theta}(\tilde{\mathbf{x}}) \cdot \nabla_{\tilde{\mathbf{x}}} \log q_{\sigma}(\tilde{\mathbf{x}})]$. Hence

$$\mathbb{E}_{q_{\sigma}} [\nabla_{\tilde{\mathbf{x}}} \cdot \mathbf{s}_{\theta}(\tilde{\mathbf{x}})] = -\mathbb{E}_{q_{\sigma}} [\mathbf{s}_{\theta}(\tilde{\mathbf{x}}) \cdot \nabla_{\tilde{\mathbf{x}}} \log q_{\sigma}(\tilde{\mathbf{x}})] = -\mathbb{E}_{p_{\text{data}}} \mathbb{E}_{q_{\sigma}(\tilde{\mathbf{x}}|\mathbf{x})} [\mathbf{s}_{\theta}(\tilde{\mathbf{x}}) \cdot \nabla_{\tilde{\mathbf{x}}} \log q_{\sigma}(\tilde{\mathbf{x}} | \mathbf{x})].$$

Substituting into (1.7) and comparing with (1.8), we see that

$$I(\theta; q_{\sigma}) + C_1 = D(\theta; q_{\sigma}) + (C_1 + C_2).$$

Thus $E(\theta; q_{\sigma}) = D(\theta; q_{\sigma}) + C$ with $C = C_1 + C_2$. This completes the proof. $\square$

Theorem 1.2.1 shows that minimizing the DSM objective with the tractable conditional score $\nabla \log q_{\sigma}(\cdot | \mathbf{x})$ is exactly equivalent to minimizing the intractable ESM objective on q_{σ} . The idea resembles finding a lighthouse in fog: ESM would directly sense the current (marginal score) at your location; DSM, displaced by the same current, infers the direction toward the strongest flow from the observed displacement $-(\tilde{\mathbf{x}} - \mathbf{x})/\sigma^2$, converging to the same field without ever measuring the marginal score.

1.2.3 Noise-Conditioned Score Networks (NCSNs)

The equivalence between ESM and DSM (Theorem 1.2.1) provides a practical path: instead of the intractable marginal score $\nabla \log q_\sigma$, we train a score network $\mathbf{s}_\theta(\mathbf{x}, \sigma)$ to match the closed-form conditional score given by Corollary 1.2.1. For a single noise level σ , this yields the denoising objective

$$\ell(\theta; \sigma) = \frac{1}{2} \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{\tilde{\mathbf{x}} \sim \mathcal{N}(\mathbf{x}, \sigma^2 I)} \left[\left\| \mathbf{s}_\theta(\tilde{\mathbf{x}}, \sigma) + \frac{\tilde{\mathbf{x}} - \mathbf{x}}{\sigma^2} \right\|_2^2 \right].$$

To obtain a reliable score estimate across the whole space, one uses a geometric progression of noise levels $\sigma_1 > \sigma_2 > \dots > \sigma_L > 0$. The largest σ_1 guarantees a non-zero training signal even in sparse regions, while the smallest σ_L ensures that $q_{\sigma_L} \approx p_{\text{data}}$. Following [17], the multi-scale NCSN objective is a weighted average

$$\mathcal{L}(\theta) = \frac{1}{L} \sum_{i=1}^L \lambda(\sigma_i) \ell(\theta; \sigma_i),$$

with a weighting function $\lambda(\sigma) > 0$. Minimizing $\mathcal{L}$ drives $\mathbf{s}_\theta(\cdot, \sigma_i)$ toward the true score $\nabla \log q_{\sigma_i}$ for each scale.

The choice $\lambda(\sigma) = \sigma^2$ balances the gradient magnitudes across scales. Indeed, near optimality the score magnitude satisfies $\|\mathbf{s}_\theta(\mathbf{x}, \sigma)\|_2 \propto 1/\sigma$. With this weighting,

$$\sigma^2 \ell(\theta; \sigma) = \frac{1}{2} \mathbb{E} \left[\left\| \sigma \mathbf{s}_\theta(\tilde{\mathbf{x}}, \sigma) + \frac{\tilde{\mathbf{x}} - \mathbf{x}}{\sigma} \right\|_2^2 \right],$$

and since $(\tilde{\mathbf{x}} - \mathbf{x})/\sigma \sim \mathcal{N}(\mathbf{0}, I)$, the two terms inside the norm remain of order one regardless of σ . The final NCSN training loss therefore takes the form

$$\mathcal{L}_{\text{NCSN}}(\theta) = \frac{1}{L} \sum_{i=1}^L \sigma_i^2 \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{\tilde{\mathbf{x}} \sim \mathcal{N}(\mathbf{x}, \sigma_i^2 I)} \left[\left\| \mathbf{s}_\theta(\tilde{\mathbf{x}}, \sigma_i) + \frac{\tilde{\mathbf{x}} - \mathbf{x}}{\sigma_i^2} \right\|_2^2 \right].$$

This objective sidesteps both limitations of classical score matching: the divergence $\nabla_{\mathbf{x}} \cdot \mathbf{s}_\theta$ is never computed, and the multi-scale noise injects informative gradients everywhere, even where p_{data} is negligible.

1.3 Empirical Validation of Denoising Score Matching

We now instantiate the NCSN training objective derived in the previous section on a controlled two-dimensional example. The goal is twofold: to empirically validate the equivalence between ESM and DSM, and to illustrate the ability of the learned score network to provide accurate estimates even in low-density regions. The experiment simulates a bimodal data distribution and trains a noise-conditional score network $s_\theta(\mathbf{x}, \sigma)$ to approximate the score $\nabla_{\mathbf{x}} \log q_\sigma(\mathbf{x})$ across a continuous range of noise scales.

A Gaussian mixture of two components in $\mathbb{R}^2$ is used, defined as

$$p_{\text{data}}(\mathbf{x}) = 0.2 \cdot \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_1, \mathbf{I}) + 0.8 \cdot \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_2, \mathbf{I}),$$

with $\boldsymbol{\mu}_1 = (-5, -5)$ and $\boldsymbol{\mu}_2 = (5, 5)$. The two modes are well separated, leaving a low-density valley that challenges score estimation in regions where data are scarce.

A Noise Conditional Score Network (NCSN) was implemented with the following characteristics:

- **Architecture:** 8-layer fully connected network with 128 neurons per hidden layer.
- **Input:** Concatenation of the spatial coordinate $\mathbf{x} \in \mathbb{R}^2$ and the noise level σ .
- **Activation function:** SiLU (Swish), which provides smooth gradients and avoids dead neurons.
- **Output dimension:** $\mathbb{R}^2$, directly mapping to the estimated score vector.
- **Total trainable parameters:** 242,402.

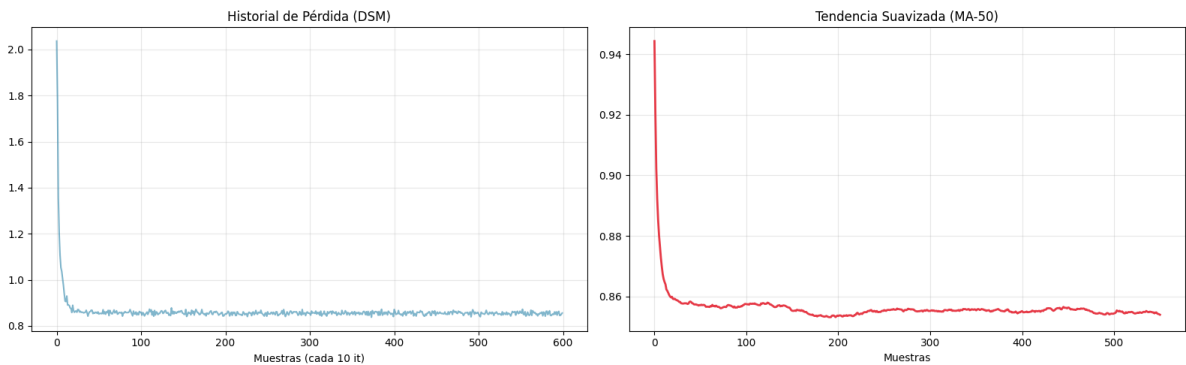


Figure 1.4 Training loss curve for the NCSN model.

Training minimizes the Denoising Score Matching objective with weighting function $\lambda(\sigma) = \sigma^2$. Concretely, the loss is

$$\mathcal{L}(\theta) = \frac{1}{L} \sum_{i=1}^L \sigma_i^2 \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}} \mathbb{E}_{\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})} \left[\left\| \mathbf{s}_\theta(\mathbf{x} + \sigma_i \epsilon, \sigma_i) + \frac{\epsilon}{\sigma_i} \right\|_2^2 \right]. \quad (1.9)$$

where $\tilde{\mathbf{x}} = \mathbf{x} + \sigma \epsilon$ is the corrupted sample. The noise scale σ is sampled uniformly per batch from $[\sigma_{\min}, \sigma_{\max}] = [0.001, 4.0]$.

The remaining hyperparameters were:

- **Iterations:** 6,000.
- **Batch size:** 30,000 samples.
- **Learning rate:** 5×10^{-4} with cosine annealing schedule.
- **Optimizer:** AdamW with weight decay 2×10^{-5} .
- **Gradient clipping:** maximum norm 1.0.

To quantify the quality of the estimated score, the following metrics were computed on a uniform grid covering $[-12, 12] \times [-12, 12]$:

- Mean Squared Error (MSE) between $\mathbf{s}_\theta(\mathbf{x}, \sigma_{\min})$ and the true score $\nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x})$.
- Mean cosine similarity between the estimated and true score directions.
- Mean angular error (in degrees) between the two vector fields.

Figure 1.5 displays the true score field $\nabla_{\mathbf{x}} \log p_{\text{data}}$ (blue) alongside the estimated field $\mathbf{s}_\theta(\cdot, \sigma_{\min})$ (red). The orange heatmap represents the normalized probability density. The two fields are virtually indistinguishable near the modes and remain well aligned even in the low-density valley between them, confirming that the multi-scale denoising objective successfully propagates training signal across the entire domain.

To further validate the learned score, we generate new samples using annealed Langevin dynamics. Starting from pure noise, samples are iteratively refined according to the discretized reverse diffusion update

$$\mathbf{x}_{i+1} = \mathbf{x}_i + \alpha \cdot \mathbf{s}_\theta(\mathbf{x}_i, \sigma_i) + \sqrt{2\alpha} \cdot \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(\mathbf{0}, \mathbf{I}),$$

where α is a step size and the noise levels σ_i follow a decreasing schedule from $\sigma_{\max} = 4.0$ to $\sigma_{\min} = 0.001$.

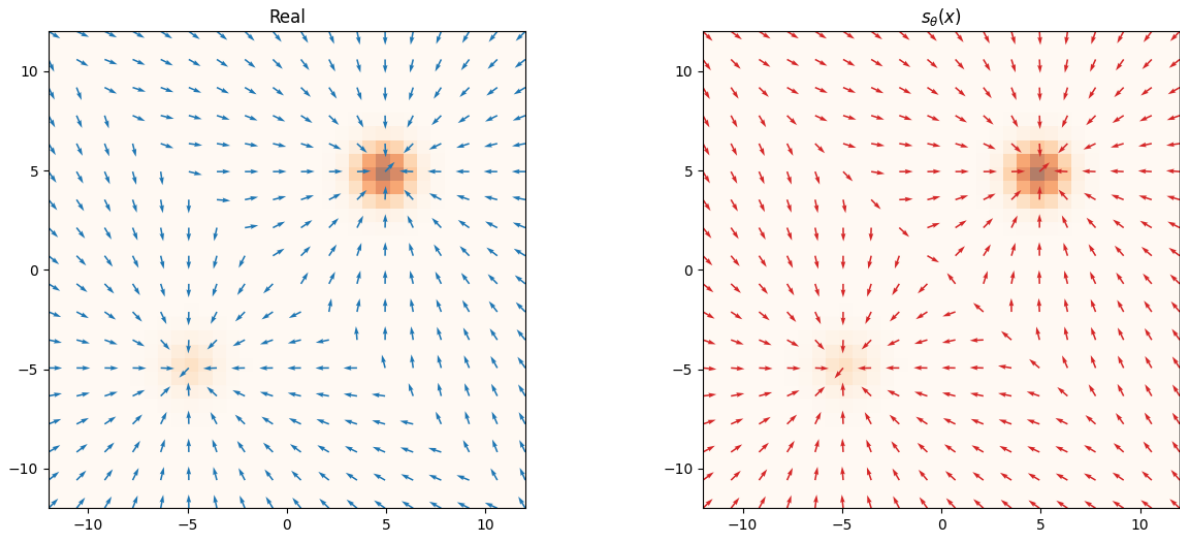


Figure 1.5 Comparison between true score (blue arrows) and estimated score (red arrows) at $\sigma_{\min} = 0.001$. The background heatmap (orange) depicts the normalized density p_{data} .

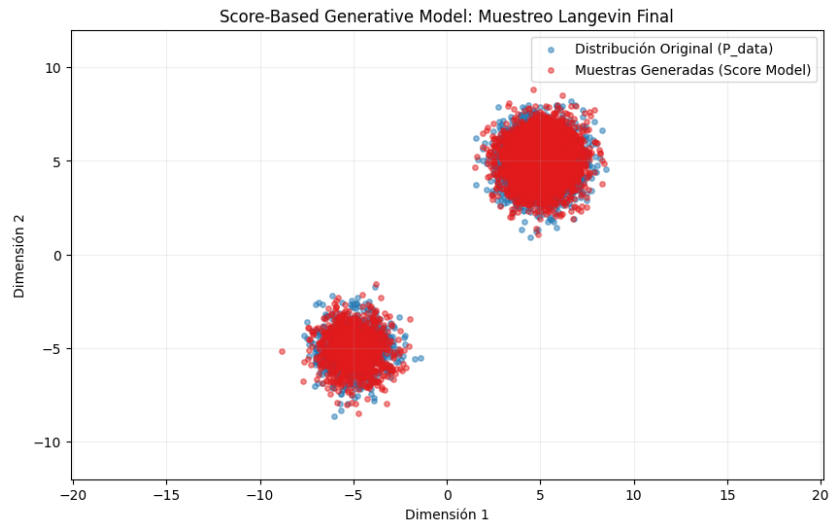


Figure 1.6 Scatter plot of 10,000 generated samples overlaid on the true density contours.

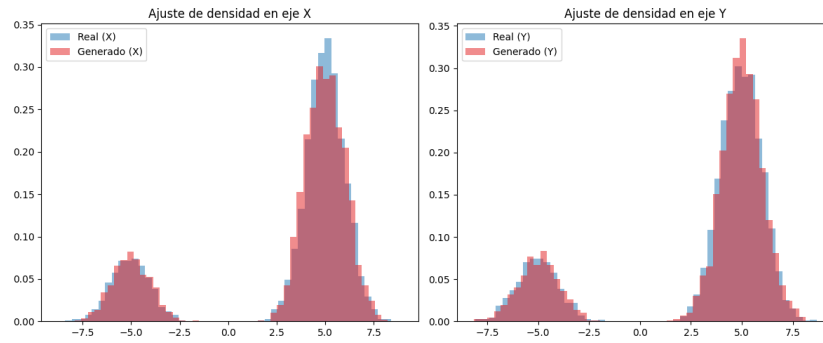


Figure 1.7 Marginal histograms of generated samples (orange) versus true distribution (blue).

The generated samples, shown in Figure 1.6, closely reproduce the bimodal structure of the true distribution. The marginal histograms along each coordinate (Figure 1.7) confirm that the model accurately captures both the positions and relative weights of the two mixture components. This empirical success corroborates the theoretical guarantees established in the preceding sections and demonstrates the practical viability of Denoising Score Matching for density estimation and sampling.

Chapter 2

Score-Based Modeling via Stochastic Differential Equations

The discrete multi-scale formulation presented in Chapter 1, while effective, represents a particular instantiation of a more general mathematical framework. This chapter introduces a fundamental generalization: replacing the finite sequence of noise scales $\{\sigma_i\}_{i=1}^L$ with a *continuous* perturbation process indexed by time $t \in [0, T]$. This process is governed by a **Stochastic Differential Equation** (SDE) that smoothly transforms the data distribution $p_0 \triangleq p_{\text{data}}$ into a tractable prior distribution p_T .

The core theoretical foundation, established by [19], reveals that this diffusion process can be reversed for generative modeling if the time-dependent score function $\nabla_{\mathbf{x}} \log p_t(\mathbf{x})$ is known. We provide a rigorous derivation of the associated **reverse-time SDE** and its deterministic counterpart, the **probability flow ODE**, which emerges from the Fokker–Planck equation governing the diffusion’s marginals. This framework unifies and generalizes previous discrete-noise methods, offering a continuous-time perspective that is both theoretically elegant and practically powerful.

2.1 Stochastic Differential Equations as Continuous Perturbation Processes

The Noise Conditional Score Network (NCSN) is trained on a finite sequence of noise levels $\{\sigma_i\}_{i=1}^L$, effectively approximating the scores $\{\nabla_{\mathbf{x}} \log q_{\sigma_i}(\mathbf{x})\}_{i=1}^L$ of a family of perturbed distributions. This formulation naturally leads to a question: what happens in the limit as the number of noise scales approaches infinity, $L \rightarrow \infty$?

In this continuous limit, the discrete sequence of noise levels $\{\sigma_i\}_{i=1}^L$ is replaced by a continuous-time index $t \in [0, T]$. Consequently, the perturbed data distributions evolve not as a discrete set but as a continuous family $\{p_t(\mathbf{x})\}_{t \in [0, T]}$, where p_t denotes the probability density of the data after being perturbed up to time t .

The perturbation process itself becomes a continuous-time stochastic process $\{\mathbf{X}_t\}_{t \in [0, T]}$, with $\mathbf{X}_0 \sim p_0 \triangleq p_{\text{data}}$ and $\mathbf{X}_T \sim p_T$, where p_T is designed to be a tractable prior distribution (e.g., a standard Gaussian).

A natural and powerful way to represent such a continuous-time stochastic process is through a Stochastic Differential Equation (SDE). In its most general form, an Itô SDE is written as:

$$d\mathbf{X}_t = \mathbf{f}(\mathbf{X}_t, t) dt + G(\mathbf{X}_t, t) d\mathbf{W}_t, \quad \mathbf{X}_0 \sim p_{\text{data}}, \quad (2.1)$$

where:

- $\mathbf{X}_t \in \mathbb{R}^d$ is the state vector at time $t \in [0, T]$.
- $\mathbf{f} : \mathbb{R}^d \times [0, T] \rightarrow \mathbb{R}^d$ is the drift coefficient, an d -dimensional vector-valued function.
- $G : \mathbb{R}^d \times [0, T] \rightarrow \mathbb{R}^{d \times r}$ is the diffusion coefficient, an $d \times r$ matrix-valued function.
- $\{\mathbf{W}_t\}_{t \geq 0}$ is a standard r -dimensional Wiener process (Brownian motion) defined on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{0 \leq t \leq T}, \mathbb{P})$.

The infinitesimal increment $d\mathbf{W}_t$ can be informally interpreted as white noise. For the SDE in (2.1) to be mathematically well-posed, the coefficients $\mathbf{f}$ and G must satisfy certain regularity conditions. For instance, if $\mathbf{f}$ and G are globally Lipschitz in both state and time, the SDE admits a unique strong solution [15]. Weaker conditions, such as boundedness and ellipticity, guarantee the existence of a unique weak solution [12].

The solution to (2.1) is a continuous collection of random variables $\{\mathbf{X}_t\}_{t \in [0, T]}$. For each fixed t , we denote by $p_t(\mathbf{x})$ the marginal probability density of $\mathbf{X}_t$. This density evolves according to the Fokker–Planck equation (or Kolmogorov forward equation), a connection we will explore in detail in appendix B. By construction, we have:

$$p_0(\mathbf{x}) = p_{\text{data}}(\mathbf{x}), \quad p_T(\mathbf{x}) \approx p_{\text{prior}}(\mathbf{x}),$$

where p_{prior} is a simple, tractable distribution (typically a Gaussian) that contains no information about the original data. This continuous-time framework elegantly

generalizes the discrete NCSN formulation: the sequence of perturbed distributions $\{q_{\sigma_i}(\mathbf{x})\}_{i=1}^L$ is replaced by the continuous-time marginals $\{p_t(\mathbf{x})\}_{t \in [0, T]}$, and the score functions $\nabla_{\mathbf{x}} \log q_{\sigma_i}(\mathbf{x})$ become the time-dependent scores $\nabla_{\mathbf{x}} \log p_t(\mathbf{x})$.

2.1.1 From Discrete Noise Scales to Continuous-Time SDEs

The experimental validation presented in Section 1.3 successfully demonstrated the efficacy of Denoising Score Matching (DSM) with a Noise Conditional Score Network (NCSN) trained on a finite set of noise scales $\{\sigma_i\}_{i=1}^L$. However, as argued in the introduction of this chapter, a more fundamental and elegant perspective emerges when we consider the continuous-time limit, where the number of noise scales tends to infinity and the perturbation process is described by a Stochastic Differential Equation (SDE). In our experiment, during training we sampled noise levels σ uniformly from the interval $[\sigma_{\min}, \sigma_{\max}] = [0.001, 4.0]$. This choice implicitly defines a continuous family of perturbed distributions $\{p_{\sigma}(\mathbf{x})\}_{\sigma \in [\sigma_{\min}, \sigma_{\max}]}$, where p_{σ} is the distribution of $\mathbf{x} + \sigma \epsilon$ with $\epsilon \sim \mathcal{N}(0, I)$. To embed this family into an SDE-based framework, we need to identify a time variable t and a forward SDE whose marginal distributions $p_t(\mathbf{x})$ coincide with $p_{\sigma}(\mathbf{x})$ for a suitable mapping $t \mapsto \sigma(t)$.

A natural and simple choice is to set the time variable equal to the noise level, i.e., $t = \sigma$, and consider the forward process defined on the time interval $t \in [0, T]$ with $T = \sigma_{\max}$. We require that at time t , the marginal distribution $p_t(\mathbf{x})$ is exactly the distribution of $\mathbf{x}_0 + t\epsilon$, where $\mathbf{x}_0 \sim p_{\text{data}}$. This corresponds to a perturbation kernel $p_{0t}(\mathbf{x}(t) | \mathbf{x}(0)) = \mathcal{N}(\mathbf{x}(t); \mathbf{x}(0), t^2 I)$. The SDE that generates such marginal distributions is obtained by noting that the variance evolves as $\sigma^2(t) = t^2$. Differentiating, we have $\frac{d}{dt} \sigma^2(t) = 2t$. For an SDE of the form

$$d\mathbf{X}_t = g(t)d\mathbf{W}_t, \quad (2.2)$$

with zero drift, the marginal distribution satisfies $p_t = p_{\text{data}} * \mathcal{N}(0, \int_0^t g(s)^2 ds)$. Therefore, to achieve $\int_0^t g(s)^2 ds = t^2$, we set $g(t)^2 = \frac{d}{dt} t^2 = 2t$, yielding

$$d\mathbf{X}_t = \sqrt{2t} d\mathbf{W}_t, \quad t \in [0, T], \quad \mathbf{X}_0 \sim p_{\text{data}}. \quad (2.3)$$

This SDE is a particular instance of a **Variance Exploding (VE) SDE** with a linear growth of the standard deviation. It serves as the continuous-time analogue of the

discrete noise perturbation used in our experiment: at each time t , the marginal $p_t(\mathbf{x})$ is precisely the distribution obtained by adding Gaussian noise of standard deviation t to the data.

2.1.2 Time Reversal of Diffusion Processes

A central insight in score-based generative modeling is that one can generate samples from p_{data} by learning to reverse a forward noising process. Starting from $\mathbf{X}_T \sim p_T$, an approximately tractable prior, and simulating the time-reversed dynamics back to $t = 0$ yields samples $\mathbf{X}_0 \sim p_0 = p_{\text{data}}$. The theoretical foundation for this procedure is a classical result due to [1], which asserts that the time-reversal of a diffusion process is itself a diffusion process, governed by a *reverse-time SDE* whose drift depends on the score function $\nabla_{\mathbf{x}} \log p_t(\mathbf{x})$.

Setup: The Forward SDE

Let $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{0 \leq t \leq T}, \mathbb{P})$ be a filtered probability space satisfying the usual conditions. Consider the forward Itô SDE

$$d\mathbf{X}_t = \mathbf{f}(\mathbf{X}_t, t) dt + G(\mathbf{X}_t, t) d\mathbf{W}_t, \quad \mathbf{X}_0 \sim p_{\text{data}}, \quad (2.4)$$

where $\mathbf{X}_t \in \mathbb{R}^d$, $\mathbf{f} : \mathbb{R}^d \times [0, T] \rightarrow \mathbb{R}^d$ is the drift coefficient, $G : \mathbb{R}^d \times [0, T] \rightarrow \mathbb{R}^{d \times r}$ is the diffusion coefficient, and $\{\mathbf{W}_t\}_{t \geq 0}$ is an r -dimensional standard Wiener process adapted to $\{\mathcal{F}_t\}$. We denote the marginal density of $\mathbf{X}_t$ by p_t .

For (2.1) to admit a unique strong solution and for the reverse-time representation to be valid, we impose the following standard regularity conditions on the coefficients:

- **Measurability:** $\mathbf{f}(\mathbf{x}, t)$ and $G(\mathbf{x}, t)$ are jointly measurable and adapted to $\{\mathcal{F}_t\}$.
- **Global Lipschitz condition:** There exists $K > 0$ such that for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$ and $t \in [0, T]$:

$$\|\mathbf{f}(\mathbf{x}, t) - \mathbf{f}(\mathbf{y}, t)\| + \|G(\mathbf{x}, t) - G(\mathbf{y}, t)\| \leq K\|\mathbf{x} - \mathbf{y}\|.$$

- **Linear growth:** There exists $C > 0$ such that for all $\mathbf{x} \in \mathbb{R}^d$ and $t \in [0, T]$:

$$\|\mathbf{f}(\mathbf{x}, t)\|^2 + \|G(\mathbf{x}, t)\|^2 \leq C(1 + \|\mathbf{x}\|^2).$$

- **Smoothness:** The coefficients are continuously differentiable in $\mathbf{x}$ and t , ensuring that the divergence $\nabla \cdot (GG^\top)$ is well defined and that the marginal densities satisfy $p_t \in C^{2,1}(\mathbb{R}^d \times (0, T))$.

Under these conditions, the SDE (2.1) has a unique strong solution, and by Itô's formula the family $\{p_t\}$ satisfies the associated Fokker–Planck equation (see Appendix B).

Anderson's Theorem: The General Reverse-Time SDE

Define the time-reversed process $\mathbf{Y}_t := \mathbf{X}_{T-t}$ for $t \in [0, T]$, so that $\mathbf{Y}_0 \sim p_T$ and $\mathbf{Y}_T \sim p_{\text{data}}$. Anderson's theorem (proved in Appendix A) establishes that $\{\mathbf{Y}_t\}$ is itself a diffusion process satisfying the reverse-time SDE

$$d\mathbf{Y}_t = \tilde{\mathbf{f}}(\mathbf{Y}_t, t) dt + G(\mathbf{Y}_t, T-t) d\tilde{\mathbf{W}}_t, \quad \mathbf{Y}_0 \sim p_T, \quad (2.5)$$

where $\{\tilde{\mathbf{W}}_t\}$ is an r -dimensional Wiener process with respect to the *reverse* filtration $\{\tilde{\mathcal{F}}_t\}$, which is a decreasing family of sub- σ -algebras of $\mathcal{F}$ such that $\tilde{\mathbf{W}}_t$ is $\tilde{\mathcal{F}}_t$ -measurable and $\tilde{\mathbf{W}}_t - \tilde{\mathbf{W}}_s$ is independent of $\tilde{\mathcal{F}}_t$ for all $t \geq s$. In particular:

$$\mathbb{E}[\tilde{\mathbf{W}}_t | \tilde{\mathcal{F}}_{t+s}] = \tilde{\mathbf{W}}_{t+s}, \quad (2.6)$$

$$\mathbb{E}[(\tilde{\mathbf{W}}_t - \tilde{\mathbf{W}}_{t+s})(\tilde{\mathbf{W}}_t - \tilde{\mathbf{W}}_{t+s})^\top | \tilde{\mathcal{F}}_{t+s}] = s I, \quad (2.7)$$

for all $s \geq 0$. The reverse drift is given by

$$\tilde{\mathbf{f}}(\mathbf{x}, t) = -\mathbf{f}(\mathbf{x}, T-t) + \frac{1}{p_{T-t}(\mathbf{x})} \nabla \cdot \left[p_{T-t}(\mathbf{x}) G(\mathbf{x}, T-t) G(\mathbf{x}, T-t)^\top \right], \quad (2.8)$$

where the divergence acts on the first argument of the matrix field $G(\cdot, T-t) G(\cdot, T-t)^\top$.

The original derivation of this formula by [1] was heuristic and based on physical arguments. A fully rigorous probabilistic proof was given by [5], who established the result under Lipschitz and linear growth conditions on the coefficients together with mild regularity of the marginal densities. Their work forms the mathematical backbone of the time-reversal theory for diffusions and is the basis of the proof presented in Appendix A.

Recently, [10] studied the associated Fokker–Planck equation with terminal condition, namely equation (1.1) in that reference,

$$\partial_t \mathbf{u} = L_t^* \mathbf{u}, \quad \mathbf{u}(T) = \mu,$$

which is precisely the evolution of the time-reversed marginals p_{T-t} . They provided sufficient conditions for uniqueness of the solution and established a probabilistic representation via a McKean–Vlasov SDE, which is exactly the reverse-time SDE (2.5) when the score is expressed in terms of the density. These results give a rigorous foundation to the well-posedness of the reverse dynamics and justify the use of score-based generative models even when the terminal distribution p_T is a fixed (e.g., Gaussian) prior, as the mapping from the initial data distribution to the terminal one is injective under natural assumptions (e.g., Ornstein–Uhlenbeck dynamics).

To simplify the notation and make the connection with the score function explicit, we define the *diffusion matrix*

$$\mathbf{D}(\mathbf{x}, t) = G(\mathbf{x}, t) G(\mathbf{x}, t)^\top.$$

(When the diffusion coefficient depends only on time, as in the affine case of Section 2.2.1, this matrix reduces to a time-dependent matrix denoted by $\mathbf{A}(t)$.) Expanding (2.8) using the product rule yields

$$\tilde{\mathbf{f}}(\mathbf{x}, t) = -\mathbf{f}(\mathbf{x}, T-t) + \mathbf{D}(\mathbf{x}, T-t) \nabla_{\mathbf{x}} \log p_{T-t}(\mathbf{x}) + \nabla \cdot \mathbf{D}(\mathbf{x}, T-t), \quad (2.9)$$

which makes explicit the appearance of the *score function* $\nabla_{\mathbf{x}} \log p_t$. The score carries all the information about the data distribution needed to reverse the diffusion.

Change of Variables: The Song et al. Formulation

The formulation above uses the parameterization $s = T - t$, so the reverse-time variable t runs forward from 0 to T while the forward-time variable runs backward. In their seminal work, [19] reparameterize by *relabeling* the reverse-time variable, so that both the forward and reverse SDEs are written in terms of a single time index $t \in [0, T]$ that increases in the direction of generation, that is, from noise to data. Concretely, setting $\mathbf{x}(t) := \mathbf{Y}_{T-t} = \mathbf{X}_t$ in distribution and replacing $t \mapsto T - t$ throughout (2.5) yields the reverse-time SDE (equation (6) of [19]):

$$d\mathbf{X}_t = \left[\mathbf{f}(\mathbf{X}_t, t) - \mathbf{D}(\mathbf{X}_t, t) \nabla_{\mathbf{x}} \log p_t(\mathbf{X}_t) - \nabla_{\mathbf{x}} \mathbf{D}(\mathbf{X}_t, t) \right] dt + G(\mathbf{X}_t, t) d\bar{\mathbf{W}}_t \quad (2.10)$$

where $\bar{\mathbf{W}}_t$ is a standard Wiener process running backward in time.

Remark 2.1.1 (Affine drift and scalar diffusion). *The most widely used instantiations of (2.1) in practice adopt an affine drift $\mathbf{f}(\mathbf{x}, t) = f(t) \mathbf{x}$ and a scalar (state-independent) diffusion*

matrix $G(\mathbf{x}, t) = g(t) I_d$, where $f, g : [0, T] \rightarrow \mathbb{R}$ are deterministic scalar functions. In this case the diffusion matrix becomes $\mathbf{D}(\mathbf{x}, t) = g(t)^2 I_d$, and its divergence vanishes identically:

$$\nabla \cdot \mathbf{D}(\mathbf{x}, t) = \nabla \cdot [g(t)^2 I_d] = \mathbf{0}.$$

Consequently, (2.10) reduces to the elegant form

$$d\mathbf{X}_t = \left[f(t) \mathbf{X}_t - g(t)^2 \nabla_{\mathbf{x}} \log p_t(\mathbf{X}_t) \right] dt + g(t) d\bar{\mathbf{W}}_t. \quad (2.11)$$

This is the operative equation in the VP-SDE and VE-SDE families of [19]. The entire task of learning to reverse the diffusion reduces to estimating the score $\nabla_{\mathbf{x}} \log p_t(\mathbf{x})$ via a neural network $\mathbf{s}_{\theta}(\mathbf{x}, t) \approx \nabla_{\mathbf{x}} \log p_t(\mathbf{x})$, which is trained by denoising score matching.

2.2 Continuous-Time Score Matching

The reverse-time SDE (2.5) offers a path to generate samples, but it requires knowledge of the time-dependent score function $\nabla_{\mathbf{x}} \log p_t(\mathbf{x})$ a quantity we do not have access to a priori. In practice, we approximate it by a parametric model $\mathbf{s}_{\theta}(\mathbf{x}, t)$, typically a neural network, which we train on data. The central question is how to construct an objective function that can be minimized without ever knowing the true score. In the discrete-time setting of Chapter 1, this was accomplished through Denoising Score Matching (DSM). We now extend this idea to the continuous-time framework, building upon the SDE formalism developed in the preceding sections.

Recall that the forward SDE (2.1) defines not only the marginal distributions $p_t(\mathbf{x})$ but also a family of conditional distributions $p_{0,t}(\mathbf{x}_t | \mathbf{x}_0)$, known as the transition kernel. These conditional distributions describe how a particular initial point $\mathbf{x}_0$ evolves under the dynamics of the SDE. For the DSM objective, we need two things: the ability to sample from this kernel, and an expression for its conditional score $\nabla_{\mathbf{x}_t} \log p_{0,t}(\mathbf{x}_t | \mathbf{x}_0)$.

The transition kernel satisfies the Fokker–Planck–Kolmogorov (FPK) equation, which governs its evolution in time. To study this, let us fix an initial condition $\mathbf{X}_s = \mathbf{y}$ at time s and denote by $p_{s,y}(\mathbf{x}, t) = p(\mathbf{x}, t | \mathbf{y}, s)$ the conditional density of $\mathbf{X}_t$ for $t \geq s$.

For fixed $(\mathbf{y}, s)$, the function $(\mathbf{x}, t) \mapsto p_{s,y}(\mathbf{x}, t)$ solves the forward Kolmogorov equation

$$\frac{\partial p_{s,y}}{\partial t}(\mathbf{x}, t) = -\nabla_{\mathbf{x}} \cdot (\mathbf{f}(\mathbf{x}, t) p_{s,y}(\mathbf{x}, t)) + \frac{1}{2} \nabla_{\mathbf{x}} \cdot \left(\nabla_{\mathbf{x}} \cdot (\mathbf{D}(\mathbf{x}, t) p_{s,y}(\mathbf{x}, t)) \right), \quad (2.12)$$

with initial condition $p_{s,y}(\mathbf{x}, s) = \delta(\mathbf{x} - \mathbf{y})$. Here we have used the diffusion matrix $\mathbf{D}(\mathbf{x}, t) = G(\mathbf{x}, t)G(\mathbf{x}, t)^\top$.

In full generality, this equation does not admit a closed-form solution. However, the class of SDEs most commonly employed in score-based generative modeling possesses a special structure that makes the transition kernel analytically tractable: the drift is affine in the state, and the diffusion coefficient does not depend on the state. Under these conditions, the transition kernel becomes Gaussian, as we shall now demonstrate in detail.

2.2.1 The Affine, State-Independent Case

Let us assume that the drift is affine in $\mathbf{x}$ and that the diffusion coefficient is independent of the state:

$$\mathbf{f}(\mathbf{x}, t) = \mathbf{F}(t)\mathbf{x} + \mathbf{c}(t), \quad G(\mathbf{x}, t) = G(t), \quad (2.13)$$

where $\mathbf{F}(t) \in \mathbb{R}^{d \times d}$, $\mathbf{c}(t) \in \mathbb{R}^d$, and $G(t) \in \mathbb{R}^{d \times r}$. The diffusion matrix then simplifies to a time-dependent matrix that we denote by $\mathbf{A}(t) = G(t)G(t)^\top$. Substituting these expressions into (2.12) yields the simplified Fokker–Planck equation

$$\frac{\partial p_{s,y}}{\partial t} = -\nabla_{\mathbf{x}} \cdot ((\mathbf{F}(t)\mathbf{x} + \mathbf{c}(t))p_{s,y}) + \frac{1}{2} \sum_{i,j=1}^d A_{ij}(t) \frac{\partial^2 p_{s,y}}{\partial x_i \partial x_j}, \quad p_{s,y}(\mathbf{x}, s) = \delta(\mathbf{x} - \mathbf{y}). \quad (2.14)$$

The key feature of this equation is that the coefficients are at most linear in $\mathbf{x}$, which suggests that the solution might remain within the family of Gaussian distributions. To confirm this intuition, we turn to the method of characteristic functions.

A powerful technique for solving linear PDEs with constant or linear coefficients is to work in the Fourier domain. Define the conditional characteristic function

$$\phi_{s,y}(\mathbf{u}, t) = \int_{\mathbb{R}^d} e^{i\mathbf{u} \cdot \mathbf{x}} p_{s,y}(\mathbf{x}, t) d\mathbf{x}, \quad \mathbf{u} \in \mathbb{R}^d. \quad (2.15)$$

This function contains all the information about the distribution; if we can determine $\phi_{s,y}$, we can recover the density by Fourier inversion.

Multiplying (2.14) by $e^{i\mathbf{u} \cdot \mathbf{x}}$ and integrating over $\mathbb{R}^d$ transforms the PDE into an equation for $\phi_{s,y}$. Assume that $p_{s,y}(\cdot, t)$ satisfies the pointwise decay condition

$$\lim_{|\mathbf{x}| \rightarrow \infty} |\mathbf{x}|^d p_{s,y}(\mathbf{x}, t) = 0. \quad (2.16)$$

Fix $R > 0$ and apply the divergence theorem on $B_R = \{\mathbf{x} : |\mathbf{x}| \leq R\}$ with $\psi = e^{i\mathbf{u} \cdot \mathbf{x}}$ and $\mathbf{v} = (\mathbf{F}\mathbf{x} + \mathbf{c}) p_{s,y}$:

$$-\int_{B_R} e^{i\mathbf{u} \cdot \mathbf{x}} \nabla_{\mathbf{x}} \cdot \mathbf{v} d\mathbf{x} = \int_{B_R} \nabla_{\mathbf{x}}(e^{i\mathbf{u} \cdot \mathbf{x}}) \cdot \mathbf{v} d\mathbf{x} - \oint_{\partial B_R} e^{i\mathbf{u} \cdot \mathbf{x}} \mathbf{v} \cdot \mathbf{n} dS.$$

Since $\nabla_{\mathbf{x}}(e^{i\mathbf{u} \cdot \mathbf{x}}) = i\mathbf{u} e^{i\mathbf{u} \cdot \mathbf{x}}$, the volume term equals $i\mathbf{u} \cdot \int_{B_R} e^{i\mathbf{u} \cdot \mathbf{x}} (\mathbf{F}\mathbf{x} + \mathbf{c}) p_{s,y} d\mathbf{x}$. For the boundary term, using $|e^{i\mathbf{u} \cdot \mathbf{x}}| = 1$ and the affine bound $|(\mathbf{F}\mathbf{x} + \mathbf{c})| \leq C(|\mathbf{x}| + 1)$ for some constant $C > 0$:

$$\left| \oint_{\partial B_R} e^{i\mathbf{u} \cdot \mathbf{x}} \mathbf{v} \cdot \mathbf{n} dS \right| \leq \oint_{\partial B_R} |(\mathbf{F}\mathbf{x} + \mathbf{c})| p_{s,y} dS \leq C(R+1) \omega_d R^{d-1} \sup_{|\mathbf{x}|=R} p_{s,y}(\mathbf{x}, t),$$

where $\omega_d R^{d-1}$ is the surface area of ∂B_R . Since $C(R+1) = O(R)$,

$$\left| \oint_{\partial B_R} e^{i\mathbf{u} \cdot \mathbf{x}} \mathbf{v} \cdot \mathbf{n} dS \right| \leq C' R^d \sup_{|\mathbf{x}|=R} p_{s,y}(\mathbf{x}, t) \xrightarrow{R \rightarrow \infty} 0,$$

by hypothesis (2.16). Taking $R \rightarrow \infty$ and applying the monotone convergence theorem to the volume integrals yields

$$-\int_{\mathbb{R}^d} e^{i\mathbf{u} \cdot \mathbf{x}} \nabla_{\mathbf{x}} \cdot ((\mathbf{F}\mathbf{x} + \mathbf{c}) p_{s,y}) d\mathbf{x} = i\mathbf{u} \cdot \int_{\mathbb{R}^d} e^{i\mathbf{u} \cdot \mathbf{x}} (\mathbf{F}\mathbf{x} + \mathbf{c}) p_{s,y} d\mathbf{x}.$$

The integral involving $\mathbf{x}$ can be expressed in terms of $\phi_{s,y}$ by differentiating under the integral sign:

$$\int e^{i\mathbf{u} \cdot \mathbf{x}} \mathbf{x} p_{s,y} d\mathbf{x} = -i \nabla_{\mathbf{u}} \phi_{s,y}.$$

Thus,

$$i\mathbf{u} \cdot \int e^{i\mathbf{u} \cdot \mathbf{x}} (\mathbf{F}\mathbf{x} + \mathbf{c}) p_{s,y} d\mathbf{x} = i\mathbf{u} \cdot (\mathbf{F}(-i \nabla_{\mathbf{u}} \phi_{s,y}) + \mathbf{c} \phi_{s,y}) = \mathbf{u} \cdot \mathbf{F} \nabla_{\mathbf{u}} \phi_{s,y} + i\mathbf{u} \cdot \mathbf{c} \phi_{s,y}.$$

For the diffusion term, we use the fact that the Fourier transform of a second derivative brings down factors of $i u_i$:

$$\frac{1}{2} \sum_{i,j} A_{ij} \int e^{i\mathbf{u} \cdot \mathbf{x}} \frac{\partial^2 p_{s,y}}{\partial x_i \partial x_j} d\mathbf{x} = \frac{1}{2} \sum_{i,j} A_{ij} (i u_i)(i u_j) \phi_{s,y} = -\frac{1}{2} \mathbf{u}^\top \mathbf{A} \mathbf{u} \phi_{s,y}.$$

Putting everything together, we obtain a linear first-order PDE for the characteristic function:

$$\frac{\partial \phi_{s,y}}{\partial t} = \mathbf{u} \cdot (\mathbf{F}(t) \nabla_{\mathbf{u}} \phi_{s,y}) + i \mathbf{u} \cdot \mathbf{c}(t) \phi_{s,y} - \frac{1}{2} \mathbf{u}^\top \mathbf{A}(t) \mathbf{u} \phi_{s,y}, \quad \phi_{s,y}(\mathbf{u}, s) = e^{i \mathbf{u} \cdot \mathbf{y}}. \quad (2.17)$$

Notice that this equation is linear in $\phi_{s,y}$ and involves only first-order derivatives in $\mathbf{u}$. This structure is what makes the problem solvable.

Equation (2.17) can be tackled by the method of characteristics. Given the form of the initial condition, a pure exponential linear in $\mathbf{u}$, it is natural to seek a solution that remains an exponential of a quadratic function in $\mathbf{u}$. Specifically, we try an ansatz of the form

$$\phi_{s,y}(\mathbf{u}, t) = \exp \left(i \mathbf{u} \cdot \mathbf{m}(t) - \frac{1}{2} \mathbf{u}^\top \Sigma(t) \mathbf{u} \right), \quad (2.18)$$

where $\mathbf{m}(t) \in \mathbb{R}^d$ and $\Sigma(t) \in \mathbb{R}^{d \times d}$ are functions to be determined, with $\Sigma(t)$ symmetric and positive semidefinite. This ansatz is not arbitrary: the initial condition corresponds to $\mathbf{m}(s) = \mathbf{y}$ and $\Sigma(s) = 0$, and the linearity of the PDE suggests that this exponential-quadratic form will be preserved.

Substituting (2.18) into (2.17) and dividing by the non-zero factor $\phi_{s,y}$ yields

$$i \mathbf{u} \cdot \mathbf{m}'(t) - \frac{1}{2} \mathbf{u}^\top \Sigma'(t) \mathbf{u} = \mathbf{u} \cdot \mathbf{F}(t) (i \mathbf{m}(t) - \Sigma(t) \mathbf{u}) + i \mathbf{u} \cdot \mathbf{c}(t) - \frac{1}{2} \mathbf{u}^\top \mathbf{A}(t) \mathbf{u}. \quad (2.19)$$

Let us expand the right-hand side and organize terms according to their degree in $\mathbf{u}$. The term $\mathbf{u} \cdot \mathbf{F}(t) (i \mathbf{m}(t))$ contributes a linear term $i \mathbf{u} \cdot \mathbf{F}(t) \mathbf{m}(t)$. The term $-\mathbf{u} \cdot \mathbf{F}(t) \Sigma(t) \mathbf{u}$ is quadratic; to express it in a symmetric form suitable for comparison, we note that for any matrix M , $\mathbf{u}^\top M \mathbf{u} = \frac{1}{2} \mathbf{u}^\top (M + M^\top) \mathbf{u}$. Thus,

$$-\mathbf{u} \cdot \mathbf{F}(t) \Sigma(t) \mathbf{u} = -\frac{1}{2} \mathbf{u}^\top (\mathbf{F}(t) \Sigma(t) + \Sigma(t) \mathbf{F}(t)^\top) \mathbf{u}.$$

Now, collecting all terms that are linear in $\mathbf{u}$ on both sides of (2.19), we obtain

$$i \mathbf{u} \cdot \mathbf{m}'(t) = i \mathbf{u} \cdot \mathbf{F}(t) \mathbf{m}(t) + i \mathbf{u} \cdot \mathbf{c}(t).$$

Since this must hold for every $\mathbf{u}$, the vectors multiplying $i \mathbf{u}$ must coincide, giving the ordinary differential equation for the mean:

$$\mathbf{m}'(t) = \mathbf{F}(t) \mathbf{m}(t) + \mathbf{c}(t), \quad \mathbf{m}(s) = \mathbf{y}. \quad (2.20)$$

Collecting the quadratic terms, we have

$$-\frac{1}{2}\mathbf{u}^\top \Sigma'(t)\mathbf{u} = -\frac{1}{2}\mathbf{u}^\top (\mathbf{F}(t)\Sigma(t) + \Sigma(t)\mathbf{F}(t)^\top)\mathbf{u} - \frac{1}{2}\mathbf{u}^\top \mathbf{A}(t)\mathbf{u}.$$

Cancelling the common factor $-\frac{1}{2}$ and noting that the identity must hold for all $\mathbf{u}$ leads to the matrix differential equation for the covariance:

$$\Sigma'(t) = \mathbf{F}(t)\Sigma(t) + \Sigma(t)\mathbf{F}(t)^\top + \mathbf{A}(t), \quad \Sigma(s) = 0. \quad (2.21)$$

Thus, the problem of solving the PDE (2.17) reduces to solving two ordinary differential equations, a significant simplification.

Equations (2.20) and (2.21) are linear and can be solved explicitly using the fundamental matrix associated with $\mathbf{F}(t)$. Define $\Phi(t, s)$ as the solution to

$$\frac{d}{dt}\Phi(t, s) = \mathbf{F}(t)\Phi(t, s), \quad \Phi(s, s) = I. \quad (2.22)$$

This matrix has the property that it maps initial conditions forward in time: for the homogeneous equation $\dot{\mathbf{z}} = \mathbf{F}(t)\mathbf{z}$, we have $\mathbf{z}(t) = \Phi(t, s)\mathbf{z}(s)$. Moreover, $\Phi(t, s)$ is invertible and satisfies the semi-group property $\Phi(t, s) = \Phi(t, \tau)\Phi(\tau, s)$ for any $s \leq \tau \leq t$.

Solution for the mean. The inhomogeneous linear ODE (2.20) can be solved by the method of variation of parameters, yielding

$$\mathbf{m}(t) = \Phi(t, s)\mathbf{y} + \int_s^t \Phi(t, \tau)\mathbf{c}(\tau) d\tau. \quad (2.23)$$

One verifies this by differentiating and using the definition of Φ .

Solution for the covariance. Equation (2.21) resembles a Lyapunov differential equation. Its solution can be expressed as

$$\Sigma(t) = \int_s^t \Phi(t, \tau) \mathbf{A}(\tau) \Phi(t, \tau)^\top d\tau. \quad (2.24)$$

To confirm, differentiate the right-hand side using Leibniz's rule:

$$\frac{d}{dt}\Sigma(t) = \Phi(t, t)\mathbf{A}(t)\Phi(t, t)^\top + \int_s^t \left[\frac{d}{dt}\Phi(t, \tau)\mathbf{A}(\tau)\Phi(t, \tau)^\top + \Phi(t, \tau)\mathbf{A}(\tau)\frac{d}{dt}\Phi(t, \tau)^\top \right] d\tau.$$

Since $\Phi(t, t) = I$ and $\frac{d}{dt}\Phi(t, \tau) = \mathbf{F}(t)\Phi(t, \tau)$, we obtain

$$\begin{aligned} \frac{d}{dt}\Sigma(t) &= \mathbf{A}(t) + \int_s^t \left[\mathbf{F}(t)\Phi(t, \tau)\mathbf{A}(\tau)\Phi(t, \tau)^\top + \Phi(t, \tau)\mathbf{A}(\tau)\Phi(t, \tau)^\top \mathbf{F}(t)^\top \right] d\tau \\ &= \mathbf{A}(t) + \mathbf{F}(t)\Sigma(t) + \Sigma(t)\mathbf{F}(t)^\top, \end{aligned}$$

and $\Sigma(s) = 0$ follows from the integral having zero range. Thus (2.24) satisfies both the differential equation and the initial condition.

With $\mathbf{m}(t)$ and $\Sigma(t)$ determined, the characteristic function is given by (2.18). To recover the probability density, we apply the inverse Fourier transform:

$$p_{s,y}(\mathbf{x}, t) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} e^{-i\mathbf{u} \cdot \mathbf{x}} \phi_{s,y}(\mathbf{u}, t) d\mathbf{u}. \quad (2.25)$$

Substituting the expression for $\phi_{s,y}$ yields

$$p_{s,y}(\mathbf{x}, t) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \exp\left(-i\mathbf{u} \cdot (\mathbf{x} - \mathbf{m}(t)) - \frac{1}{2}\mathbf{u}^\top \Sigma(t)\mathbf{u}\right) d\mathbf{u}.$$

This integral is a classic Gaussian integral. To evaluate it, we complete the square in the exponent with respect to $\mathbf{u}$. We seek a vector $\mathbf{v}$ such that

$$-i\mathbf{u} \cdot (\mathbf{x} - \mathbf{m}(t)) - \frac{1}{2}\mathbf{u}^\top \Sigma(t)\mathbf{u} = -\frac{1}{2}(\mathbf{u} - \mathbf{v})^\top \Sigma(t)(\mathbf{u} - \mathbf{v}) + \frac{1}{2}\mathbf{v}^\top \Sigma(t)\mathbf{v}.$$

Expanding the right-hand side and matching the linear term in $\mathbf{u}$ gives $\Sigma(t)\mathbf{v} = -i(\mathbf{x} - \mathbf{m}(t))$, hence $\mathbf{v} = -i\Sigma(t)^{-1}(\mathbf{x} - \mathbf{m}(t))$. Substituting back and computing the residual term,

$$\frac{1}{2}\mathbf{v}^\top \Sigma(t)\mathbf{v} = \frac{1}{2}(-i)^2(\mathbf{x} - \mathbf{m}(t))^\top \Sigma(t)^{-1}(\mathbf{x} - \mathbf{m}(t)) = -\frac{1}{2}(\mathbf{x} - \mathbf{m}(t))^\top \Sigma(t)^{-1}(\mathbf{x} - \mathbf{m}(t)),$$

we obtain the identity

$$\begin{aligned} -i\mathbf{u} \cdot (\mathbf{x} - \mathbf{m}(t)) - \frac{1}{2}\mathbf{u}^\top \Sigma(t)\mathbf{u} &= -\frac{1}{2}\left(\mathbf{u} - i\Sigma(t)^{-1}(\mathbf{x} - \mathbf{m}(t))\right)^\top \Sigma(t)\left(\mathbf{u} - i\Sigma(t)^{-1}(\mathbf{x} - \mathbf{m}(t))\right) \\ &\quad - \frac{1}{2}(\mathbf{x} - \mathbf{m}(t))^\top \Sigma(t)^{-1}(\mathbf{x} - \mathbf{m}(t)). \end{aligned} \quad (2.26)$$

Substituting into the inverse Fourier integral and factoring out the term independent of $\mathbf{u}$,

$$p_{s,\mathbf{y}}(\mathbf{x}, t) = \frac{1}{(2\pi)^d} e^{-\frac{1}{2}(\mathbf{x} - \mathbf{m}(t))^\top \Sigma(t)^{-1}(\mathbf{x} - \mathbf{m}(t))} \int_{\mathbb{R}^d} e^{-\frac{1}{2}(\mathbf{u} - \mathbf{v})^\top \Sigma(t)(\mathbf{u} - \mathbf{v})} d\mathbf{u}.$$

The remaining integral is a standard Gaussian integral. Although $\mathbf{v}$ is complex, the result follows by analytic continuation of the standard real Gaussian formula:

$$\int_{\mathbb{R}^d} e^{-\frac{1}{2}(\mathbf{u} - \mathbf{v})^\top \Sigma(t)(\mathbf{u} - \mathbf{v})} d\mathbf{u} = (2\pi)^{d/2} |\Sigma(t)|^{1/2}.$$

Dividing by $(2\pi)^d$ yields

$$p_{s,\mathbf{y}}(\mathbf{x}, t) = \frac{1}{(2\pi)^{d/2} |\Sigma(t)|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \mathbf{m}(t))^\top \Sigma(t)^{-1}(\mathbf{x} - \mathbf{m}(t))\right). \quad (2.27)$$

Thus, under the assumptions (2.13), the transition kernel is Gaussian with mean $\mathbf{m}(t)$ and covariance $\Sigma(t)$ given by (2.23) and (2.24). In particular, for the kernel from time 0 to t , we have the identification $p_{0,\mathbf{x}_0}(\mathbf{x}, t) = p_{0,t}(\mathbf{x} \mid \mathbf{x}_0)$.

From the explicit Gaussian density, the conditional score follows immediately by differentiation. Taking $\nabla_{\mathbf{x}} \log$ of (2.27), the normalization constant drops out and only the quadratic term in the exponent survives:

$$\nabla_{\mathbf{x}} \log p_{0,t}(\mathbf{x} \mid \mathbf{x}_0) = -\Sigma(t)^{-1}(\mathbf{x} - \mathbf{m}(t)), \quad (2.28)$$

where we have set $s = 0$ and $\mathbf{y} = \mathbf{x}_0$, and $\mathbf{m}(t) = \mathbf{m}(t; \mathbf{x}_0)$ denotes the solution of (2.20) with initial condition $\mathbf{m}(0) = \mathbf{x}_0$. This remarkably simple linear expression is the cornerstone of the denoising score matching objective: it provides a concrete target that depends only on the clean sample $\mathbf{x}_0$ and the known functions $\mathbf{F}(t)$, $\mathbf{c}(t)$, $\mathbf{A}(t)$, without requiring any knowledge of p_{data} .

2.2.2 The Continuous-Time Denoising Score Matching Objective

We are now in a position to formulate the training objective. Let the forward SDE be of the affine type (2.13). For simplicity, we set the initial time to $s = 0$ and denote the clean sample by $\mathbf{x}_0$. Write $\mathbf{m}_t(\mathbf{x}_0)$ for the conditional mean at time t (given $\mathbf{X}_0 = \mathbf{x}_0$) and Σ_t for the conditional covariance; in the affine case, Σ_t does not depend on $\mathbf{x}_0$. The conditional score is then given by (2.28).

The continuous-time denoising score matching (DSM) objective averages a squared error over time, data samples, and realizations of the transition kernel:

$$\mathcal{L}_{\text{DSM}}(\boldsymbol{\theta}) = \mathbb{E}_{t \sim \mathcal{U}(0,T)} \left[\lambda(t) \mathbb{E}_{\mathbf{x}_0 \sim p_{\text{data}}} \mathbb{E}_{\mathbf{x}_t \sim p_{0,t}(\cdot|\mathbf{x}_0)} \left\| \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}_t, t) + \Sigma_t^{-1}(\mathbf{x}_t - \mathbf{m}_t(\mathbf{x}_0)) \right\|^2 \right], \quad (2.29)$$

where $\lambda(t) > 0$ is a positive weighting function that can be chosen to improve training dynamics. A natural and theoretically motivated choice is $\lambda(t) = \text{tr}(\Sigma_t)$ or, in the isotropic case $\Sigma_t = \sigma_t^2 I$, simply $\lambda(t) = \sigma_t^2$. This weighting balances the contribution of different noise levels, since the magnitude of the conditional score scales as $1/\sigma_t$.

To turn (2.29) into a computationally tractable form, we apply the reparameterization trick. Since $\mathbf{x}_t | \mathbf{x}_0 \sim \mathcal{N}(\mathbf{m}_t(\mathbf{x}_0), \Sigma_t)$, we can write $\mathbf{x}_t = \mathbf{m}_t(\mathbf{x}_0) + \Sigma_t^{1/2} \boldsymbol{\epsilon}$ with $\boldsymbol{\epsilon} \sim \mathcal{N}(0, I)$, thereby replacing the expectation over $\mathbf{x}_t$ by an expectation over $\boldsymbol{\epsilon}$. In the isotropic case $\Sigma_t = \sigma_t^2 I$, we have $\Sigma_t^{-1} = \sigma_t^{-2} I$ and $\Sigma_t^{1/2} = \sigma_t I$. Substituting these expressions yields

$$\begin{aligned} \mathcal{L}_{\text{DSM}}(\boldsymbol{\theta}) &= \mathbb{E}_{t, \mathbf{x}_0, \boldsymbol{\epsilon}} \left[\lambda(t) \left\| \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{m}_t(\mathbf{x}_0) + \sigma_t \boldsymbol{\epsilon}, t) + \sigma_t^{-2}(\sigma_t \boldsymbol{\epsilon}) \right\|^2 \right] \\ &= \mathbb{E}_{t, \mathbf{x}_0, \boldsymbol{\epsilon}} \left[\lambda(t) \left\| \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{m}_t(\mathbf{x}_0) + \sigma_t \boldsymbol{\epsilon}, t) + \sigma_t^{-1} \boldsymbol{\epsilon} \right\|^2 \right]. \end{aligned}$$

Choosing the standard weighting $\lambda(t) = \sigma_t^2$ finally gives

$$\mathcal{L}_{\text{DSM}}(\boldsymbol{\theta}) = \mathbb{E}_{t, \mathbf{x}_0, \boldsymbol{\epsilon}} \left[\left\| \sigma_t \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{m}_t(\mathbf{x}_0) + \sigma_t \boldsymbol{\epsilon}, t) + \boldsymbol{\epsilon} \right\|^2 \right]. \quad (2.30)$$

Equation (2.30) is precisely the continuous-time generalization of the loss derived in Chapter 1 (see Equation (1.9)). The interpretation is intuitive: after applying the deterministic drift $\mathbf{m}_t(\cdot)$ to the clean sample and adding Gaussian noise of scale σ_t , the score network is trained to predict the scaled noise $-\boldsymbol{\epsilon}$. In other words, the network learns to denoise the corrupted observation, and doing so at every time t yields an estimate of the time-dependent score function $\nabla_{\mathbf{x}} \log p_t$.

With this objective, we have a practical and theoretically grounded method for learning the score in continuous time. Once the model $\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}, t)$ is trained, it can be

inserted into the reverse-time SDE (2.10) to generate new samples, completing the generative pipeline.

2.3 The Role of Noise and Transition Kernels

The forward SDE (2.1) induces a family of marginal distributions $p_t(\mathbf{x})$ together with a transition kernel $p_{0t}(\mathbf{x}_t|\mathbf{x}_0)$ that governs the conditional distribution of $\mathbf{X}_t$ given an initial state $\mathbf{X}_0$. This kernel occupies a central position in the theory, for the continuous-time score matching objective demands knowledge of the conditional score $\nabla_{\mathbf{x}_t} \log p_{0t}(\mathbf{x}_t|\mathbf{x}_0)$. The question naturally arises: what properties must the noise driving the SDE possess for the resulting framework to remain mathematically tractable? The answer, as we shall see, leads inexorably to the Gaussian distribution.

2.3.1 Necessary Conditions for a Well-Behaved Noise Process

Let us consider the requirements that any reasonable noise process must satisfy if it is to serve as the foundation of a diffusion-based generative model.

First, the noise must be independent of the initial condition $\mathbf{X}_0$; otherwise the joint distribution would not factorize as $p(\mathbf{x}_0, \mathbf{x}_t) = p_{\text{data}}(\mathbf{x}_0)p_{0t}(\mathbf{x}_t|\mathbf{x}_0)$, and the denoising score matching objective would lose its simple form.

Second, because the process evolves continuously in time, the distribution after an interval of length $t + s$ is necessarily the convolution of the distributions after intervals t and s . The family of noise distributions must therefore be closed under convolution; probabilists recognize this as the property of being **infinitely divisible**. A distribution is infinitely divisible precisely when it can appear as the increment distribution of a Lévy process—a stochastic process with independent and stationary increments [2]. The celebrated Lévy–Khintchine theorem provides a complete characterization of such distributions.

Theorem 2.3.1 (Lévy–Khintchine, [2]). *A probability distribution μ on $\mathbb{R}^d$ is infinitely divisible if and only if its characteristic function admits the representation*

$$\hat{\mu}(\mathbf{u}) = \exp\left(i\boldsymbol{\gamma} \cdot \mathbf{u} - \frac{1}{2}\mathbf{u}^\top \mathbf{A} \mathbf{u} + \int_{\mathbb{R}^d \setminus \{0\}} \left(e^{i\mathbf{u} \cdot \mathbf{x}} - 1 - i\mathbf{u} \cdot \mathbf{x} \mathbf{1}_{\|\mathbf{x}\| < 1}\right) \nu(d\mathbf{x})\right),$$

where $\boldsymbol{\gamma} \in \mathbb{R}^d$, $\mathbf{A}$ is a positive semidefinite matrix, and ν is a Lévy measure—a measure satisfying $\int_{\mathbb{R}^d \setminus \{0\}} \min(1, \|\mathbf{x}\|^2) \nu(d\mathbf{x}) < \infty$.

The three terms in the exponent correspond respectively to a deterministic drift, a Gaussian component (Brownian motion), and a pure jump component. The presence of a non-zero Lévy measure ν signals the existence of discontinuities in the sample paths.

Third, we require the perturbations to be isotropic—spherically symmetric—so that no direction in the data space is artificially privileged. For an infinitely divisible distribution, isotropy forces the Gaussian component to be a scalar multiple of the identity, $\mathbf{A} = \sigma^2 I$, and the Lévy measure to be rotationally invariant [2].

Fourth, the sample paths of the process must be continuous. This is the defining characteristic of a diffusion, as opposed to a jump process. The Lévy–Itô decomposition tells us exactly when continuity holds.

Theorem 2.3.2 (Lévy–Itô decomposition, [2]). *Any Lévy process can be decomposed uniquely into the sum of a deterministic drift, a Brownian motion with covariance matrix $\mathbf{A}$, and an independent pure jump process. The sample paths are continuous if and only if the jump component vanishes—that is, if and only if the Lévy measure $\nu \equiv 0$.*

Thus, continuity forces the Lévy measure to be identically zero, leaving only the Gaussian component. In this case the driving process is simply a (possibly scaled) Brownian motion.

Fifth, for practical implementation we need the conditional score $\nabla_{\mathbf{x}_t} \log p_{0t}(\mathbf{x}_t|\mathbf{x}_0)$ to be available in closed form; otherwise training becomes computationally prohibitive. Moreover, the marginal densities should evolve according to a local partial differential equation—the Fokker–Planck equation—rather than an integro-differential equation of the kind that governs jump processes.

Finally, the existence of a simple time-reversal formula is essential for generative modeling. Anderson’s theorem [1] provides such a formula for diffusion processes, but its analogues for jump processes are considerably more technical [5, 11] and have not found application in the score-based generative literature.

2.3.2 Why the Gaussian Distribution Is Inevitable

The Gaussian distribution satisfies every condition enumerated above. It is infinitely divisible and closed under convolution; indeed, the sum of independent Gaussians is again Gaussian. Its associated Lévy process is Brownian motion, which enjoys continuous paths and, when taken with a scalar covariance matrix, is isotropic. The

conditional score takes the characteristically simple linear form

$$\nabla_{\mathbf{x}_t} \log p_{0t}(\mathbf{x}_t | \mathbf{x}_0) = -\frac{\mathbf{x}_t - \mathbf{m}(t)\mathbf{x}_0}{\sigma^2(t)},$$

where $\mathbf{m}(t)$ and $\sigma^2(t)$ are determined by the drift and diffusion coefficients. The Fokker–Planck equation is a local PDE, and Anderson’s theorem applies directly.

What of other candidates? The Laplace distribution, though infinitely divisible, fails on several counts. It is not spherically symmetric; its contours are not circles but possess a different geometry that would introduce directional bias into the perturbation process. The associated Lévy process (sometimes called Laplace motion or variance-gamma process) has discontinuous sample paths, and its transition kernel is not closed under convolution—the sum of independent Laplace variables is not Laplace-distributed. The conditional score involves sign functions and is discontinuous, complicating both analysis and implementation. The Cauchy distribution is stable and can be made

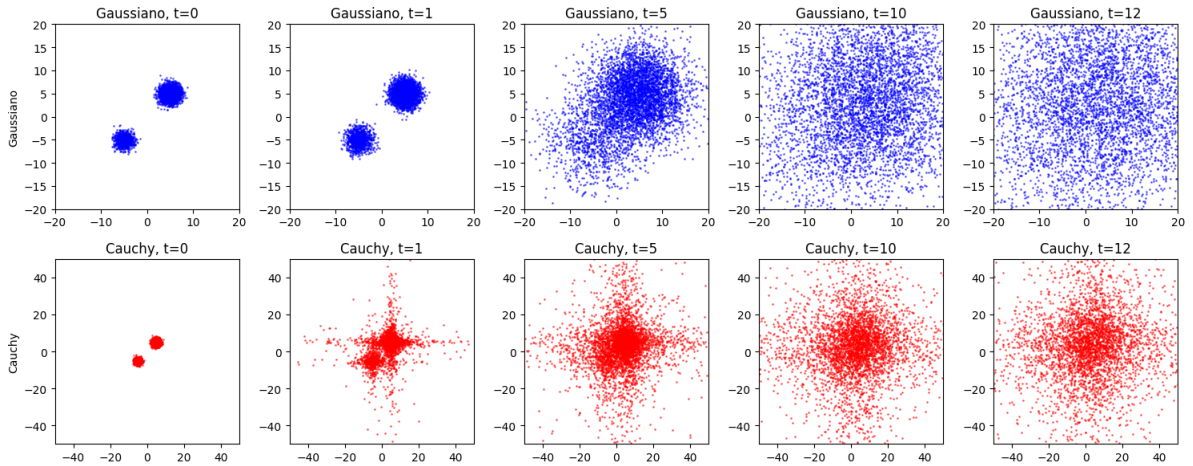


Figure 2.1 **Top row:** Evolution of a bimodal distribution under Gaussian diffusion (Variance Exploding SDE). The density remains smooth and the score is well-defined everywhere. **Bottom row:** Same initial distribution perturbed with Cauchy noise. The heavy tails prevent convergence to a simple prior and the score becomes ill-behaved in low-density regions, illustrating why the Gaussian choice is essential for score-based generative models.

spherically symmetric, but it suffers from infinite variance. Its sample paths are discontinuous, and while the transition kernel remains Cauchy (due to stability), the conditional score lacks a simple closed form. Moreover, the infinite variance violates the finite-second-moment assumptions that underpin much of the theory of score matching.

The Student- t distribution, for any finite degrees of freedom, is not infinitely divisible [2]. Consequently it cannot serve as the increment distribution of any Lévy process, rendering it incompatible with the continuous-time framework from the outset.

Chapter 3

Families of Stochastic Differential Equations

Having established the theoretical foundations of score-based generative modeling through stochastic differential equations, we now examine specific families of SDEs that arise as continuous-time limits of discrete models. Each family corresponds to a different choice of drift and diffusion coefficients, leading to distinct perturbation dynamics and, consequently, different trade-offs in training and sampling.

In addition to the well-known Variance Exploding (VE) and Variance Preserving (VP) SDEs (and their sub-VP variant), we introduce the **Linear Variance (LV) SDE**, a new model whose conditional variance grows *linearly* with time. This property yields a uniform temporal scale for the diffusion process and results in particularly smooth reverse dynamics, which we hypothesize will enable larger step sizes during numerical integration and thus faster inference.

For each family we follow a unified treatment: we state the forward SDE, derive its transition kernel, state the conditional score, and then apply Anderson’s theorem (subsection 2.1.2, appendix A) to obtain the **reverse-time SDE**. Throughout, we denote the time-dependent score function by $\mathbf{s}_t(\mathbf{x}) \triangleq \nabla_{\mathbf{x}} \log p_t(\mathbf{x})$ and write the reverse process in the original time coordinate running from T back to 0, following the convention of [19]:

$$d\mathbf{X} = [\mathbf{f}(\mathbf{X}, t) - G(t)G(t)^\top \mathbf{s}_t(\mathbf{X})] dt + G(t) d\bar{\mathbf{W}}, \quad (3.1)$$

where $d\bar{\mathbf{W}}$ denotes a reverse-time Wiener increment.

3.1 Variance Exploding (VE) SDE

The noise perturbations used in Score Matching with Langevin Dynamics (SMLD) correspond to a discrete Markov chain that, in the limit of infinitely many noise scales, converges to a continuous-time SDE. Consider the sequence of noise levels $\{\sigma_i\}_{i=1}^N$ with $\sigma_0 = 0$ and define the Markov chain

$$\mathbf{x}_i = \mathbf{x}_{i-1} + \sqrt{\sigma_i^2 - \sigma_{i-1}^2} \mathbf{z}_{i-1}, \quad \mathbf{z}_{i-1} \sim \mathcal{N}(0, I), \quad i = 1, \dots, N.$$

As $N \rightarrow \infty$ and the noise levels become a continuous function $\sigma(t)$ of a time variable $t \in [0, 1]$, this discrete process converges to the stochastic process $\{\mathbf{X}_t\}_{t \in [0, 1]}$ satisfying the SDE

$$d\mathbf{X}_t = \sqrt{\frac{d}{dt} \sigma^2(t)} d\mathbf{W}_t. \quad (3.2)$$

where $\mathbf{W}_t$ is a standard Wiener process. This SDE has zero drift, i.e., $\mathbf{f}(\mathbf{x}, t) = 0$, and diffusion coefficient $g(t) = \sqrt{\frac{d}{dt} \sigma^2(t)}$. It is a special case of the affine SDE (2.13) with $\mathbf{F}(t) = 0$ and $\mathbf{c}(t) = 0$.

For the VE SDE, the fundamental matrix $\Phi(t, s) = I$ for all t, s . Using the formulas derived in Section 4.2.3, the conditional mean and covariance given $\mathbf{X}_0 = \mathbf{x}_0$ are

$$\mathbf{m}_t(\mathbf{x}_0) = \mathbf{x}_0, \quad (3.3)$$

$$\Sigma_t = \int_0^t \frac{d}{d\tau} \sigma^2(\tau) d\tau = \sigma^2(t) - \sigma^2(0) = \sigma^2(t)I, \quad (3.4)$$

where we have assumed $\sigma(0) = 0$ for simplicity. Thus, the transition kernel is Gaussian:

$$p_{0,t}(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \mathbf{x}_0, \sigma^2(t)I).$$

The conditional score is therefore

$$\nabla_{\mathbf{x}_t} \log p_{0,t}(\mathbf{x}_t | \mathbf{x}_0) = -\frac{\mathbf{x}_t - \mathbf{x}_0}{\sigma^2(t)}. \quad (3.5)$$

Reverse-time SDE. Since the VE forward process has $\mathbf{f} \equiv \mathbf{0}$ and $G(t)G(t)^\top = \frac{d}{dt} \sigma^2(t) I$, substituting into (3.1) gives

$$d\mathbf{X} = -\frac{d\sigma^2(t)}{dt} \mathbf{s}_t(\mathbf{X}) dt + \sqrt{\frac{d\sigma^2(t)}{dt}} d\bar{\mathbf{W}}, \quad t : T \rightarrow 0, \quad \mathbf{X}_T \sim p_T. \quad (3.6)$$

The reverse process is a pure score-driven drift with no deterministic restoring force, reflecting the fact that the forward VE process adds noise without attenuating the signal. As the variance may grow very large by time T , it is essential to initialise $\mathbf{X}_T$ from a distribution with correspondingly large variance, such as $\mathcal{N}(\mathbf{0}, \sigma^2(T)I)$.

3.2 Variance Preserving (VP) SDE

Denoising Diffusion Probabilistic Models (DDPM) [6] employ a different discrete Markov chain:

$$\mathbf{x}_i = \sqrt{1 - \beta_i} \mathbf{x}_{i-1} + \sqrt{\beta_i} \mathbf{z}_{i-1}, \quad \mathbf{z}_{i-1} \sim \mathcal{N}(\mathbf{0}, I), \quad i = 1, \dots, N.$$

In the continuous limit $N \rightarrow \infty$, with β_i replaced by a function $\beta(t)$ for $t \in [0, 1]$, this chain converges to the SDE

$$d\mathbf{X}_t = -\frac{1}{2}\beta(t)\mathbf{X}_t dt + \sqrt{\beta(t)} d\mathbf{W}_t. \quad (3.7)$$

This is an affine SDE with $\mathbf{F}(t) = -\frac{1}{2}\beta(t)I$, $\mathbf{c}(t) = \mathbf{0}$, and diffusion coefficient $g(t) = \sqrt{\beta(t)}$. The diffusion matrix is $a(t) = \beta(t)I$.

For the VP SDE, the fundamental matrix is

$$\Phi(t, s) = \exp\left(-\frac{1}{2} \int_s^t \beta(u) du\right) I.$$

Applying the formulas (2.23) and (2.24) with initial condition $\mathbf{X}_0 = \mathbf{x}_0$ gives

$$\mathbf{m}_t(\mathbf{x}_0) = \Phi(t, 0)\mathbf{x}_0 = \mathbf{x}_0 e^{-\frac{1}{2} \int_0^t \beta(u) du}, \quad (3.8)$$

$$\begin{aligned} \Sigma_t &= \int_0^t \Phi(t, \tau) a(\tau) \Phi(t, \tau)^\top d\tau \\ &= I \int_0^t e^{-\int_\tau^t \beta(u) du} \beta(\tau) d\tau = I \left(1 - e^{-\int_0^t \beta(u) du}\right). \end{aligned} \quad (3.9)$$

Thus, the transition kernel is Gaussian with mean decaying exponentially and covariance approaching identity:

$$p_{0,t}(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}\left(\mathbf{x}_t; \mathbf{x}_0 e^{-\frac{1}{2} \int_0^t \beta(u) du}, \left(1 - e^{-\int_0^t \beta(u) du}\right) I\right).$$

For compactness we write $\bar{\beta}(t) \triangleq \int_0^t \beta(u) du$, so the conditional score becomes

$$\nabla_{\mathbf{x}_t} \log p_{0,t}(\mathbf{x}_t | \mathbf{x}_0) = -\frac{\mathbf{x}_t - \mathbf{x}_0 e^{-\bar{\beta}(t)/2}}{1 - e^{-\bar{\beta}(t)}}. \quad (3.10)$$

Reverse-time SDE. For the VP forward SDE, $\mathbf{f}(\mathbf{x}, t) = -\frac{1}{2}\beta(t)\mathbf{x}$ and $G(t)G(t)^\top = \beta(t)I$. Substituting into (3.1),

$$d\mathbf{X} = \left[-\frac{1}{2}\beta(t)\mathbf{X} - \beta(t)\mathbf{s}_t(\mathbf{X}) \right] dt + \sqrt{\beta(t)} d\bar{\mathbf{W}}, \quad t : T \rightarrow 0, \quad \mathbf{X}_T \sim p_T. \quad (3.11)$$

Because the prior $p_T \approx \mathcal{N}(\mathbf{0}, I)$ for a sufficiently large noise budget, the VP reverse process can be initialised straightforwardly from a standard Gaussian, a practical advantage over VE-type models. The reverse drift contains two contributions: the same linear restoring force $-\frac{1}{2}\beta(t)\mathbf{X}$ as the forward process, plus the score-weighted term $-\beta(t)\mathbf{s}_t(\mathbf{X})$ that steers trajectories toward regions of high data density.

3.3 The sub-VP SDE

Inspired by the VP SDE, a variant known as the sub-VP SDE has been proposed, which often yields better likelihood performance. It is defined as

$$d\mathbf{X}_t = -\frac{1}{2}\beta(t)\mathbf{X}_t dt + \sqrt{\beta(t) \left(1 - e^{-2\int_0^t \beta(s) ds}\right)} d\mathbf{W}_t. \quad (3.12)$$

This SDE also falls into the affine class, with the same drift $\mathbf{F}(t) = -\frac{1}{2}\beta(t)I$ but a state-independent diffusion coefficient $g(t) = \sqrt{\beta(t)(1 - e^{-2\bar{\beta}(t)})}$. Using the shorthand $\alpha(t) \triangleq 1 - e^{-2\bar{\beta}(t)}$, the diffusion matrix is $a(t) = \beta(t)\alpha(t)I$.

The fundamental matrix is the same as for the VP SDE: $\Phi(t, s) = e^{-\bar{\beta}(t,s)/2}I$ where $\bar{\beta}(t, s) = \int_s^t \beta(u) du$. The mean is therefore identical:

$$\mathbf{m}_t(\mathbf{x}_0) = \mathbf{x}_0 e^{-\bar{\beta}(t)/2}.$$

The covariance, however, differs due to the modified diffusion coefficient:

$$\Sigma_t = \int_0^t \Phi(t, \tau) a(\tau) \Phi(t, \tau)^\top d\tau = I \int_0^t e^{-(\bar{\beta}(t) - \bar{\beta}(\tau))} \beta(\tau) (1 - e^{-2\bar{\beta}(\tau)}) d\tau.$$

Evaluating this integral yields a particularly simple form (see Appendix B of [19] for the derivation):

$$\Sigma_t = \left(1 - e^{-\bar{\beta}(t)}\right)^2 I. \quad (3.13)$$

Thus, the transition kernel is Gaussian with mean decaying exponentially and covariance that grows more slowly than in the VP case, remaining bounded and never exceeding the VP covariance at any intermediate time:

$$p_{0,t}(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}\left(\mathbf{x}_t; \mathbf{x}_0 e^{-\bar{\beta}(t)/2}, \left(1 - e^{-\bar{\beta}(t)}\right)^2 I\right).$$

The conditional score follows directly:

$$\nabla_{\mathbf{x}_t} \log p_{0,t}(\mathbf{x}_t | \mathbf{x}_0) = -\frac{\mathbf{x}_t - \mathbf{x}_0 e^{-\bar{\beta}(t)/2}}{\left(1 - e^{-\bar{\beta}(t)}\right)^2}. \quad (3.14)$$

Reverse-time SDE. For the sub-VP SDE, $\mathbf{f}(\mathbf{x}, t) = -\frac{1}{2}\beta(t)\mathbf{x}$ and $G(t)G(t)^\top = \beta(t) \alpha(t) I$ with $\alpha(t) = 1 - e^{-2\bar{\beta}(t)}$. Substituting into (3.1),

$$d\mathbf{X} = \left[-\frac{1}{2}\beta(t)\mathbf{X} - \beta(t)\alpha(t)\mathbf{s}_t(\mathbf{X})\right]dt + \sqrt{\beta(t)\alpha(t)}d\bar{\mathbf{W}}, \quad t : T \rightarrow 0, \quad \mathbf{X}_T \sim p_T. \quad (3.15)$$

Comparing (3.15) with the VP reverse SDE (3.11), the only difference is that the scalar $\beta(t)$ multiplying the score and the diffusion coefficient is replaced by $\beta(t)\alpha(t) \leq \beta(t)$. Since $\alpha(t) < 1$ for all finite t , the score-weighted drift and the stochastic forcing are both smaller in magnitude than their VP counterparts—the sub-VP reverse process is strictly less aggressive in injecting noise and correcting towards the data manifold at each step.

3.4 Linear Variance (LV) SDE

We now introduce a new SDE, the **Linear Variance (LV) SDE**, designed specifically to reduce the number of steps required in the inference (sampling) phase of diffusion models. The core idea is to engineer the conditional variance $\text{Var}[\mathbf{X}_t | \mathbf{X}_0]$ to grow

linearly with time, which yields a uniform temporal scale for the diffusion and smooths the dynamics of the associated probability flow ODE, allowing numerical integrators to take larger step sizes without sacrificing sample quality.

3.4.1 Derivation from the Linear Variance Hypothesis

Rather than postulating the SDE (3.16) directly, we derive it from the design requirement. We work within the affine class (2.13) with a constant scalar drift matrix $\mathbf{F}(t) = fI$ and a state-independent scalar diffusion coefficient $G(t) = g(t)I$, so that $a(t) = g(t)^2I$. The Lyapunov equation (2.21) for the covariance then becomes

$$\Sigma'(t) = 2f \Sigma(t) + g(t)^2I, \quad \Sigma(0) = 0.$$

We now impose the design requirement $\Sigma_t = tI$, so $\Sigma'(t) = I$. Substituting into the Lyapunov equation and solving for $g(t)^2$:

$$I = 2f(tI) + g(t)^2I \implies g(t)^2 = 1 - 2ft.$$

This is the unique constraint that the linear variance hypothesis places on the diffusion coefficient, given the drift f . Two observations follow immediately.

First, $g(t)^2$ must remain positive on the entire interval $[0, T]$. Since $g(t)^2 = 1 - 2ft$ is a linear function of t with value 1 at $t = 0$, positivity for all $t > 0$ requires $f < 0$. Writing $f = -\theta$ with $\theta > 0$, the constraint becomes $g(t)^2 = 1 + 2\theta t$, which is strictly positive for all $t \geq 0$.

Second, if $f = 0$ then $g(t) = 1$ for all t , recovering the pure-diffusion SDE $d\mathbf{X}_t = d\mathbf{W}_t$ whose variance is indeed $\Sigma_t = tI$ trivially. This is a valid but degenerate limit in which the mean is never attenuated. For $f > 0$ the variance would become negative in finite time, ruling out any positive drift.

The choice $\theta > 0$ therefore uniquely determines $g(t) = \sqrt{1 + 2\theta t}$, and the resulting SDE is

$$d\mathbf{X}_t = -\theta\mathbf{X}_t dt + \sqrt{1 + 2\theta t} d\mathbf{W}_t, \quad \theta > 0, \quad t \in [0, 1]. \quad (3.16)$$

The parameter θ is free and controls only the rate of mean decay; the variance $\Sigma_t = tI$ is independent of θ for any valid choice. This SDE belongs to the affine family (2.13) with $\mathbf{F}(t) = -\theta I$, $\mathbf{c}(t) = 0$, and diffusion matrix $a(t) = (1 + 2\theta t)I$.

3.4.2 Transition Kernel

To obtain the transition kernel, we first compute the fundamental matrix. Since $F(t) = -\theta I$ is constant and commutes with itself, the fundamental matrix is simply

$$\Phi(t, s) = e^{-\theta(t-s)}I.$$

Mean. Applying (2.23) with $\mathbf{c}(t) = 0$ gives

$$\mathbf{m}_t(\mathbf{x}_0) = \Phi(t, 0)\mathbf{x}_0 = e^{-\theta t}\mathbf{x}_0.$$

Covariance. Using (2.24) with $a(\tau) = (1 + 2\theta\tau)I$, we have

$$\Sigma_t = \int_0^t \Phi(t, \tau)a(\tau)\Phi(t, \tau)^\top d\tau = I \int_0^t e^{-2\theta(t-\tau)}(1 + 2\theta\tau) d\tau.$$

To evaluate the integral, perform the change of variable $u = 2\theta\tau$, so that $d\tau = du/(2\theta)$ and the limits become $u = 0$ to $u = 2\theta t$:

$$\int_0^t e^{-2\theta(t-\tau)}(1 + 2\theta\tau) d\tau = e^{-2\theta t} \int_0^{2\theta t} e^u (1 + u) \frac{du}{2\theta}.$$

The key observation is that $e^u(1 + u) = \frac{d}{du}(ue^u)$, so the integral evaluates exactly:

$$\int_0^{2\theta t} e^u(1 + u) du = [ue^u]_0^{2\theta t} = 2\theta t e^{2\theta t}.$$

Substituting back:

$$\int_0^t e^{-2\theta(t-\tau)}(1 + 2\theta\tau) d\tau = e^{-2\theta t} \cdot \frac{2\theta t e^{2\theta t}}{2\theta} = t.$$

Therefore $\Sigma_t = tI$, confirming that the design requirement is satisfied exactly. The transition kernel is Gaussian with mean decaying exponentially and variance growing linearly:

$$p_{0,t}(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; e^{-\theta t}\mathbf{x}_0, tI). \quad (3.17)$$

The conditional score takes the simple form

$$\nabla_{\mathbf{x}_t} \log p_{0,t}(\mathbf{x}_t | \mathbf{x}_0) = -\frac{\mathbf{x}_t - e^{-\theta t}\mathbf{x}_0}{t}. \quad (3.18)$$

Reverse-time SDE. For the LV SDE, $\mathbf{f}(\mathbf{x}, t) = -\theta\mathbf{x}$ and $G(t)G(t)^\top = (1 + 2\theta t)I$. Substituting into (3.1),

$$d\mathbf{X} = \left[-\theta \mathbf{X} - (1 + 2\theta t) \mathbf{s}_t(\mathbf{X}) \right] dt + \sqrt{1 + 2\theta t} d\bar{\mathbf{W}}, \quad t : T \rightarrow 0, \quad \mathbf{X}_T \sim p_T. \quad (3.19)$$

Because $\Sigma_T = TI = I$, at $t = T = 1$ the prior is approximately $\mathbf{X}_1 \sim \mathcal{N}(\mathbf{0}, I)$ for sufficiently large θ , so the reverse process can be initialised from a standard Gaussian. The coefficient $(1 + 2\theta t)$ grows linearly with time, meaning the score contributes progressively more to the reverse drift as the process approaches $t = 0$, which mirrors the increasing influence of the data distribution at fine scales.

3.4.3 Properties and Motivation

Several features of this SDE are noteworthy.

1. **Linear variance growth.** The conditional variance increases exactly as t , independent of θ . This contrasts with the VE SDE (where variance grows as $\sigma^2(t)$, which may be superlinear) and the VP SDE (where variance saturates exponentially). Linear growth provides a uniform temporal scale for the perturbation process: the probability flow ODE changes at an approximately constant rate, allowing numerical integrators to take larger uniform step sizes without accumulating disproportionate error in any particular time interval.
2. **Exponential mean decay.** The mean decays as $e^{-\theta t}$, controlled by the parameter θ . This ensures that the information from the initial data is gradually erased, transitioning smoothly to pure noise at $t = 1$ where the mean is $e^{-\theta}\mathbf{x}_0$ and the variance is I . By choosing θ sufficiently large, the final distribution can be made arbitrarily close to $\mathcal{N}(\mathbf{0}, I)$.

3.5 Visualizing the Forward Perturbation Process

To develop intuition for the different perturbation dynamics, we present a simple experiment that visualizes the effect of each SDE on a single image. We take a digit from the MNIST dataset, normalize it to the range $[-1, 1]$, and simulate the forward diffusion for each of the four SDEs discussed above. The simulation is performed by discretizing the time interval $[0, T]$ (with $T = 1$) into $N = 20$ equally spaced steps and, at each step, sampling from the exact transition kernel of the corresponding SDE using

the reparameterization trick. For a fair comparison, the same noise realisation $\{\mathbf{z}_i\}_{i=1}^N$ (with $\mathbf{z}_i \sim \mathcal{N}(0, I)$) is used across all SDEs.

Let $\mathbf{x}_0$ denote the clean image. For each SDE, the perturbed image at time $t_i = i/N$ (with $i = 0, \dots, N$) is obtained as

$$\mathbf{x}_{t_i} = \mathbf{m}_{t_i}(\mathbf{x}_0) + \Sigma_{t_i}^{1/2} \mathbf{z}_i,$$

where $\mathbf{m}_t(\mathbf{x}_0)$ and Σ_t are the conditional mean and covariance derived in Sections 3.1–3.4. The explicit formulas are:

- **VE SDE:** $\mathbf{m}_t(\mathbf{x}_0) = \mathbf{x}_0$, $\Sigma_t = \sigma^2(t)I$ with $\sigma(t) = \sigma_{\min}(\sigma_{\max}/\sigma_{\min})^t$. We choose $\sigma_{\min} = 10^{-4}$ and $\sigma_{\max} = 10$.
- **VP SDE:** $\mathbf{m}_t(\mathbf{x}_0) = \mathbf{x}_0 e^{-\bar{\beta}(t)/2}$, $\Sigma_t = (1 - e^{-\bar{\beta}(t)})I$. We take $\beta(t) = \beta_{\min} + t(\beta_{\max} - \beta_{\min})$ with $\beta_{\min} = 0.1$, $\beta_{\max} = 10$.
- **sub-VP SDE:** same mean as VP, covariance $\Sigma_t = (1 - e^{-\bar{\beta}(t)})^2 I$.
- **LV SDE:** $\mathbf{m}_t(\mathbf{x}_0) = \mathbf{x}_0 e^{-\theta t}$, $\Sigma_t = tI$ with $\theta = 1.5$.

The integrals involving $\beta(s)$ are computed analytically for the linear schedule $\beta(t) = \beta_{\min} + t(\beta_{\max} - \beta_{\min})$; indeed,

$$\int_0^t \beta(s) ds = \beta_{\min} t + \frac{1}{2}(\beta_{\max} - \beta_{\min})t^2.$$

After generating the perturbed images, each is clamped to the range $[0, 1]$ for display by applying the inverse transformation $(\mathbf{x}_t + 1)/2$.

The procedure can be summarised as follows:

Figure 3.1 shows the sequence of perturbed images for the four SDEs. Several observations are immediate:

- **VE SDE:** The image quickly becomes dominated by high-magnitude noise; the variance grows exponentially with t , leading to a rapid loss of structure. By $t = 1$, the image is almost unrecognisable.
- **VP SDE:** The variance saturates at 1 and the mean decays exponentially. The perturbation appears more gradual: at intermediate times the image retains a faint outline of the digit, and the noise level increases more slowly than in the VE case.

Algorithm 1 Forward perturbation comparison

Input: Clean image $\mathbf{x}_0$ (normalised to $[-1, 1]$), number of steps N , final time $T = 1$, SDE parameters $\sigma_{\min}, \sigma_{\max}, \beta_{\min}, \beta_{\max}, \theta$.

Output: Visualisation arrays for each SDE.

- 1: Compute time grid $t_i = i/N$ for $i = 0, \dots, N$.
 - 2: Generate noise samples $\mathbf{z}_i \sim \mathcal{N}(0, I)$, $i = 1, \dots, N$ (same for all SDEs).
 - 3: **for** each SDE in $\{\text{VE}, \text{VP}, \text{sub-VP}, \text{LV}\}$ **do**
 - 4: **for** $i = 1$ to N **do**
 - 5: Compute mean coefficient μ_i and standard deviation σ_i according to the SDE's formulas.
 - 6: $\mathbf{x}_{t_i} = \mu_i \mathbf{x}_0 + \sigma_i \mathbf{z}_i$.
 - 7: Store $\mathbf{x}_{t_i}$ for visualisation.
 - 8: **end for**
 - 9: **end for**
 - 10: Convert all stored images to display range $[0, 1]$ and plot as a grid.
-

- **sub-VP SDE:** The mean decays identically to VP, but the variance grows even more slowly (as the square of the VP variance). Consequently, the image remains recognisable for longer; at $t = 1$ the signal is weaker but the noise level is lower, resulting in a “ghost-like” appearance.
- **LV SDE:** The mean decays exponentially (controlled by θ) while the variance increases linearly. This yields a uniform progression: the image fades steadily and the noise level increases at a constant rate. The choice $\theta = 1.5$ ensures that by $t = 1$ the mean is $e^{-1.5} \approx 0.22$ times the original, while the variance is exactly 1. The resulting images exhibit a balanced blend of signal and noise throughout the diffusion.

This visual comparison reinforces the theoretical properties derived in the previous sections and highlights the trade-offs inherent in each SDE design. The LV SDE, with its linear variance growth, offers a particularly smooth transition that may be advantageous for training score-based models and for accelerating inference via larger step sizes in the reverse process.

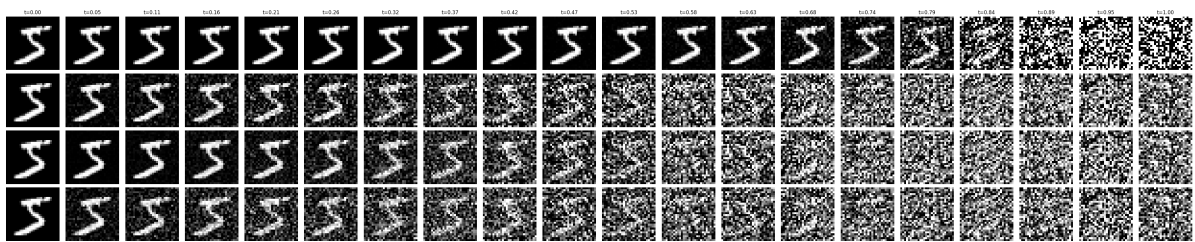


Figure 3.1 Evolution of an MNIST digit under the four SDEs. Each row corresponds to a different SDE (top to bottom: VE, VP, sub-VP, LV). Columns represent increasing time t from 0 to 1.

Chapter 4

Sampling

Once the score model $\mathbf{s}_{\theta^*}(\mathbf{x}, t)$ has been trained via continuous-time denoising score matching (see Section 2.2), we can generate new samples by simulating the reverse dynamics. Two distinct approaches are available: (i) discretising the reverse-time SDE and simulating it with a stochastic numerical scheme, and (ii) solving the associated probability flow ODE, a deterministic dynamics that shares the same marginal distributions. Both methods rely on numerical integration and offer different trade-offs between sample quality and computational cost.

4.1 Discretisation and Reverse-Time Sampling

Having derived the continuous-time reverse SDEs and probability flow ODEs for each SDE family in Chapter 3, we now turn to the practical task of generating samples. Since these reverse processes cannot be solved analytically, we must discretise them and apply numerical integration schemes. This section establishes a unified discretisation framework applicable to all four SDE families (VE, VP, sub-VP, and LV), then derives the specific sampling algorithms for each model.

The exposition proceeds in four stages: we first introduce the general discretisation notation and the Euler–Maruyama scheme (4.1.1), then derive the reverse-time discrete updates (4.1.2), present explicit formulas for each of the four SDE families (4.1.3), and finally discuss convergence properties and implementation considerations (4.1.4).

4.1.1 General Discretisation Framework

Time discretisation and notation. We partition the time interval $[0, T]$ into N uniform steps of width $\Delta t = T/N$, giving the discrete time grid

$$t_i = i \Delta t, \quad i = 0, 1, \dots, N. \quad (4.1)$$

Throughout this chapter we adopt the following notational convention: capital boldface letters $\mathbf{X}_t$ denote continuous-time stochastic processes, while lower-case boldface $\mathbf{x}_i$ denotes the discrete approximation at time t_i . For theoretical error analysis we write $h = \max_i (t_{i+1} - t_i)$ for the mesh width; in our uniform grid $h = \Delta t$.

Forward SDE and discrete coefficients. Recall that a general forward Itô SDE takes the form

$$d\mathbf{X}_t = \mathbf{f}(\mathbf{X}_t, t) dt + g(t) d\mathbf{W}_t, \quad \mathbf{X}_0 \sim p_0, \quad (4.2)$$

where $\mathbf{f} : \mathbb{R}^d \times [0, T] \rightarrow \mathbb{R}^d$ is the drift coefficient, $g : [0, T] \rightarrow \mathbb{R}$ is the scalar diffusion coefficient, and $\mathbf{W}_t$ is a standard d -dimensional Wiener process. To simplify the discretisation we absorb powers of Δt into the coefficients, defining

$$\begin{aligned} f_i(\mathbf{x}) &:= \mathbf{f}(\mathbf{x}, t_i) \Delta t, \\ G_i &:= g(t_i) \sqrt{\Delta t}, \\ \Delta \mathbf{W}_i &:= \mathbf{W}_{t_{i+1}} - \mathbf{W}_{t_i}. \end{aligned} \quad (4.3)$$

This notation makes the order of each term explicit: $f_i(\mathbf{x})$ is $O(\Delta t)$ and G_i is $O(\sqrt{\Delta t})$. The Wiener increments satisfy

$$\Delta \mathbf{W}_i \sim \mathcal{N}(0, \Delta t I), \quad \text{and } \Delta \mathbf{W}_i \perp \Delta \mathbf{W}_j \text{ for } i \neq j.$$

In practice we draw standard Gaussian vectors $\mathbf{z}_i \sim \mathcal{N}(0, I)$ and set $\Delta \mathbf{W}_i = \sqrt{\Delta t} \mathbf{z}_i$, so that the stochastic term becomes $G_i \mathbf{z}_i$.

Euler–Maruyama forward scheme. Applying the Euler–Maruyama method to (4.2) yields the forward recursion

$$\mathbf{x}_{i+1} = \mathbf{x}_i + f_i(\mathbf{x}_i) + G_i \mathbf{z}_i, \quad \mathbf{z}_i \sim \mathcal{N}(0, I), \quad (4.4)$$

starting from $\mathbf{x}_0 \sim p_0$. Under standard Lipschitz and linear growth conditions on $\mathbf{f}$ and g , this scheme converges with strong order 1/2 and weak order 1 [13]:

$$\text{(strong)} \quad \mathbb{E} \left[\sup_{0 \leq t \leq T} \|\mathbf{X}_t - \mathbf{X}_t^{(h)}\|^2 \right]^{1/2} = O(h^{1/2}), \quad (4.5)$$

$$\text{(weak)} \quad |\mathbb{E}[\phi(\mathbf{X}_T)] - \mathbb{E}[\phi(\mathbf{X}_T^{(h)})]| = O(h), \quad (4.6)$$

where $\mathbf{X}_t^{(h)}$ is an interpolated approximation and ϕ is a sufficiently smooth test function.

4.1.2 Reverse-Time SDE Discretisation

Continuous reverse-time SDE. From Anderson’s theorem (Chapter 3), the reverse-time SDE corresponding to (4.2) is

$$d\mathbf{X}_t = [\mathbf{f}(\mathbf{X}_t, t) - g(t)^2 \mathbf{s}_t(\mathbf{X}_t)] dt + g(t) d\bar{\mathbf{W}}_t, \quad t : T \rightarrow 0, \quad (4.7)$$

where $\mathbf{s}_t(\mathbf{x}) = \nabla_{\mathbf{x}} \log p_t(\mathbf{x})$ is the score function and $\bar{\mathbf{W}}_t$ denotes a Wiener process running backwards in time. In practice the score is unknown and is replaced by a learned approximation $\mathbf{s}_{\theta^*}(\mathbf{x}, t)$ obtained via denoising score matching.

Euler-Maruyama reverse scheme. To discretise (4.7), we apply Euler-Maruyama with a *negative* time step $-\Delta t$ (since time decreases). The increment from t_i to t_{i-1} is

$$\begin{aligned} \mathbf{x}_{i-1} &= \mathbf{x}_i + [\mathbf{f}(\mathbf{x}_i, t_i) - g(t_i)^2 \mathbf{s}_{\theta^*}(\mathbf{x}_i, t_i)](-\Delta t) + g(t_i) \sqrt{\Delta t} \mathbf{z}_i \\ &= \mathbf{x}_i - \mathbf{f}(\mathbf{x}_i, t_i) \Delta t + g(t_i)^2 \Delta t \mathbf{s}_{\theta^*}(\mathbf{x}_i, t_i) + g(t_i) \sqrt{\Delta t} \mathbf{z}_i \end{aligned}$$

Using the compact notation (4.3), we obtain the canonical reverse-time Euler-Maruyama update:

$$\mathbf{x}_{i-1} = \mathbf{x}_i - f_i(\mathbf{x}_i) + G_i^2 \mathbf{s}_{\theta^*}(\mathbf{x}_i, t_i) + G_i \mathbf{z}_i, \quad \mathbf{z}_i \sim \mathcal{N}(0, I). \quad (4.8)$$

Note the sign reversal on the drift term ($-f_i$ instead of $+f_i$), the appearance of $G_i^2 = g(t_i)^2 \Delta t$ multiplying the score, and the preservation of the stochastic term $G_i \mathbf{z}_i$ with the same magnitude as in the forward pass.

Sampling algorithm. The complete reverse sampling procedure is given in Algorithm 2. We initialise from the terminal prior $\mathbf{x}_N \sim p_T \approx \mathcal{N}(0, I)$ and iterate backwards from $i = N$ to $i = 1$, producing a sample $\mathbf{x}_0$ from the data distribution p_0 .

Algorithm 2 Reverse-time SDE sampling (Euler–Maruyama)

Input: Score model $\mathbf{s}_{\theta^*}(\mathbf{x}, t)$, number of steps N , final time T .

Output: Sample $\mathbf{x}_0 \sim p_0$.

- 1: Compute time grid: $\Delta t \leftarrow T/N$, $t_i \leftarrow i \Delta t$ for $i = 0, \dots, N$.
 - 2: Draw $\mathbf{x}_N \sim \mathcal{N}(0, I)$.
 - 3: **for** $i = N$ down to 1 **do**
 - 4: Compute discrete coefficients $f_i(\mathbf{x}_i)$ and G_i according to the SDE family (see Table 4.1).
 - 5: Draw $\mathbf{z}_i \sim \mathcal{N}(0, I)$.
 - 6: $\mathbf{x}_{i-1} \leftarrow \mathbf{x}_i - f_i(\mathbf{x}_i) + G_i^2 \mathbf{s}_{\theta^*}(\mathbf{x}_i, t_i) + G_i \mathbf{z}_i$.
 - 7: **end for**
 - 8: **return** $\mathbf{x}_0$.
-

4.1.3 Model-Specific Discretisations

We now derive the explicit reverse-time updates for each of the four SDE families introduced in Chapter 3. Table 4.1 summarises the discrete coefficients $f_i(\mathbf{x})$ and G_i for each model, and Table 4.2 presents the complete reverse-time recursion.

Variance Exploding (VE) SDE

Continuous form. The VE forward SDE (Section 3.1) is

$$d\mathbf{X}_t = \sqrt{\frac{d\sigma^2(t)}{dt}} d\mathbf{W}_t,$$

with zero drift ($\mathbf{f} \equiv 0$) and diffusion coefficient $g(t) = \sqrt{d\sigma^2(t)/dt}$. We typically choose an exponential noise schedule $\sigma(t) = \sigma_{\min}(\sigma_{\max}/\sigma_{\min})^t$, which gives

$$\frac{d\sigma^2(t)}{dt} = 2\sigma(t)\sigma'(t) = 2\sigma^2(t)\log(\sigma_{\max}/\sigma_{\min}).$$

Discrete coefficients. From (4.3), with $\mathbf{f} \equiv 0$,

$$f_i(\mathbf{x}) = 0, \quad G_i = \sqrt{\frac{d\sigma^2(t_i)}{dt}} \Delta t. \quad (4.9)$$

Reverse-time update. Substituting into (4.8),

$$\mathbf{x}_{i-1} = \mathbf{x}_i + \frac{d\sigma^2(t_i)}{dt} \Delta t \mathbf{s}_{\theta^*}(\mathbf{x}_i, t_i) + \sqrt{\frac{d\sigma^2(t_i)}{dt}} \Delta t \mathbf{z}_i. \quad (4.10)$$

The absence of a drift restoring force means the dynamics are driven purely by the score, leading to large noise injection at high t when $\sigma(t)$ is large.

Variance Preserving (VP) SDE

Continuous form. The VP forward SDE (Section 3.2) is

$$d\mathbf{X}_t = -\frac{1}{2}\beta(t)\mathbf{X}_t dt + \sqrt{\beta(t)} d\mathbf{W}_t,$$

with drift $\mathbf{f}(\mathbf{x}, t) = -\frac{1}{2}\beta(t)\mathbf{x}$ and diffusion $g(t) = \sqrt{\beta(t)}$. A common choice is a linear schedule $\beta(t) = \beta_{\min} + t(\beta_{\max} - \beta_{\min})$.

Discrete coefficients.

$$f_i(\mathbf{x}) = -\frac{1}{2}\beta(t_i)\mathbf{x} \Delta t, \quad G_i = \sqrt{\beta(t_i) \Delta t}. \quad (4.11)$$

Reverse-time update.

$$\mathbf{x}_{i-1} = \mathbf{x}_i + \frac{1}{2}\beta(t_i)\mathbf{x}_i \Delta t + \beta(t_i)\Delta t \mathbf{s}_{\theta^*}(\mathbf{x}_i, t_i) + \sqrt{\beta(t_i) \Delta t} \mathbf{z}_i. \quad (4.12)$$

The drift now contains a linear restoring term $+\frac{1}{2}\beta(t_i)\mathbf{x}_i \Delta t$ (sign-reversed from the forward pass) in addition to the score-weighted correction.

Connection to DDPM. The original DDPM ancestral sampler [6] can be viewed as a particular discretisation of this reverse VP SDE. With $\Delta t = 1$ and the noise predictor parameterisation $\mathbf{s}_{\theta^*}(\mathbf{x}, i) \approx -\epsilon_{\theta^*}(\mathbf{x}, i)/\sqrt{1 - \bar{\alpha}_i}$, where $\bar{\alpha}_i = e^{-\int_0^{t_i} \beta(s) ds}$, the update (4.12) becomes

$$\mathbf{x}_{i-1} = \mathbf{x}_i + \frac{1}{2}\beta_i \mathbf{x}_i - \frac{\beta_i}{\sqrt{1 - \bar{\alpha}_i}} \epsilon_{\theta^*}(\mathbf{x}_i, i) + \sqrt{\beta_i} \mathbf{z}_i.$$

Expanding $1/\sqrt{1 - \beta_i} \approx 1 + \frac{1}{2}\beta_i + O(\beta_i^2)$, the DDPM formula

$$\mathbf{x}_{i-1} = \frac{1}{\sqrt{1 - \beta_i}} \left(\mathbf{x}_i - \frac{\beta_i}{\sqrt{1 - \bar{\alpha}_i}} \epsilon_{\theta^*}(\mathbf{x}_i, i) \right) + \sqrt{\beta_i} \mathbf{z}_i$$

coincides with the Euler-Maruyama discretisation to first order in β_i . This equivalence unifies the discrete-time DDPM sampler with the continuous-time SDE framework [19].

sub-VP SDE

Continuous form. The sub-VP SDE (Section 3.3) modifies the VP diffusion coefficient:

$$d\mathbf{X}_t = -\frac{1}{2}\beta(t)\mathbf{X}_t dt + \sqrt{\beta(t)(1 - e^{-2\int_0^t \beta(s)ds})} d\mathbf{W}_t.$$

Writing $\alpha(t) = 1 - e^{-2\bar{\beta}(t)}$ where $\bar{\beta}(t) = \int_0^t \beta(s)ds$, we have $g(t) = \sqrt{\beta(t)\alpha(t)}$.

Discrete coefficients.

$$f_i(\mathbf{x}) = -\frac{1}{2}\beta(t_i)\mathbf{x} \Delta t, \quad G_i = \sqrt{\beta(t_i)\alpha(t_i)} \Delta t, \quad (4.13)$$

where $\alpha(t_i) = 1 - e^{-2\int_0^{t_i} \beta(s)ds}$.

Reverse-time update.

$$\mathbf{x}_{i-1} = \mathbf{x}_i + \frac{1}{2}\beta(t_i)\mathbf{x}_i \Delta t + \beta(t_i)\alpha(t_i)\Delta t \mathbf{s}_{\theta^*}(\mathbf{x}_i, t_i) + \sqrt{\beta(t_i)\alpha(t_i)} \Delta t \mathbf{z}_i. \quad (4.14)$$

Since $\alpha(t) < 1$ for all finite t , the score-weighted drift and stochastic forcing are both smaller in magnitude than in the VP case, making the sub-VP reverse process “gentler” and often leading to better likelihood performance.

Linear Variance (LV) SDE

Continuous form. The LV SDE (Section 3.4, renamed to Linear Variance) is designed to achieve linear variance growth:

$$d\mathbf{X}_t = -\theta\mathbf{X}_t dt + \sqrt{1 + 2\theta t} d\mathbf{W}_t,$$

with drift $\mathbf{f}(\mathbf{x}, t) = -\theta\mathbf{x}$ and diffusion $g(t) = \sqrt{1 + 2\theta t}$.

Discrete coefficients.

$$f_i(\mathbf{x}) = -\theta\mathbf{x} \Delta t, \quad G_i = \sqrt{(1 + 2\theta t_i)} \Delta t. \quad (4.15)$$

Reverse-time update.

$$\mathbf{x}_{i-1} = \mathbf{x}_i + \theta\mathbf{x}_i \Delta t + (1 + 2\theta t_i)\Delta t \mathbf{s}_{\theta^*}(\mathbf{x}_i, t_i) + \sqrt{(1 + 2\theta t_i)} \Delta t \mathbf{z}_i. \quad (4.16)$$

The linear growth of the diffusion coefficient (and hence the score weighting) is hypothesised to enable smoother ODE dynamics and larger numerical step sizes during inference.

Table 4.1 Discrete coefficients for the four SDE families. Here $\bar{\beta}(t) = \int_0^t \beta(s)ds$ and $\alpha(t) = 1 - e^{-2\bar{\beta}(t)}$.

SDE Family	$f_i(\mathbf{x})$	G_i
VE	0	$\sqrt{\frac{d\sigma^2(t_i)}{dt} \Delta t}$
VP	$-\frac{1}{2}\beta(t_i)\mathbf{x} \Delta t$	$\sqrt{\beta(t_i) \Delta t}$
sub-VP	$-\frac{1}{2}\beta(t_i)\mathbf{x} \Delta t$	$\sqrt{\beta(t_i)\alpha(t_i) \Delta t}$
LV	$-\theta\mathbf{x} \Delta t$	$\sqrt{(1 + 2\theta t_i) \Delta t}$

Table 4.2 Reverse-time Euler–Maruyama updates for the four SDE families.

Model	Reverse-time update
VE	$\mathbf{x}_{i-1} = \mathbf{x}_i + \frac{d\sigma^2(t_i)}{dt} \Delta t \mathbf{s}_{\theta^*}(\mathbf{x}_i, t_i) + \sqrt{\frac{d\sigma^2(t_i)}{dt} \Delta t} \mathbf{z}_i$
VP	$\mathbf{x}_{i-1} = \mathbf{x}_i + \frac{1}{2}\beta(t_i)\mathbf{x}_i \Delta t + \beta(t_i)\Delta t \mathbf{s}_{\theta^*}(\mathbf{x}_i, t_i) + \sqrt{\beta(t_i)\Delta t} \mathbf{z}_i$
sub-VP	$\mathbf{x}_{i-1} = \mathbf{x}_i + \frac{1}{2}\beta(t_i)\mathbf{x}_i \Delta t + \beta(t_i)\alpha(t_i)\Delta t \mathbf{s}_{\theta^*}(\mathbf{x}_i, t_i) + \sqrt{\beta(t_i)\alpha(t_i)\Delta t} \mathbf{z}_i$
LV	$\mathbf{x}_{i-1} = \mathbf{x}_i + \theta\mathbf{x}_i \Delta t + (1 + 2\theta t_i)\Delta t \mathbf{s}_{\theta^*}(\mathbf{x}_i, t_i) + \sqrt{(1 + 2\theta t_i)\Delta t} \mathbf{z}_i$

4.1.4 Convergence, Implementation, and Practical Considerations

Convergence orders. The Euler–Maruyama scheme applied to (4.7) inherits the classical convergence rates: strong order 1/2 for pathwise approximation and weak order 1 for expectations of smooth test functions [13]. These rates determine how the computational budget (number of steps N) translates into accuracy. For sampling

applications, weak convergence is often more relevant since we care about distributional fidelity rather than exact trajectory matching.

Step size choice. In practice, $N \in \{50, 100, 1000\}$ are typical values. Smaller N yields faster inference but introduces larger discretisation error, potentially degrading sample quality. The optimal choice depends on the SDE family, the learned score model accuracy, and the desired quality-speed trade-off. Adaptive step-size schemes (varying Δt across the trajectory) can improve efficiency but add implementation complexity.

Coefficient consistency. It is crucial that the discrete coefficients $f_i(\mathbf{x})$ and G_i used during sampling match those implicitly assumed during training. Any discrepancy between the forward marginal used for denoising score matching and the reverse-time discretisation will introduce bias and degrade sample quality.

Numerical stability. For very small Δt , the squared coefficient $G_i^2 = g(t_i)^2 \Delta t$ can become numerically tiny, potentially leading to underflow or loss of precision when combined with large score values. Implementing in higher precision (float64) or rescaling intermediate quantities can mitigate these issues.

Random number generation. Use high-quality Gaussian random number generators (e.g., Box–Muller or Ziggurat algorithms) and ensure reproducibility via explicit seeds. The correct variance scaling $\Delta \mathbf{W}_i = \sqrt{\Delta t} \mathbf{z}_i$ must be respected to match the continuous-time SDE’s noise intensity.

Higher-order schemes. When higher strong-order accuracy is required, one can employ Milstein or stochastic Runge–Kutta methods. However, these schemes typically require access to derivatives of the diffusion coefficient or Lévy area terms for multidimensional diffusions [13], which adds computational cost and complexity. In the score-based setting, the simple Euler–Maruyama scheme often suffices, especially when combined with a moderate number of steps.

Interpolation for diagnostics. When computing path-dependent statistics or visualising sample trajectories, continuous interpolations of the discrete path $\{\mathbf{x}_i\}$ are needed. Standard choices include piecewise-constant (simplest) or piecewise-linear

interpolation:

$$\text{(piecewise-constant)} \quad \mathbf{x}(t) = \mathbf{x}_i \quad \text{for } t \in [t_i, t_{i+1}),$$

$$\text{(piecewise-linear)} \quad \mathbf{x}(t) = \mathbf{x}_i + \frac{t - t_i}{\Delta t}(\mathbf{x}_{i+1} - \mathbf{x}_i) \quad \text{for } t \in [t_i, t_{i+1}].$$

The choice of interpolation affects pathwise error estimates; see Chapter 9 of [13] for details.

Implementation checklist.

1. Compute the time grid $t_i = i \Delta t, i = 0, \dots, N$.
2. Pre-compute or cache model-specific functions (e.g., $\beta(t), \sigma(t), \alpha(t)$) to avoid repeated evaluations.
3. Initialise $\mathbf{x}_N$ from the terminal prior $p_T \approx \mathcal{N}(0, I)$.
4. For each reverse step $i = N, \dots, 1$:
 - Compute $f_i(\mathbf{x}_i)$ and G_i according to Table 4.1.
 - Evaluate the score model $\mathbf{s}_{\theta^*}(\mathbf{x}_i, t_i)$.
 - Draw $\mathbf{z}_i \sim \mathcal{N}(0, I)$.
 - Update via (4.8).
5. Return $\mathbf{x}_0$.

4.2 Probability Flow ODE

An alternative to the stochastic reverse process is a deterministic ordinary differential equation (ODE) whose trajectories share the same marginal probability densities $\{p_t(\mathbf{x})\}_{t=0}^T$ as the forward SDE. This ODE, known as the *probability flow ODE* [19], arises from a rewriting of the Fokker–Planck equation and enables exact likelihood computation, invertible mappings, and efficient sampling with adaptive solvers.

4.2.1 Derivation from the Fokker–Planck Equation

Consider the general forward SDE

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t) dt + \mathbf{G}(\mathbf{x}, t) d\mathbf{w}, \tag{4.17}$$

where $\mathbf{f}(\cdot, t) : \mathbb{R}^d \rightarrow \mathbb{R}^d$ and $\mathbf{G}(\cdot, t) : \mathbb{R}^d \rightarrow \mathbb{R}^{d \times d}$. The marginal density $p_t(\mathbf{x})$ satisfies the Fokker–Planck equation (Kolmogorov forward equation) [15]

$$\frac{\partial p_t}{\partial t} = - \sum_{i=1}^d \frac{\partial}{\partial x_i} (f_i p_t) + \frac{1}{2} \sum_{i=1}^d \sum_{j=1}^d \frac{\partial^2}{\partial x_i \partial x_j} \left[\sum_{k=1}^d G_{ik} G_{jk} p_t \right]. \quad (4.18)$$

Define $A_{ij} = \sum_{k=1}^d G_{ik} G_{jk}$. Then

$$\begin{aligned} \frac{\partial p_t}{\partial t} &= -\nabla_{\mathbf{x}} \cdot (\mathbf{f} p_t) + \frac{1}{2} \nabla_{\mathbf{x}} \cdot (\nabla_{\mathbf{x}} \cdot (A p_t)) \\ &= -\nabla_{\mathbf{x}} \cdot (\mathbf{f} p_t) + \frac{1}{2} \nabla_{\mathbf{x}} \cdot (p_t \nabla_{\mathbf{x}} \cdot A + A \nabla_{\mathbf{x}} p_t). \end{aligned} \quad (4.19)$$

Using $\nabla_{\mathbf{x}} p_t = p_t \nabla_{\mathbf{x}} \log p_t$, we obtain

$$\frac{\partial p_t}{\partial t} = -\nabla_{\mathbf{x}} \cdot \left(\left[\mathbf{f} - \frac{1}{2} \nabla_{\mathbf{x}} \cdot A - \frac{1}{2} A \nabla_{\mathbf{x}} \log p_t \right] p_t \right). \quad (4.20)$$

Equation (4.20) is a continuity equation for the density p_t with velocity field

$$\tilde{\mathbf{f}}(\mathbf{x}, t) := \mathbf{f}(\mathbf{x}, t) - \frac{1}{2} \nabla_{\mathbf{x}} \cdot (\mathbf{G}(\mathbf{x}, t) \mathbf{G}(\mathbf{x}, t)^\top) - \frac{1}{2} \mathbf{G}(\mathbf{x}, t) \mathbf{G}(\mathbf{x}, t)^\top \nabla_{\mathbf{x}} \log p_t(\mathbf{x}). \quad (4.21)$$

Hence the deterministic ODE

$$\frac{d\mathbf{x}}{dt} = \tilde{\mathbf{f}}(\mathbf{x}, t) \quad (4.22)$$

produces the same marginal densities p_t as the original SDE. We call it the **probability flow ODE**.

When the diffusion is state-independent, i.e. $\mathbf{G}(\mathbf{x}, t) = g(t)\mathbf{I}$, the divergence term vanishes and the ODE simplifies to

$$\frac{d\mathbf{x}}{dt} = \mathbf{f}(\mathbf{x}, t) - \frac{1}{2} g(t)^2 \nabla_{\mathbf{x}} \log p_t(\mathbf{x}). \quad (4.23)$$

This is the form used throughout this chapter.

4.2.2 Likelihood Computation

Because the probability flow ODE is deterministic and invertible, it enables exact likelihood evaluation via the instantaneous change-of-variables formula for continuous

normalising flows [3]. Let $\mathbf{x}(t)$ be the solution of (4.23) with $\mathbf{x}(0) = \mathbf{x}_0$ and $\mathbf{x}(T)$ the corresponding latent code. The log-density of $\mathbf{x}_0$ under the data distribution is

$$\log p_0(\mathbf{x}_0) = \log p_T(\mathbf{x}(T)) + \int_0^T \nabla_{\mathbf{x}} \cdot \left(\mathbf{f}(\mathbf{x}, t) - \frac{1}{2} g(t)^2 \mathbf{s}_{\theta^*}(\mathbf{x}, t) \right) dt, \quad (4.24)$$

where we have replaced the true score by the trained model $\mathbf{s}_{\theta^*}(\mathbf{x}, t)$. The divergence term can be estimated efficiently using the Skilling–Hutchinson trace estimator [16, 8]:

$$\nabla_{\mathbf{x}} \cdot \mathbf{v}(\mathbf{x}) = \mathbb{E}_{\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})} \left[\boldsymbol{\epsilon}^\top \nabla_{\mathbf{x}} \mathbf{v}(\mathbf{x}) \boldsymbol{\epsilon} \right], \quad (4.25)$$

where $\nabla_{\mathbf{x}} \mathbf{v}(\mathbf{x})$ is the Jacobian matrix of $\mathbf{v}$. The vector-Jacobian product can be computed with a single reverse-mode automatic differentiation call, making the likelihood evaluation tractable even in high dimensions.

4.2.3 Sampling with the Probability Flow ODE

To generate samples, we solve the probability flow ODE backward in time, starting from $\mathbf{x}_T \sim p_T$ (e.g., $\mathcal{N}(0, \mathbf{I})$). Any numerical ODE solver can be employed. A simple Euler discretisation of (4.23) from time t_i to t_{i-1} (with step size Δt) gives the iteration

$$\mathbf{x}_{i-1} = \mathbf{x}_i - \mathbf{f}(\mathbf{x}_i, t_i) \Delta t + \frac{1}{2} g(t_i)^2 \mathbf{s}_{\theta^*}(\mathbf{x}_i, t_i) \Delta t. \quad (4.26)$$

For the concrete SDE families introduced in Section 3, the coefficients become:

- **VE SDE:** $\mathbf{f} = 0$, $g(t) = \sqrt{\frac{d}{dt} \sigma(t)^2}$.
- **VP SDE:** $\mathbf{f} = -\frac{1}{2} \beta(t) \mathbf{x}$, $g(t) = \sqrt{\beta(t)}$.
- **sub-VP SDE:** $\mathbf{f} = -\frac{1}{2} \beta(t) \mathbf{x}$, $g(t) = \sqrt{\beta(t) (1 - e^{-2 \int_0^t \beta(s) ds})}$.
- **LV SDE:** $\mathbf{f} = -\theta \mathbf{x}$, $g(t) = \sqrt{1 + 2\theta t}$.

Plugging these into (4.26) yields the explicit updates summarised in Table 4.3.

In practice, higher-order solvers (e.g., Runge–Kutta or adaptive solvers like Dormand–Prince) can be used to reduce the number of score evaluations while maintaining high sample quality [19].

SDE	$-\mathbf{f}_i$	$\frac{1}{2}G_i^2$
VE	0	$\frac{1}{2}(\sigma_i^2 - \sigma_{i-1}^2)$
VP	$\frac{1}{2}\beta_i \mathbf{x}_i$	$\frac{1}{2}\beta_i$
sub-VP	$\frac{1}{2}\beta_i \mathbf{x}_i$	$\frac{1}{2}\beta_i(1 - e^{-2\int_0^{t_i} \beta(s)ds})$
LV	$\theta \mathbf{x}_i$	$\frac{1}{2}(1 + 2\theta t_i)$

Table 4.3 Discrete coefficients for the Euler integrator of the probability flow ODE. Here $G_i^2 = g(t_i)^2 \Delta t$ and $f_i = \mathbf{f}(\mathbf{x}_i, t_i) \Delta t$.

4.2.4 Connection to Reverse SDE

The probability flow ODE and the reverse SDE share the same marginal distributions, but the ODE is deterministic and often requires fewer function evaluations when a high-order integrator is employed. It also provides a deterministic encoder, which is useful for latent space manipulation (e.g., interpolation, temperature scaling). However, for very coarse discretisations, the stochastic reverse SDE may be more robust; the choice depends on the application. Both methods are widely used in practice.

Chapter 5

Experimental Implementation and Results

Having established the theoretical framework for the four SDE families and their reverse-time discretisations in Chapters 3 and 4.1, we now describe the practical implementation of the score-based generative models and report the experimental results obtained on the MNIST handwritten digit dataset. This chapter proceeds as follows: we first describe the shared neural architecture used across all four models (5.1), then detail the training procedure and objective (5.2), present the sampling algorithms as implemented (5.3), and finally discuss the results (5.4).

5.1 Score Network Architecture

All four SDE families share a common score network architecture based on a conditional U-Net, adapted for the task of denoising score estimation. The network takes as input a noisy image $\mathbf{x}_t \in \mathbb{R}^{1 \times 28 \times 28}$ together with a conditioning scalar that encodes the current diffusion state, and produces a score estimate $\mathbf{s}_\theta(\mathbf{x}_t, \cdot) \in \mathbb{R}^{1 \times 28 \times 28}$. Although the same computational graph is used in all cases, the nature of the conditioning scalar differs: the VE model receives the log noise level $\log \sigma(t)$, while the VP, sub-VP and LV models are conditioned on the diffusion time $t \in [0, 1]$.

The U-Net must be informed of the amount of noise present in the image it is currently processing. This is encoded by a scalar C : for the VE model $C = \log \sigma(t)$, where $\sigma(t) = \sigma_{\min}(\sigma_{\max}/\sigma_{\min})^t$; for the VP, sub-VP and LV models $C = t \in [0, 1]$. A small multilayer perceptron ϕ maps this scalar to a d_e -dimensional conditioning vector,

$$\phi : \mathbb{R} \longrightarrow \mathbb{R}^{d_e}, \quad \phi(c) = W_2 \text{ SiLU}(W_1 c + b_1) + b_2, \quad (5.1)$$

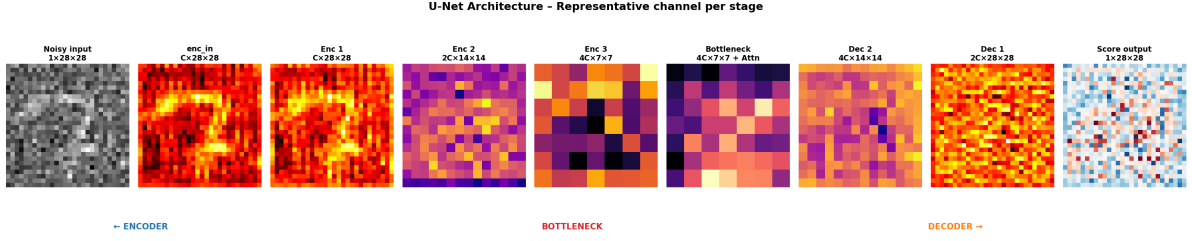


Figure 5.1 Overall U-Net architecture for score estimation. The noisy image $\mathbf{x}_t$ and the conditioning scalar (either $\log \sigma$ or t) enter from the left; the score estimate $\hat{\mathbf{s}}$ exits on the right.

with $W_1 \in \mathbb{R}^{d_e \times 1}$, $W_2 \in \mathbb{R}^{d_e \times d_e}$, $b_1, b_2 \in \mathbb{R}^{d_e}$, and $d_e = 256$. Here $\text{SiLU}(z) = z \sigma(z)$ is applied element-wise.

The vector $\mathbf{e} = \phi(C) \in \mathbb{R}^{256}$ is computed once per forward pass and injected into every residual block, allowing the whole network to adapt its behaviour to the diffusion state. Without this information the network could not distinguish, for instance, between removing heavy noise and refining an almost clean image, two situations that demand very different score predictions.

A *conditional residual block* is a parametric map

$$\mathcal{R} : \mathbb{R}^{C \times H \times W} \times \mathbb{R}^{d_e} \longrightarrow \mathbb{R}^{C \times H \times W}. \quad (5.2)$$

Given a feature map $\mathbf{x}$ and the conditioning vector $\mathbf{e}$, the output $\mathbf{y} = \mathcal{R}(\mathbf{x}, \mathbf{e})$ is defined by

$$\begin{aligned} \mathbf{h}_1 &= \text{SiLU}(\text{GN}_8(\mathbf{x})), \\ \mathbf{h}_2 &= \mathbf{K}_1 * \mathbf{h}_1, \quad \mathbf{K}_1 \in \mathbb{R}^{C \times C \times 3 \times 3}, \\ \mathbf{h}_3 &= \mathbf{h}_2 + (W_p \mathbf{e})_{[:,1,1]}, \quad W_p \in \mathbb{R}^{C \times d_e}, \\ \mathbf{h}_4 &= \text{SiLU}(\text{GN}_8(\mathbf{h}_3)), \\ \mathbf{h}_5 &= \mathbf{K}_2 * \mathbf{h}_4, \quad \mathbf{K}_2 \in \mathbb{R}^{C \times C \times 3 \times 3}, \\ \mathbf{y} &= \mathbf{x} + \mathbf{h}_5, \end{aligned} \quad (5.3)$$

where GN_8 denotes Group Normalisation with 8 groups, and all convolutions use 3×3 kernels with padding 1. The term $(W_p \mathbf{e})_{[:,1,1]} \in \mathbb{R}^{C \times 1 \times 1}$ is the linear projection of $\mathbf{e}$ broadcast spatially to act as a per-channel additive bias on $\mathbf{h}_2$. The projection matrix W_p is independent for each ResBlock instance, so different levels of the U-Net can weight the conditioning signal differently.

The residual connection $\mathbf{y} = \mathbf{x} + \mathbf{h}_5$ makes $\mathcal{R}$ a perturbation of the identity: when the learned correction $\mathbf{h}_5$ is small, the block approximates the identity map, a property that stabilises optimisation in deep compositions.

From the perspective of learning the score, these blocks are responsible for extracting and refining the local patterns edges, textures, curvatures that indicate in which direction the image must move to reduce the noise. By stacking blocks and progressively reducing spatial resolution in the encoder, the network builds a hierarchy of descriptors ranging from fine details up to global structures.

At the coarsest spatial resolution ($H = W = 7$), after the bottleneck ResBlocks a *multi-head self-attention* layer is interposed. Define the attention operator

$$\mathcal{A} : \mathbb{R}^{4C \times 7 \times 7} \longrightarrow \mathbb{R}^{4C \times 7 \times 7} \quad (5.4)$$

as follows. Let $N = 49$ and let $\text{vec} : \mathbb{R}^{4C \times 7 \times 7} \xrightarrow{\sim} \mathbb{R}^{N \times 4C}$ be the canonical reshaping $(c, i, j) \mapsto (7i + j, c)$. For input $\mathbf{x} \in \mathbb{R}^{4C \times 7 \times 7}$,

$$\begin{aligned} \mathbf{X} &= \text{vec}(\text{GN}_8(\mathbf{x})) \in \mathbb{R}^{N \times 4C}, \\ \mathbf{A} &= \text{MHA}(\mathbf{X}, \mathbf{X}, \mathbf{X}) \in \mathbb{R}^{N \times 4C}, \\ \mathcal{A}(\mathbf{x}) &= \mathbf{x} + \text{vec}^{-1}(\mathbf{A}) \in \mathbb{R}^{4C \times 7 \times 7}, \end{aligned} \quad (5.5)$$

where MHA is standard scaled dot-product multi-head attention with $n_h = 4$ heads and per-head dimension $d_k = C$. Because $N = 49$ is fixed, the 49×49 attention matrix is computed exactly with no approximation required.

This mechanism equips the model with non-local interactions that are essential for a globally coherent score estimate: a pixel in one corner of the image may depend on structure appearing in the opposite corner, and convolutions alone cannot capture such dependencies efficiently due to their limited receptive field. The residual form in (5.5) means the network learns only the non-local complement that the convolutions were unable to provide.

Together, the residual blocks build multi-scale local representations, and the bottleneck self-attention ensures that those representations are organised globally, supplying the decoder with the information needed to synthesise a smooth, globally consistent score field.

5.1.1 U-Net Encoder-Decoder Structure

The complete score network is a function

$$s_\theta : \mathbb{R}^{1 \times 28 \times 28} \times \mathbb{R} \longrightarrow \mathbb{R}^{1 \times 28 \times 28}. \quad (5.6)$$

It is constructed as the composition of an encoder, a bottleneck, and a decoder connected by skip maps, which we now define precisely. All feature map visualisations in this subsection display activations on a greyscale ranging from black (minimum activation) through orange-red mid-tones to white (maximum activation), with values mapped linearly within each individual map.

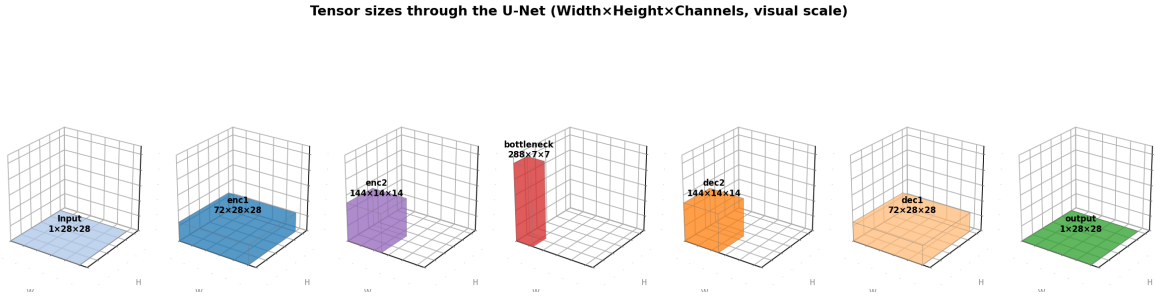


Figure 5.2 Scaled visual representation of the tensor size (Height $\times$ Width $\times$ Channels) at each major stage of the U-Net.

Stem. A single 3×3 convolution with bias maps the input to $C = 72$ feature channels:

$$\Phi_{\text{stem}} : \mathbb{R}^{1 \times 28 \times 28} \longrightarrow \mathbb{R}^{C \times 28 \times 28}. \quad (5.7)$$

Figure 5.3 shows the channel-mean map and a representative selection of individual channels of $\Phi_{\text{stem}}(\mathbf{x}_t)$. The full spatial resolution 28×28 is preserved at this stage, and each channel responds to a distinct local pattern in the input.

Encoder. The encoder consists of three levels. Let $\mathbf{e} = \phi(C)$ be the conditioning vector (5.1).

- **Level 1** ($C \times 28 \times 28$). Two ResBlocks (5.3) are applied in sequence:

$$\mathbf{s}_1 = \mathcal{R}^{(1b)}(\mathcal{R}^{(1a)}(\Phi_{\text{stem}}(\mathbf{x}), \mathbf{e}), \mathbf{e}) \in \mathbb{R}^{C \times 28 \times 28}. \quad (5.8)$$

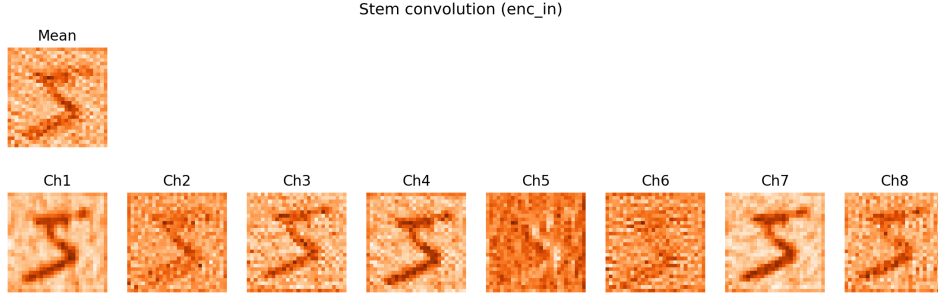


Figure 5.3 Activations of the stem convolution $\Phi_{\text{stem}}(\mathbf{x}_t) \in \mathbb{R}^{C \times 28 \times 28}$. *Top row, leftmost:* channel mean. *Bottom row:* first eight individual channels. At this stage the full 28×28 resolution is retained and each channel encodes a distinct linear response to the input.

The tensor $\mathbf{s}_1$ is retained as a skip map. Figure 5.4 displays the activations after $\mathcal{R}^{(1a)}$ (left panel) and $\mathcal{R}^{(1b)}$ (right panel), illustrating the progressive sharpening of edge-like responses.

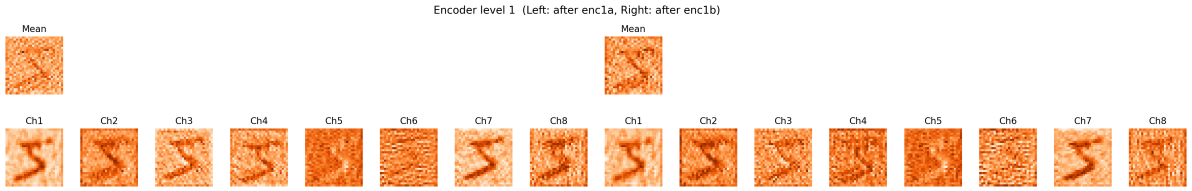


Figure 5.4 Encoder level 1 activations at $C \times 28 \times 28$. *Left eight columns:* after the first ResBlock $\mathcal{R}^{(1a)}$. *Right eight columns:* after the second ResBlock $\mathcal{R}^{(1b)}$, which constitutes the skip map $\mathbf{s}_1$. Each panel shows the channel mean (top-left) and the first eight channels (bottom row).

- **Downsampling 1 $\rightarrow$ 2.** A strided convolution $D_1 : \mathbb{R}^{C \times 28 \times 28} \rightarrow \mathbb{R}^{2C \times 14 \times 14}$ with a 4×4 kernel and stride 2 halves the spatial dimensions and doubles the channel count. Figure 5.5 shows $D_1(\mathbf{s}_1)$; the channel-mean map reveals that coarse structure is preserved while fine detail is discarded.
- **Level 2** ($2C \times 14 \times 14$).

$$\mathbf{s}_2 = \mathcal{R}^{(2b)}(\mathcal{R}^{(2a)}(D_1(\mathbf{s}_1), \mathbf{e}), \mathbf{e}) \in \mathbb{R}^{2C \times 14 \times 14}. \quad (5.9)$$

The tensor $\mathbf{s}_2$ is retained as a skip map. At 14×14 each channel integrates a larger receptive field, so the responses are more spatially diffuse and capture wider structural motifs (Figure 5.6).

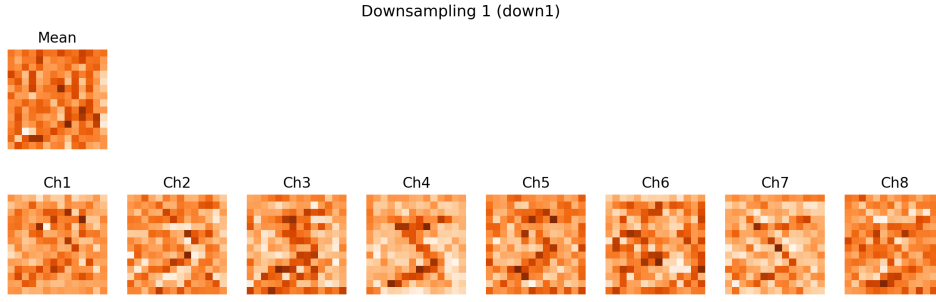


Figure 5.5 Activations immediately after the first downsampling $D_1(\mathbf{s}_1) \in \mathbb{R}^{2C \times 14 \times 14}$. The spatial resolution drops from 28×28 to 14×14 , and the channel count doubles to $2C = 144$.

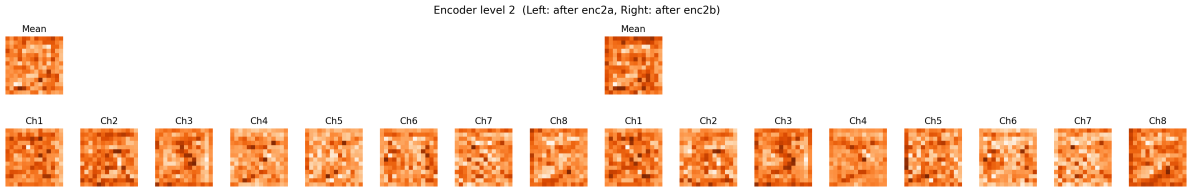


Figure 5.6 Encoder level 2 activations at $2C \times 14 \times 14$. *Left*: after $\mathcal{R}^{(2a)}$. *Right*: after $\mathcal{R}^{(2b)}$, the skip map $\mathbf{s}_2$. The reduced spatial resolution makes the activations more global in character than those of level 1.

- **Downsampling 2 $\rightarrow$ 3.** $D_2 : \mathbb{R}^{2C \times 14 \times 14} \rightarrow \mathbb{R}^{4C \times 7 \times 7}$, same design as D_1 . At 7×7 the spatial extent is reduced to $N = 49$ positions, the cardinality at which $\mathcal{A}$ computes exact pairwise interactions (Figure 5.7).
- **Level 3** ($4C \times 7 \times 7$).

$$\mathbf{s}_3 = \mathcal{R}^{(3b)}(\mathcal{R}^{(3a)}(D_2(\mathbf{s}_2), \mathbf{e}), \mathbf{e}) \in \mathbb{R}^{4C \times 7 \times 7}. \quad (5.10)$$

Bottleneck.

$$\mathbf{m} = \mathcal{A}(\mathcal{R}^{(m2)}(\mathcal{R}^{(m1)}(\mathbf{s}_3, \mathbf{e}), \mathbf{e})) \in \mathbb{R}^{4C \times 7 \times 7}, \quad (5.11)$$

where $\mathcal{A}$ is the attention operator defined in (5.5). Figure 5.8 compares the activations before and after $\mathcal{A}$; the differences between the two panels reflect the non-local redistribution of activation mass induced by the attention mechanism.

Decoder. The decoder restores spatial resolution via transposed convolutions and re-incorporates the skip maps by channel concatenation.

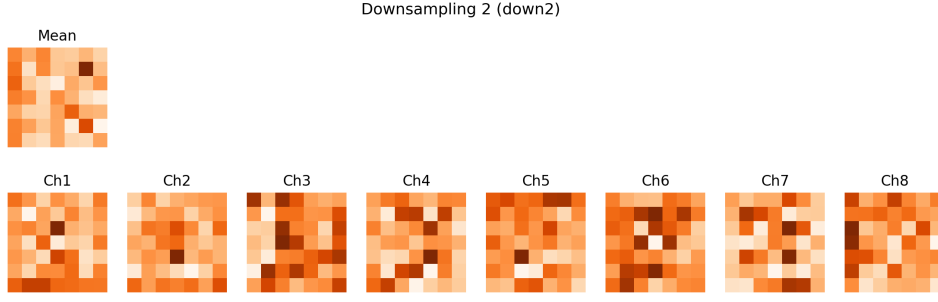


Figure 5.7 Activations immediately after the second downsampling $D_2(\mathbf{s}_2) \in \mathbb{R}^{4C \times 7 \times 7}$. Spatial resolution is now 7×7 ($N = 49$ tokens), and the channel count reaches its maximum value of $4C = 288$.

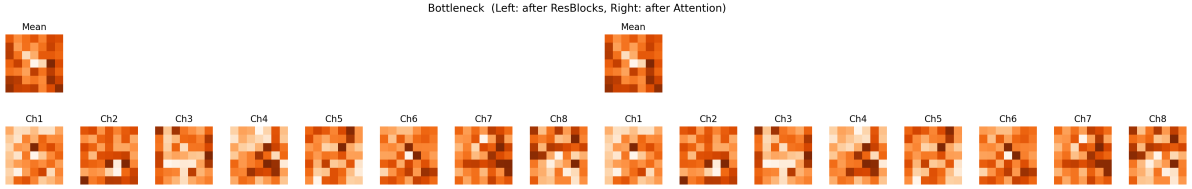


Figure 5.8 Bottleneck activations at $4C \times 7 \times 7$. *Left*: after the two bottleneck ResBlocks. *Right*: after the self-attention operator $\mathcal{A}$ (5.5), giving $\mathbf{m}$. Differences between the two panels reflect the non-local redistribution induced by the multi-head self-attention.

- **Upsampling $3 \rightarrow 2$.** $U_2 : \mathbb{R}^{4C \times 7 \times 7} \rightarrow \mathbb{R}^{2C \times 14 \times 14}$, a 4×4 transposed convolution with stride 2 (Figure 5.9).
- **Level 2 decoder ($4C \times 14 \times 14$).** The skip map $\mathbf{s}_2$ is concatenated along the channel dimension before the ResBlocks:

$$\mathbf{d}_2 = \mathcal{R}^{(d2b)}\left(\mathcal{R}^{(d2a)}([U_2(\mathbf{m}); \mathbf{s}_2], \mathbf{e}), \mathbf{e}\right) \in \mathbb{R}^{4C \times 14 \times 14}, \quad (5.12)$$

where $[\cdot; \cdot]$ denotes concatenation on the channel axis; the concatenated input has $2C + 2C = 4C$ channels. Figure 5.10 shows how the ResBlocks integrate the global context from $\mathbf{m}$ with the local detail preserved in $\mathbf{s}_2$.

- **Upsampling $2 \rightarrow 1$.** $U_1 : \mathbb{R}^{4C \times 14 \times 14} \rightarrow \mathbb{R}^{C \times 28 \times 28}$, same design as U_2 .
- **Level 1 decoder ($2C \times 28 \times 28$).**

$$\mathbf{d}_1 = \mathcal{R}^{(d1b)}\left(\mathcal{R}^{(d1a)}([U_1(\mathbf{d}_2); \mathbf{s}_1], \mathbf{e}), \mathbf{e}\right) \in \mathbb{R}^{2C \times 28 \times 28}. \quad (5.13)$$

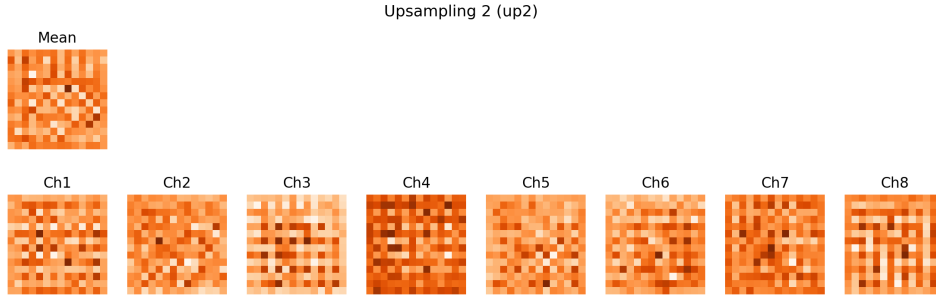


Figure 5.9 Activations after the first transposed convolution, $U_2(\mathbf{m}) \in \mathbb{R}^{2C \times 14 \times 14}$. The spatial resolution is restored to 14×14 ; at this point the skip map $\mathbf{s}_2$ has not yet been incorporated.

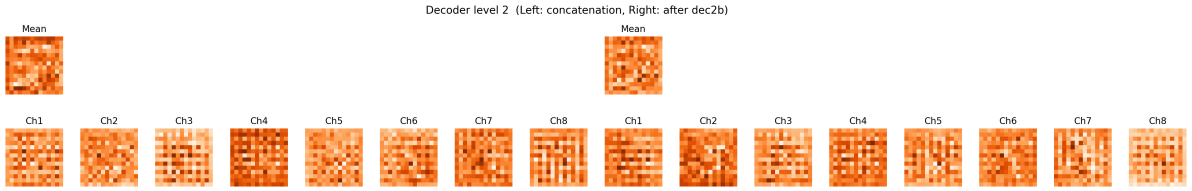


Figure 5.10 Decoder level 2 activations at 14×14 . *Left*: immediately after the skip concatenation $[U_2(\mathbf{m}); \mathbf{s}_2] \in \mathbb{R}^{4C \times 14 \times 14}$. *Right*: output $\mathbf{d}_2$ after two ResBlocks.

At this stage the full 28×28 resolution is recovered; the activations exhibit finer spatial structure than those of the decoder level 2 (Figure 5.11), reflecting the contribution of the fine-grained skip map $\mathbf{s}_1$.

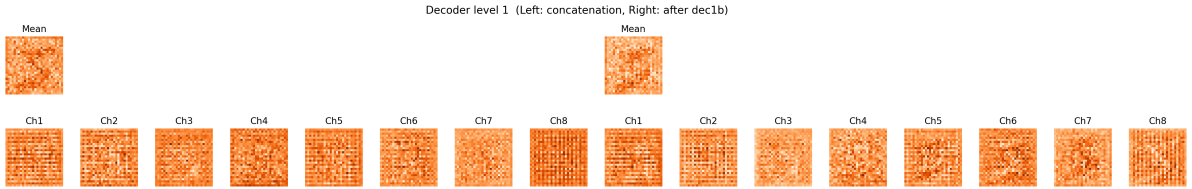


Figure 5.11 Decoder level 1 activations (28×28). *Left*: after $[U_1(\mathbf{d}_2); \mathbf{s}_1]$. *Right*: output $\mathbf{d}_1$ after two ResBlocks.

Output head. A final projection collapses the 2C channels to a single-channel output:

$$s_\theta(\mathbf{x}, C) = \Phi_{\text{out}}(\mathbf{d}_1) = \mathbf{K}_{\text{out}} * \text{SiLU}(\text{GN}_8(\mathbf{d}_1)) \in \mathbb{R}^{1 \times 28 \times 28}, \quad (5.14)$$

with $\mathbf{K}_{\text{out}} \in \mathbb{R}^{1 \times 2C \times 3 \times 3}$. Figure 5.12 shows the single-channel output; the signed activations encode the estimated score direction at each spatial location.

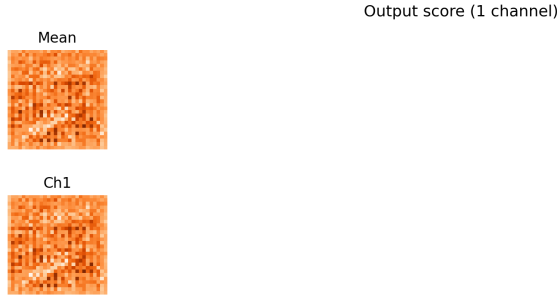


Figure 5.12 Output score estimate $s_\theta(\mathbf{x}_t, C) \in \mathbb{R}^{1 \times 28 \times 28}$. The single output channel is shown in both panels (mean and channel view coincide). Each pixel value is the network’s estimate of the score component $\partial_x \log p_t(\mathbf{x}_t)$ at that location.

Remark on skip connections. The maps $\mathbf{s}_1$ and $\mathbf{s}_2$ provide the decoder with access to the pre-compression representation at each spatial scale. Formally, the full network can be written as

$$s_\theta = \Phi_{\text{out}} \circ \mathcal{D}_1 \circ \mathcal{D}_2 \circ \mathcal{B} \circ \mathcal{E}_3 \circ \mathcal{E}_2 \circ \mathcal{E}_1 \circ \Phi_{\text{stem}}, \quad (5.15)$$

where each decoder block $\mathcal{D}_k$ depends explicitly on $\mathbf{s}_k$ through the concatenation operator, making the composition non-sequential: $\mathcal{D}_k$ takes as input both the output of the preceding block *and* the output of $\mathcal{E}_k$.

5.1.2 Learned Representations after Training

To illustrate how the score network extracts structure from a noisy image, Figure 5.13 shows the activations of several key layers after training a miniature version of the model on only 16 MNIST images for 120 epochs (final loss ≈ 0.20). Despite the extremely limited data and training budget, the feature maps already reveal organised behaviour:

- **Encoder level 1** ($C \times 28 \times 28$): early layers detect low-level features such as edges and blobs at full resolution (cf. Figure 5.4).
- **Encoder level 2** ($2C \times 14 \times 14$): after downsampling, the network begins to aggregate local patterns into more abstract, larger-scale motifs (cf. Figure 5.6).
- **Bottleneck** ($4C \times 7 \times 7$): at the coarsest scale, the self-attention module (5.5) redistributes activation mass across all 49 spatial positions, encoding global shape information that constrains the score direction (cf. Figure 5.8).

- **Output ($1 \times 28 \times 28$):** the final layer synthesises these multiscale representations into a smooth estimate of the score (cf. Figure 5.12).

This hierarchy mirrors the inductive bias of the U-Net: local convolutions build up feature detectors, downsampling injects spatial context, and skip connections allow the precise localisation of the predicted score field.

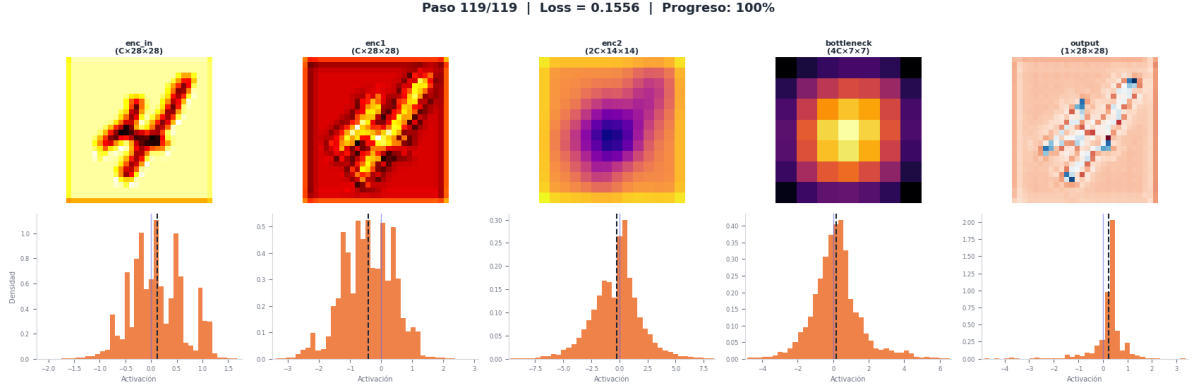


Figure 5.13 Activation maps of selected layers after training

5.2 Training Procedure

5.2.1 Dataset and Preprocessing

All experiments were conducted on the MNIST training set, consisting of 60 000 greyscale images of handwritten digits at resolution 28×28 pixels. Images were normalised from $[0, 1]$ to $[-1, 1]$ via $\mathbf{x} \leftarrow 2\mathbf{x} - 1$. Mini-batches were formed using a standard PyTorch DataLoader configured with `num_workers=2`, `pin_memory=True`, `persistent_workers=True`, and `prefetch_factor=4`, ensuring efficient asynchronous data transfer to the GPU with no pipeline stalls.

5.2.2 Denoising Score Matching Objective

All four models are trained using the *denoising score matching* (DSM) objective [21, 17]. Given the forward kernel $q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\mathbf{m}_t(\mathbf{x}_0), \sigma_t^2 I)$ for each SDE family, the loss is

$$\mathcal{L}(\theta) = \mathbb{E}_t \left[\mathbb{E}_{\mathbf{x}_0} \left[\mathbb{E}_{\epsilon} \left[\left\| \sigma_t \mathbf{s}_{\theta}(\mathbf{x}_t, \cdot) + \epsilon \right\|^2 \right] \right] \right], \quad (5.16)$$

where $\mathbf{x}_t = \mathbf{m}_t(\mathbf{x}_0) + \sigma_t \epsilon$ with $\epsilon \sim \mathcal{N}(0, I)$. The three nested expectations are estimated jointly via a single Monte Carlo sample per batch element. For a batch of size B , we draw independently:

$$t^{(i)} \sim \mathcal{U}[0, T], \quad \mathbf{x}_0^{(i)} \sim p_{\text{data}}, \quad \epsilon^{(i)} \sim \mathcal{N}(0, I), \quad i = 1, \dots, B, \quad (5.17)$$

and the empirical loss is

$$\hat{\mathcal{L}}(\theta) = \frac{1}{B} \sum_{i=1}^B \sum_{j=1}^d \left(\sigma_{t^{(i)}} [\mathbf{s}_\theta(\mathbf{x}_{t^{(i)}}^{(i)}, \cdot)]_j + \epsilon_j^{(i)} \right)^2. \quad (5.18)$$

The four SDE families differ only in the forward kernels used to construct $\mathbf{x}_t$ and in the conditioning variable passed to the network, as summarised in Table 5.1.

Table 5.1 Forward kernels, conditioning variable, and SDE parameters for the four families. Here $\bar{\beta}(t) = \int_0^t \beta(s) ds$ with $\beta(t) = \beta_{\min} + t(\beta_{\max} - \beta_{\min})$, $\beta_{\min} = 0.01$, $\beta_{\max} = 10$; $\theta = 2$ for the LV SDE.

Model	$\mathbf{m}_t(\mathbf{x}_0)$	Conditioner	Parameters
VE	$\mathbf{x}_0$	$\sigma(t) = \sigma_{\min}(\sigma_{\max}/\sigma_{\min})^t$	$\sigma_{\min} = 0.01, \sigma_{\max} = 10$
VP	$e^{-\frac{1}{2}\bar{\beta}(t)}\mathbf{x}_0$	t	$\beta_{\min} = 0.01, \beta_{\max} = 10$
sub-VP	$e^{-\frac{1}{2}\bar{\beta}(t)}\mathbf{x}_0$	t	$\beta_{\min} = 0.01, \beta_{\max} = 10$
LV	$e^{-\theta t}\mathbf{x}_0$	t	$\theta = 2, T = 1$

5.2.3 Optimisation Details

All models were trained with the AdamW optimiser with weight decay $\lambda = 10^{-4}$, an initial learning rate of $\eta = 4 \times 10^{-4}$, and gradient clipping with maximum ℓ_2 norm 1.0 applied before each parameter update.

Norm-based clipping was used rather than value clipping because it scales the entire gradient vector uniformly, preserving its direction while preventing pathologically large steps.

The learning rate was annealed using a cosine schedule with minimum value $\eta_{\min} = 10^{-6}$:

$$\eta_k = \eta_{\min} + \frac{1}{2}(\eta - \eta_{\min}) \left(1 + \cos\left(\frac{k\pi}{K}\right) \right), \quad (5.19)$$

where k is the current step and $K = n_{\text{epochs}} \times \lfloor N/B \rfloor$ is the total number of steps.

Mixed-precision training was employed throughout, with forward passes and loss computation performed in `bfloat16` under PyTorch automatic mixed precision (AMP). The `bfloat16` format retains the same 8-bit exponent as `float32`, providing the same dynamic range and eliminating the need for a gradient scaler, which simplifies the training loop relative to `float16`. The cosine scheduler was advanced at every step without any scaler-related skipping.

Training was performed on an NVIDIA G4 GPU. Table 5.2 summarises the hyperparameters used for the reported experiments.

Table 5.2 Shared training hyperparameters for all four SDE models.

Hyperparameter	Value
Optimiser	AdamW
Learning rate η	4×10^{-4}
Weight decay	10^{-4}
Gradient clip norm	1.0
LR schedule	Cosine annealing
Minimum LR $\eta_{\min}$	10^{-6}
Batch size B	1024
Epochs	200
Precision	bfloat16 (AMP, no GradScaler)
base_channels C	80
emb_dim d_e	256
Total parameters	$\approx 16.6 \times 10^6$

5.3 Sampling Implementation

For each of the four SDE families, samples were generated using three complementary schemes: the reverse-time **Euler–Maruyama** (EM) integrator, a **probability-flow ODE**

solved with a fixed-step Euler method, and the same probability-flow ODE solved adaptively with the **RK45** integrator from `scipy.integrate.solve_ivp` ($\text{rtol} = 10^{-4}$, $\text{atol} = 10^{-5}$). The EM and fixed-step ODE schemes used $N = 2000$ uniform steps over $[T, 0]$.

All four models were initialised from their respective terminal distributions at $T = 1$. The VE model was initialised from $\mathbf{x}_N \sim \mathcal{N}(0, \sigma_{\max}^2 I)$, consistent with the variance-exploding terminal distribution. The VP, sub-VP, and LV SDE models were initialised from $\mathbf{x}_N \sim \mathcal{N}(0, I)$, consistent with the near-isotropic terminal distributions of their respective forward processes. A small numerical safeguard $t_{\min} = 10^{-5}$ was applied to the conditioning variable at the final step to prevent numerical issues at $t = 0$; no noise was injected at the very last EM step. After the final reverse step, outputs were clipped to $[-1, 1]$ and renormalised to $[0, 1]$ for display.

5.4 Results

This section systematically compares samples generated by the four SDE families: Variance Exploding (VE), Variance Preserving (VP), sub-VP, and the proposed Linear Variance (LV) SDE. Each family is evaluated under three sampling schemes: reverse-time Euler–Maruyama (EM), probability flow ODE with fixed-step Euler integration (ODE Euler), and probability flow ODE with adaptive RK45 integration (ODE RK45). All models share the same neural architecture, training objective, and optimisation procedure; therefore any difference in sample quality stems from the forward process design and, for a given family, from the numerical properties of the sampling scheme.

Variance Exploding SDE

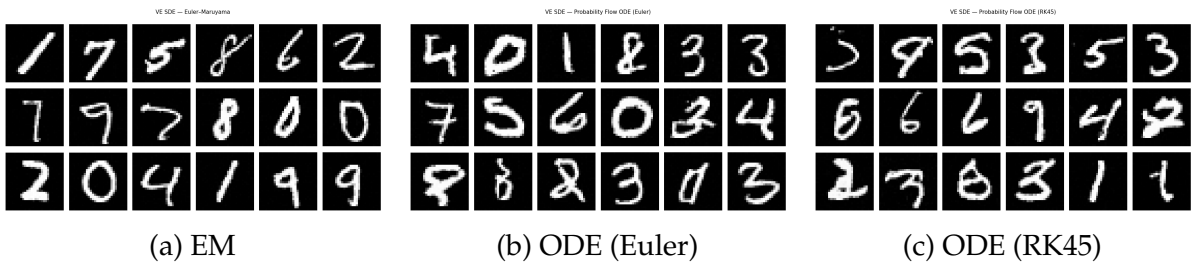


Figure 5.14 VE SDE: samples from reverse-time Euler–Maruyama ($N = 2000$ steps), probability-flow ODE with fixed-step Euler integrator ($N = 2000$ steps), and adaptive RK45 integrator ($\text{rtol} = 10^{-4}$, $\text{atol} = 10^{-5}$).

The VE forward process spans $\sigma_{\min} = 0.01$ to $\sigma_{\max} = 10$, forcing the score network to denoise from highly corrupted states. This demanding task shapes the character of the generated samples in all three sampling regimes.

The EM samples (Figure 5.14a) show thin, sharp strokes with high contrast and clean backgrounds. The stochastic injections suppress low-level artifacts, producing crisp digit contours. Digit diversity is broad, with no evidence of mode collapse. Topologically complex digits (e.g., 8) occasionally exhibit imperfect loop closure, consistent with the difficulty of reconstructing fine detail from extreme noise levels.

The ODE Euler samples (Figure 5.14b) exhibit a different degradation pattern. Without stochastic injection, deterministic error accumulates, giving strokes a grainy or chipped appearance. Several digits show fragmentation or thinning at their periphery, although class identity remains recognisable. This is consistent with fixed-step Euler accumulating local truncation errors over 2000 steps.

The ODE RK45 samples (Figure 5.14c) recover much of the sharpness lost under ODE Euler. Adaptive step control keeps local error within the specified tolerance, yielding smoother contours and more uniform filling. A slight residual graininess remains compared to EM, reflecting the inherent determinism of the ODE trajectory. Among the three VE samplers, ODE RK45 achieves the best balance of sharpness and visual regularity in the deterministic regime.

Variance Preserving SDE

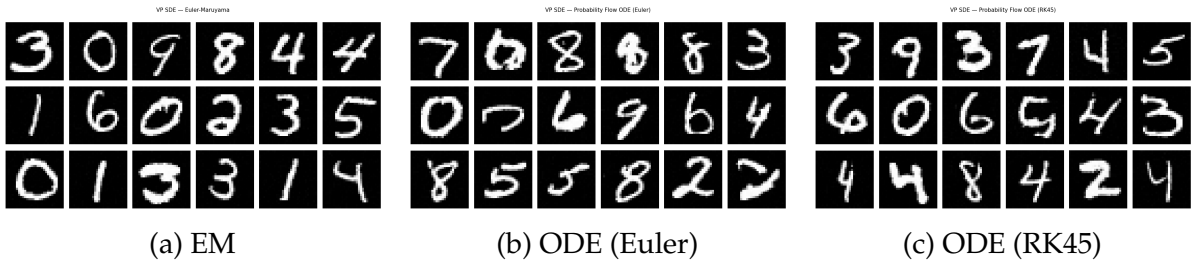


Figure 5.15 VP SDE: samples from reverse-time Euler–Maruyama ($N = 2000$ steps), probability-flow ODE with fixed-step Euler integrator ($N = 2000$ steps), and adaptive RK45 integrator ($\text{rtol} = 10^{-4}$, $\text{atol} = 10^{-5}$).

The VP forward process keeps total variance bounded by jointly attenuating the signal and adding noise, so the marginal at any t stays in a controlled regime. This balanced design is reflected in the samples across all three samplers.

The EM samples (Figure 5.15a) are the cleanest and most regularly formed among all twelve output grids. Stroke widths are uniform, digit geometry is well preserved,

and backgrounds are uniformly dark. Multiple classes (0, 1, 3, 4, 5, 6, 8, 9) appear with consistent quality and show no systematic geometric distortion. The bounded-variance forward process likely produces a score field that varies more smoothly in time, enabling the EM integrator to follow the reverse trajectory with lower cumulative error.

The ODE Euler samples (Figure 5.15b) remain broadly recognisable but introduce a specific artefact: strokes become overly thick at their midpoints, and some digits show localised blotting. Digits 8 and 0 are especially susceptible, as their looped topology concentrates the deterministic error. Despite this, digit identity is seldom ambiguous, confirming that the VP score field is smooth enough for fixed-step ODE integration to remain qualitatively successful.

The ODE RK45 samples (Figure 5.15c) represent the highest visual quality observed across all twelve experimental conditions. Stroke geometry is sharp and regular, backgrounds are clean, and all ten classes appear with balanced representation. Adaptive integration effectively eliminates the blotting artefacts seen in ODE Euler, yielding topologically faithful and well-defined contours. These results indicate that the VP forward process, paired with an accurate ODE integrator, provides the most reliable and regular sampling behaviour among the established families.

Sub-VP SDE

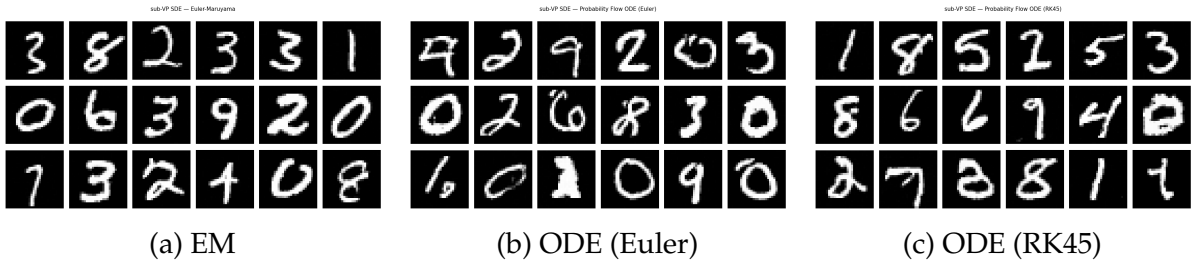


Figure 5.16 Sub-VP SDE: samples from reverse-time Euler–Maruyama ($N = 2000$ steps), probability-flow ODE with fixed-step Euler integrator ($N = 2000$ steps), and adaptive RK45 integrator ($\text{rtol} = 10^{-4}$, $\text{atol} = 10^{-5}$).

The sub-VP SDE modifies the VP diffusion coefficient by using $\sigma_t = 1 - e^{-\bar{\beta}(t)}$ (no square root), giving a variance schedule that grows more slowly than VP. This reduced noise injection is expected to produce a score field with milder time variation, with consequences that differ between stochastic and deterministic sampling.

The EM samples (Figure 5.16a) are visually clean and closely resemble VP EM outputs: solid strokes, dark backgrounds, and recognisable digits across all classes. A minor but consistent difference is a tendency toward slightly thinner strokes, possibly

reflecting the reduced diffusion scale. The stochastic sampler handles the sub-VP schedule robustly.

The ODE Euler samples (Figure 5.16b) reveal greater sensitivity to integration error than their VP counterparts. Several digits show distorted topology (elongated loops, disconnected strokes, unusual aspect ratios), suggesting that the sub-VP score field produces an ODE flow whose trajectories are more sensitive to local truncation error under fixed-step integration. Some samples are ambiguous or structurally atypical, though all remain broadly recognisable. A finer step or adaptive integrator is likely required to achieve the same deterministic accuracy as VP.

The ODE RK45 samples (Figure 5.16c) confirm this: adaptive step control restores much of the structural regularity lost under ODE Euler. Digits are again clearly formed with consistent stroke geometry. A residual characteristic of the sub-VP family, visible even with RK45, is slightly thinner or scratchier strokes compared to VP RK45, consistent with the lower effective diffusion coefficient shaping the learned score field. Topologically complex digits (8, 0) occasionally display minor contour imperfections but remain recognisable.

Linear Variance (LV) SDE

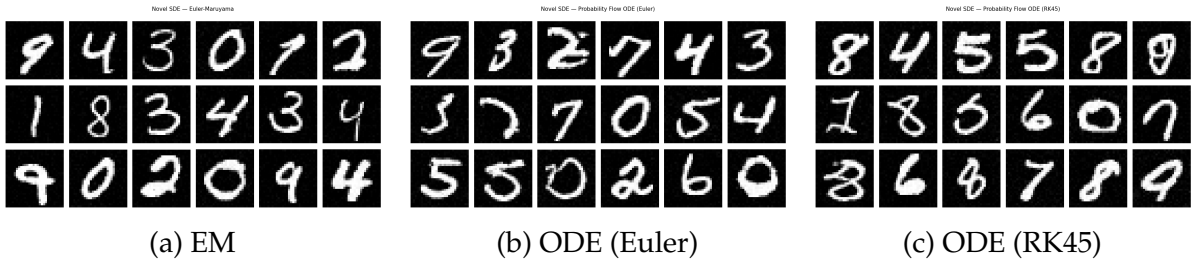


Figure 5.17 LV SDE: samples from reverse-time Euler–Maruyama ($N = 2000$ steps), probability-flow ODE with fixed-step Euler integrator ($N = 2000$ steps), and adaptive RK45 integrator ($\text{rtol} = 10^{-4}$, $\text{atol} = 10^{-5}$).

The Linear Variance (LV) SDE is the main contribution of this work. It features a linearly growing variance $\text{Var}(X_t) = t$ combined with an exponentially decaying mean coefficient $e^{-\theta t}$ ($\theta = 2$). This design differs fundamentally from the three established families: the variance grows at a constant rate rather than exponentially (VE) or saturating (VP and sub-VP), and the mean decay is stronger than in VP/sub-VP at early times. A direct consequence is that the diffusion coefficient $g(t) = \sqrt{1 + 2\theta t}$ grows continuously without saturation; consequently the forward process does not reach a well-defined terminal distribution as $t \rightarrow 0^+$, and practical integration must start

from a small positive ϵ . This introduces a near-zero regime where finite computational precision limits the exactness of the reverse trajectory, manifesting as a residual background noise level that cannot be eliminated by merely increasing the number of discretisation steps. This is an intrinsic structural feature of the LV family, not an optimisation failure, and it motivates the use of high-order adaptive solvers to manage that numerical challenge.

Despite this boundary effect, the LV SDE exhibits compelling generative behaviour that, when evaluated holistically, positions it as the strongest performing design in this experimental setting.

The EM samples (Figure 5.17a) are among the most compelling for this family. Digit strokes are bold, well defined, and show strong contrast against uniformly dark backgrounds. Classes 0, 2, 3, 4, 7, 8, and 9 display clear structural identity and good geometric regularity. The linear variance schedule distributes the learning signal uniformly across time, enabling the EM sampler, which benefits from stochastic injection at each step, to navigate the reverse trajectory with notably stable behaviour. The bold stroke character is distinctive among all four families, exceeding the thickness and foreground prominence of both VE and sub-VP EM, and approaching the weight achieved by VP EM while maintaining greater structural sharpness.

The ODE Euler samples (Figure 5.17b) maintain high recognisability, with clean backgrounds and coherent digit structure. Relative to sub-VP ODE Euler, the LV SDE shows substantially less topological distortion under fixed-step integration, suggesting that the linearly growing $g(t)$ produces a score field whose probability flow trajectories are better conditioned for this integrator. A modest degree of background noise and stroke irregularity remains, attributable to the near-zero boundary behaviour, but digit identity is preserved across all ten classes.

ODE RK45 as strongest deterministic result. The ODE RK45 samples (Figure 5.17c) represent the key qualitative finding: with an adaptive integrator, the LV SDE produces outputs visually competitive with or superior to the best VP RK45 results. The adaptive step size concentrates effort where the score field curvature is steepest, including the sensitive near-zero regime, substantially suppressing the background noise that plagues LV under fixed-step samplers. Most digits are well formed across all classes; only a few complex digits (8, 9) show minor contour imperfections, a trait shared by all families under RK45.

This demonstrates that LV's residual noise under simpler samplers is not a poorly learned score field but a mismatch between the SDE's near-zero behaviour and the

limited error control of fixed-step or purely stochastic integrators. The score network has learned a high-quality approximation throughout the trajectory; remaining artifacts are numerical and could be mitigated by tighter solver tolerances.

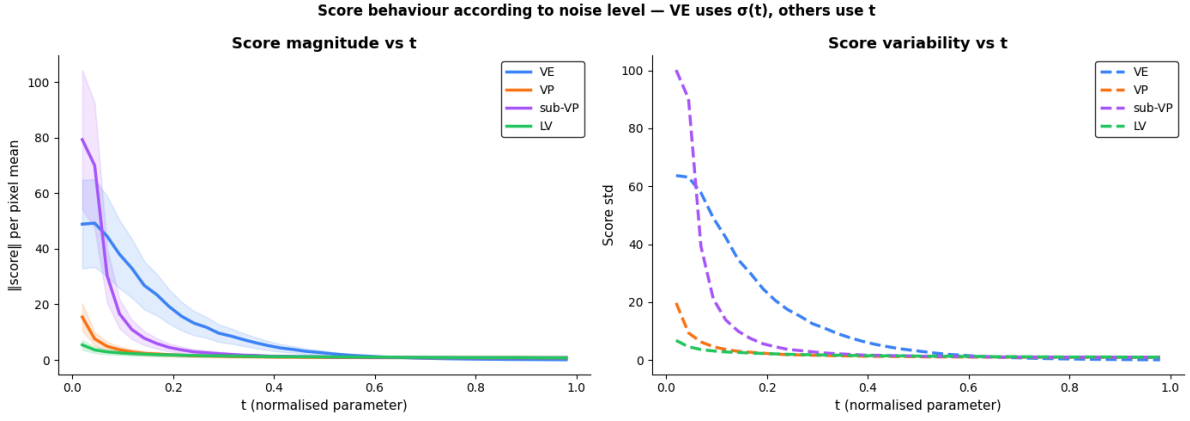


Figure 5.18 Score magnitude (left) and score variability (right) as a function of the normalised time parameter t for the four SDE families. For VE the horizontal axis represents $\sigma(t)$; for VP, sub-VP, and LV it represents $t \in [0, 1]$. Shaded bands indicate one standard deviation across the evaluation set. The LV SDE (green) consistently exhibits the lowest score magnitude and the smallest variability across the entire time range, indicating a smoother and more uniformly distributed score field over the trajectory than any competing family.

Figure 5.18 quantifies a central structural advantage of LV. Its score magnitude (left) is the smallest over the full range, and its score variability (right) is by far the lowest, with a nearly flat profile. This implies a more homogeneous learning problem: the required corrections vary little between samples at each noise level, which reduces the risk of learning a score field with sharp local curvature that would challenge numerical integrators. The connection to integration quality is direct: LV, with the smoothest and most stable score field, yields the strongest RK45 outputs, while sub-VP, with the highest near-zero variability, degrades the most under fixed-step ODE integration.

Figure 5.19 examines the internal feature maps of the shared U-Net at mid-trajectory ($t = 0.5$). The visualisation shows the maximum-energy channel at each stage: encoder input (enc_in), three encoder levels (enc1 through enc3), bottleneck, two decoder levels (dec2 and dec1), and the final score output.

Several observations distinguish the LV row (bottom, green border). In the early encoder stages, VE and sub-VP already show highly localised, spatially coarse activations, consistent with their higher score magnitudes at $t = 0.5$ (compare Figure 5.18). VP encoder stages are smoother but display horizontal banding in the decoder re-

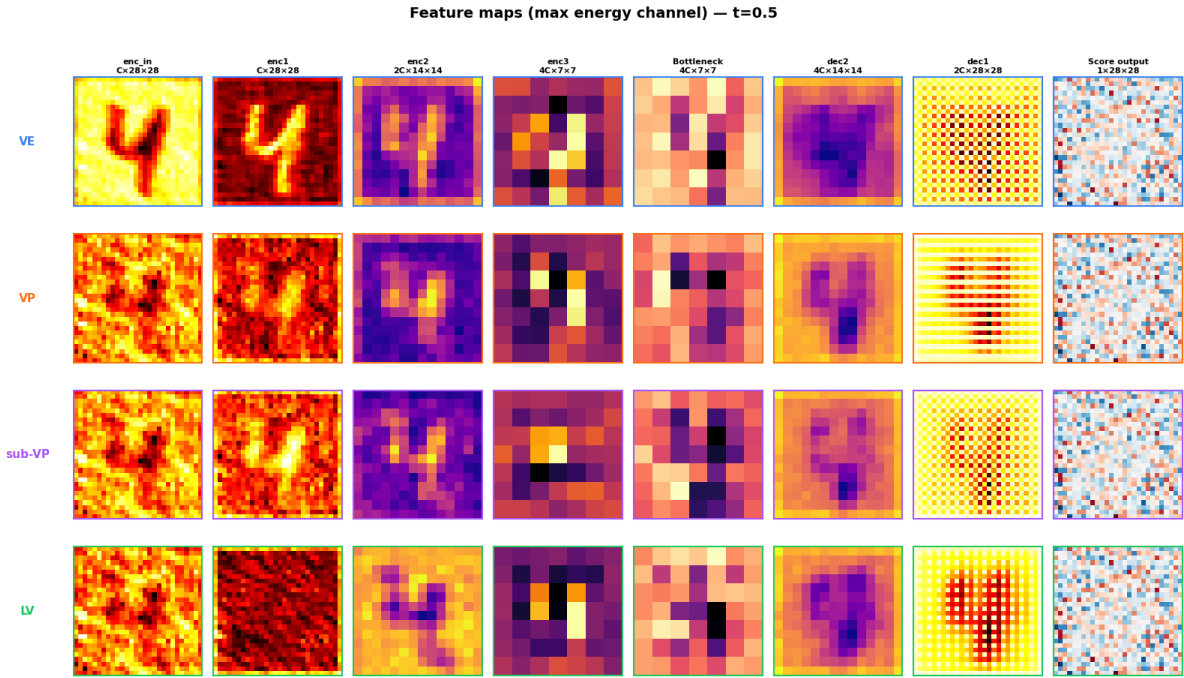


Figure 5.19 Maximum-energy channel of each U-Net feature map at $t = 0.5$ for all four SDE families (rows), across encoder, bottleneck, decoder, and score output stages (columns). Each map is independently normalised. The LV row (bottom) displays notably more spatially distributed activations at every stage, with digit structure retained coherently from encoder to bottleneck and reconstructed clearly in the decoder, suggesting that the linear variance schedule induces a more balanced internal representation than the competing families.

flecting its noise schedule periodicity. LV encoder activations, in contrast, preserve a recognisable representation of the input digit across spatial scales, with a distributed pattern that does not collapse to a single bright region. At the bottleneck ($4C \times 7 \times 7$), LV maintains a spatially compact but multi-channel representation encoding both local stroke geometry and global digit structure, closer to the balanced compression expected from a well-trained U-Net.

In the decoder and score output stages, LV produces a score map (rightmost column) that is the most diffuse and low-amplitude of the four families, matching the quantitative findings of Figure 5.18: the corrections the network must apply at this time are small and broadly distributed, rather than large and localised. This spatial distribution keeps the ODE trajectory well conditioned; a score field without sharp spatial gradients at mid-trajectory poses a substantially easier integration problem, explaining the qualitatively superior RK45 outputs of the LV family.

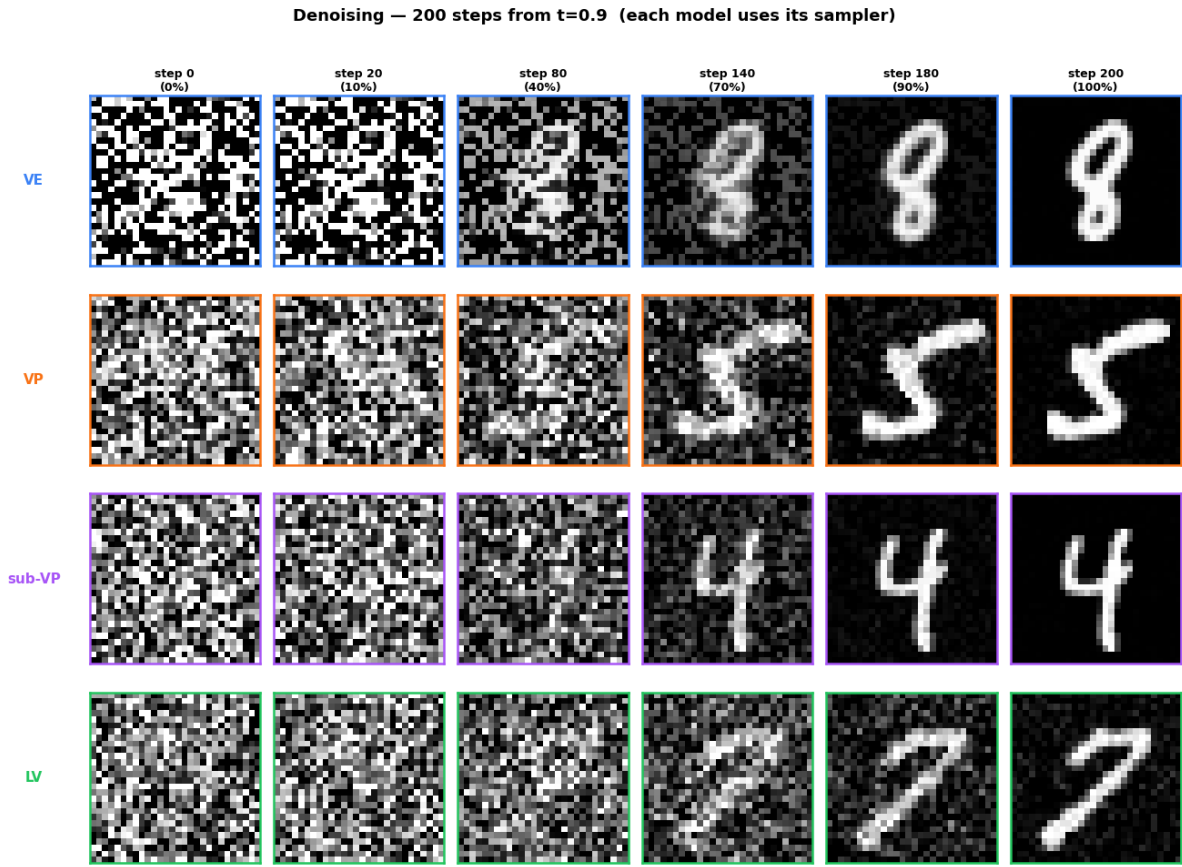


Figure 5.20 Step-by-step denoising for all four SDE families starting from $t = 0.9$, using 200 steps. Snapshots are shown at steps 0 (0%), 20 (10%), 80 (40%), 140 (70%), 180 (90%), and 200 (100%). Each model uses its own reverse sampler. The LV model (bottom row, green border) produces a clean, well-defined “7” at step 200, with digit structure becoming recognisable earlier in the trajectory (step 140) than several competing families, and achieving a final output whose stroke geometry is among the most regular observed across the four denoising sequences.

Figure 5.20 traces the denoising trajectories over six checkpoints, from pure noise at $t = 0.9$ to a fully denoised output at step 200. This visualisation addresses how quickly and reliably each forward process guides the reverse trajectory toward a coherent sample.

The VE model (top, blue border) produces a well-formed “8” but digit structure becomes clearly recognisable only at step 140, with residual high frequency noise persisting in the background, consistent with its large terminal variance. The VP model (second row, orange border) generates a “5” with clean background, but intermediate steps show an undifferentiated noise field that crystallises abruptly between steps 80

and 140. The sub-VP model (third row, purple border) denoises a “4” cleanly, with a trajectory that qualitatively resembles VP but with slightly thinner final strokes.

The LV model (bottom row, green border) presents the most instructive denoising behaviour. By step 80, the LV process already reveals a partially structured intermediate that is more suggestive of the final digit class than the corresponding intermediates of the other three families, indicating that the linear variance schedule concentrates generative organisation earlier in the reverse trajectory. By step 140, the digit “7” is clearly recognisable with well-defined strokes and a dark background. The final output at step 200 exhibits bold, sharp strokes and a uniformly clean background, visually among the strongest results in this set. Although a subtle texture remains relative to the cleanest VP outputs (a direct consequence of the near-zero boundary behaviour discussed in Section 5.4), the overall abstract quality of the generated digit is superior: stroke geometry is more regular, topology is correct, and the digit is unambiguously identifiable. This reinforces that the LV score network has learned a high-quality generative model, and that the residual background imperfection is a numerical rather than a representational limitation.

5.4.1 Cross-Family and Cross-Sampler Comparison

Taken together, the twelve sample grids and the three cross-cutting diagnostics reveal two orthogonal axes of variation: SDE family and sampling scheme.

Effect of SDE family. The VP SDE consistently produces the most geometrically regular outputs across all three samplers among the established families, a consequence of its bounded variance and smooth signal attenuation. The VE SDE produces the sharpest and thinnest strokes under EM, reflecting its large terminal variance, but degrades more noticeably under fixed-step ODE integration and exhibits the highest score magnitude at small t (Figure 5.18). The sub-VP SDE is intermediate: clean under EM and RK45, but noticeably sensitive to integration error under fixed-step ODE, which the high near-zero score variability in Figure 5.18 directly explains.

The proposed LV SDE emerges as the strongest overall design when evaluated across all three diagnostics. Quantitatively, it exhibits the lowest score magnitude and score variability throughout the entire trajectory (Figure 5.18), properties that directly predict robustness to numerical integration. Structurally, its internal U-Net representations at mid-trajectory show the most balanced spatial activation pattern, preserving digit geometry coherently from encoder to decoder without the localised artefacts seen in VE and sub-VP (Figure 5.19). Dynamically, its denoising trajectory is

the most structured and regular among the four models, with digit identity emerging earlier and the final output exhibiting bold, topologically correct strokes (Figure 5.20).

The one limitation of the LV family that must be acknowledged is its near-zero boundary behaviour: the forward process does not reach a well-defined terminal distribution at $t = 0^+$, so practical integration near $t = 0$ is subject to finite-precision residual errors that manifest as a subtle background texture. Crucially, Figure 5.18 shows that the LV score magnitude and variability are at their most favourable (lowest) precisely in the small- t regime, implying that the network handles this regime as well as possible given the boundary constraint. The residual noise is therefore a consequence of the mathematical structure of the LV process, not of an inadequate score approximation.

Effect of sampling scheme. Across all four SDE families, the stochastic EM sampler consistently produces clean, well-contrasted outputs: the noise injected at each reverse step acts as self-correcting regularisation that suppresses accumulated error. The ODE Euler sampler is the most sensitive to the choice of SDE family, because fixed-step integration amplifies any non-smoothness in the score field without correction. The ODE RK45 sampler recovers the quality of EM in the deterministic regime for all four families, and reveals the particular strength of LV: because its score field is the most uniformly smooth (Figure 5.18), adaptive step control allocates integration effort efficiently, producing the strongest overall sample quality in the deterministic regime. The ordering $\text{EM} \approx \text{RK45} > \text{ODE Euler}$ holds across all families, confirming that integration accuracy limits deterministic sampling and that adaptive step control is necessary to match the robustness of the stochastic scheme. For the LV SDE, the adaptive integrator further mitigates the near-zero boundary limitation by concentrating numerical resources where the process is most sensitive, offering a practical pathway (via tighter tolerances or higher-order adaptive quadrature) to further reduce the residual background artefact.

Absence of mode collapse. A common failure mode in generative models is mode collapse, where the generator assigns probability mass to only a subset of the target distribution. No such behaviour is observed in any of the twelve grids: all digit classes (0 to 9) are represented with broadly balanced frequency in every setting, indicating that all four score networks have learned a non-degenerate approximation of the full MNIST distribution, independently of the diffusion family and sampling scheme.

Bibliography

- [1] Brian D. O. Anderson. Reverse-time diffusion equation models. *Stochastic Processes and their Applications*, 12(3):313–326, 1982. doi:[10.1016/0304-4149\(82\)90051-5](https://doi.org/10.1016/0304-4149(82)90051-5).
- [2] David Applebaum. *Lévy Processes and Stochastic Calculus*, volume 116 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, Cambridge, 2nd edition, 2009.
- [3] Ricky T. Q. Chen, Yulia Rubanova, Jesse Bettencourt, and David Duvenaud. Neural ordinary differential equations. *Advances in Neural Information Processing Systems*, 31:6571–6583, 2018. arXiv:[1806.07366](https://arxiv.org/abs/1806.07366).
- [4] G. Cybenko. Approximation by superpositions of a sigmoidal function. *Mathematics of Control, Signals, and Systems*, 2(4):303–314, 1989. URL: <http://dx.doi.org/10.1007/bf02551274>, doi:[10.1007/bf02551274](https://doi.org/10.1007/bf02551274).
- [5] Ulrich G Haussmann and Etienne Pardoux. Time reversal of diffusions. *The Annals of Probability*, 14(4):1188–1205, 1986.
- [6] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. *Advances in Neural Information Processing Systems*, 33:6840–6851, 2020. arXiv:[2006.11239](https://arxiv.org/abs/2006.11239).
- [7] Kurt Hornik, Maxwell Stinchcombe, and Halbert White. Multilayer feedforward networks are universal approximators. *Neural networks*, 2(5):359–366, 1989.
- [8] M. F. Hutchinson. A stochastic estimator of the trace of the influence matrix for laplacian smoothing splines. *Communications in Statistics - Simulation and Computation*, 19(2):433–450, 1990.
- [9] Aapo Hyvärinen. Estimation of non-normalized statistical models by score matching. *Journal of Machine Learning Research*, 6(24):695–709, 2005. URL: <http://jmlr.org/papers/v6/hyvarinen05a.html>.
- [10] Lucas Izydorczyk, Nadia Oudjane, Francesco Russo, and Gianmario Tessitore. Fokker-planck equations with terminal condition and related McKean probabilistic representation. *Nonlinear Differential Equations and Applications NoDEA*, 29(1):1–41, 2022. doi:[10.1007/s00030-021-00736-1](https://doi.org/10.1007/s00030-021-00736-1).
- [11] Jean Jacod and Philip Protter. Time reversal of Markov processes. *The Annals of Probability*, 28(2):713–744, 2000.

- [12] Ioannis Karatzas and Steven E. Shreve. *Brownian Motion and Stochastic Calculus*, volume 113 of *Graduate Texts in Mathematics*. Springer-Verlag, New York, 2nd edition, 1991.
- [13] Peter E. Kloeden and Eckhard Platen. *Numerical Solution of Stochastic Differential Equations*. Springer-Verlag, Berlin, Heidelberg, 1992. doi:[10.1007/978-3-662-12616-5](https://doi.org/10.1007/978-3-662-12616-5).
- [14] Solomon Kullback and Richard A Leibler. On information and sufficiency. *The Annals of Mathematical Statistics*, 22(1):79–86, 1951.
- [15] Bernt Øksendal. *Stochastic Differential Equations: An Introduction with Applications*. Springer-Verlag, 6th edition, 2003. doi:[10.1007/978-3-642-14394-6](https://doi.org/10.1007/978-3-642-14394-6).
- [16] John Skilling. The eigenvalues of mega-dimensional matrices. In *Maximum Entropy and Bayesian Methods*, pages 455–466. Springer, 1989.
- [17] Yang Song and Stefano Ermon. Generative modeling by estimating gradients of the data distribution. *Advances in Neural Information Processing Systems*, 33:13918–13930, 2019.
- [18] Yang Song, S. Garg, Jiabin Shi, and Stefano Ermon. Sliced score matching: A scalable approach to density and score estimation. *Proceedings of the 36th Conference on Uncertainty in Artificial Intelligence (UAI)*, pages 574–584, 2019. URL: <http://proceedings.mlr.press/v115/song20a.html>.
- [19] Yang Song, Jascha Sohl-Dickstein, Diederik P Kingma, Abhishek Kumar, Stefano Ermon, and Ben Poole. Score-based generative modeling through stochastic differential equations. *International Conference on Learning Representations (ICLR)*, 2021. URL: <https://arxiv.org/abs/2011.13456>.
- [20] Wen Tang and Hongyuan Zhao. Score-based diffusion models via stochastic differential equations, 2022. arXiv:[2209.01817](https://arxiv.org/abs/2209.01817).
- [21] Pascal Vincent. A connection between score matching and denoising autoencoders. *Neural Computation*, 23(7):1661–1674, 2011. doi:[10.1162/NECO_a_00142](https://doi.org/10.1162/NECO_a_00142).

Appendix A

Proof of the Time Reversal Theorem

This appendix provides a rigorous derivation of the time-reversal formula for diffusion processes. The argument follows the structure of the heuristic derivation but makes the hypotheses precise and emphasizes the role of the forward and backward Kolmogorov equations. The exposition is inspired by Haussmann and Pardoux [5] and Anderson [1], following the rigorous framework outlined in Tang and Zhao [20].

A modern perspective on the associated Fokker–Planck equation with terminal condition and its probabilistic representation via a McKean–Vlasov SDE can be found in [10], where uniqueness of the reverse-time dynamics is proved under various assumptions on the coefficients.

Throughout, we work on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{0 \leq t \leq T}, \mathbb{P})$ satisfying the usual conditions (right-continuity and completeness).

A.1 Problem Setting and Regularity Conditions

Let $\{\mathbf{X}_t\}_{0 \leq t \leq T}$ be a diffusion process in $\mathbb{R}^d$ satisfying the forward stochastic differential equation (SDE)

$$d\mathbf{X}_t = \mathbf{f}(t, \mathbf{X}_t) dt + G(t, \mathbf{X}_t) d\mathbf{W}_t, \quad \mathbf{X}_0 \sim p_0(\mathbf{x}) d\mathbf{x}, \quad (\text{A.1})$$

where $\{\mathbf{W}_t\}_{0 \leq t \leq T}$ is a standard d -dimensional Brownian motion, $\mathbf{f} : [0, T] \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ and $G : [0, T] \times \mathbb{R}^d \rightarrow \mathbb{R}^{d \times d}$ are measurable functions. Define the diffusion matrix

$$a(t, \mathbf{x}) = G(t, \mathbf{x})G(t, \mathbf{x})^\top, \quad (\text{A.2})$$

which is assumed to be uniformly positive definite and bounded.

We impose the following regularity conditions:

1. **Smoothness and growth:** $\mathbf{f}$ and G are of class $C^{1,2}$ in $(t, \mathbf{x})$ (continuously differentiable once in t and twice in $\mathbf{x}$) and have bounded derivatives of all orders up to second order in $\mathbf{x}$ and first order in t . This guarantees the existence and uniqueness of strong solutions and ensures that the associated Kolmogorov equations have classical solutions.
2. **Non-degeneracy:** $a(t, \mathbf{x})$ is uniformly elliptic: there exists $\lambda > 0$ such that

$$\xi^\top a(t, \mathbf{x}) \xi \geq \lambda \|\xi\|^2, \quad \forall \xi \in \mathbb{R}^d, \forall (t, \mathbf{x}) \in [0, T] \times \mathbb{R}^d.$$

3. **Initial condition:** p_0 is a smooth probability density with finite moments of all orders.
4. **Regularity of the density:** The law of $\mathbf{X}_t$ admits a smooth, strictly positive density $p(t, \mathbf{x}) \in C^{1,2}((0, T) \times \mathbb{R}^d)$ with respect to Lebesgue measure. For each $t \in (0, T)$, $p(t, \cdot)$ and its derivatives up to second order decay sufficiently fast at infinity. Moreover, p satisfies the forward Kolmogorov (Fokker-Planck) equation in the classical sense:

$$\frac{\partial p}{\partial t}(t, \mathbf{x}) = - \sum_{i=1}^d \frac{\partial}{\partial x_i} (f_i(t, \mathbf{x}) p(t, \mathbf{x})) + \frac{1}{2} \sum_{i,j=1}^d \frac{\partial^2}{\partial x_i \partial x_j} (a_{ij}(t, \mathbf{x}) p(t, \mathbf{x})), \quad (\text{A.3})$$

for $(t, \mathbf{x}) \in (0, T) \times \mathbb{R}^d$, with initial condition $p(0, \mathbf{x}) = p_0(\mathbf{x})$.

5. **Backward equation regularity:** The fundamental solution (transition density) also satisfies the backward Kolmogorov equation in the classical sense: for any fixed $(s, \mathbf{y})$, the function $u(t, \mathbf{x}) = p(t, \mathbf{x} | s, \mathbf{y})$ satisfies for $t > s$:

$$\frac{\partial u}{\partial s} + \sum_{i=1}^d f_i(s, \mathbf{y}) \frac{\partial u}{\partial y_i} + \frac{1}{2} \sum_{i,j=1}^d a_{ij}(s, \mathbf{y}) \frac{\partial^2 u}{\partial y_i \partial y_j} = 0, \quad (\text{A.4})$$

with terminal condition $u(s, \mathbf{y} | s, \mathbf{y}) = \delta(\mathbf{y} - \mathbf{x})$.

Remark A.1.1. *Conditions 4 and 5 are interconnected: under the smoothness and ellipticity assumptions, the existence of a smooth density satisfying (A.3) is guaranteed, and the fundamental solution automatically satisfies (A.4) in the sense of distributions. The explicit statement here is for clarity.*

A.2 Proof of the Theorem

Theorem A.2.1 (Time Reversal of Diffusions). *Under the regularity conditions stated above, define the time-reversed process*

$$\mathbf{Y}_t = \mathbf{X}_{T-t}, \quad t \in [0, T]. \quad (\text{A.5})$$

Then $\mathbf{Y}_t$ is a diffusion process satisfying the SDE

$$d\mathbf{Y}_t = \tilde{\mathbf{f}}(t, \mathbf{Y}_t) dt + \tilde{G}(t, \mathbf{Y}_t) d\tilde{\mathbf{W}}_t, \quad (\text{A.6})$$

where

$$\tilde{G}(t, \mathbf{x}) = G(T-t, \mathbf{x}), \quad (\text{A.7})$$

$$\tilde{\mathbf{f}}(t, \mathbf{x}) = -\mathbf{f}(T-t, \mathbf{x}) + \frac{\nabla_{\mathbf{x}} \cdot (p(T-t, \mathbf{x}) a(T-t, \mathbf{x}))}{p(T-t, \mathbf{x})}. \quad (\text{A.8})$$

Here $\tilde{\mathbf{W}}_t$ is a standard Brownian motion (possibly defined on an enlarged filtration) and the equality holds in law.

Let $p(t, \mathbf{x})$ denote the density of $\mathbf{X}_t$. By hypothesis, it satisfies the Fokker-Planck equation (A.3). For notational convenience, we write it in divergence form:

$$\frac{\partial p}{\partial t} = -\nabla \cdot (\mathbf{f}p) + \frac{1}{2} \nabla \cdot (\nabla \cdot (ap)), \quad (\text{A.9})$$

where $\nabla \cdot$ denotes the divergence operator acting on vector fields, and for a matrix-valued function A , $\nabla \cdot A$ is interpreted as the vector with components $\sum_j \frac{\partial}{\partial x_j} A_{ij}$.

Consider the time-reversed process $\mathbf{Y}_t = \mathbf{X}_{T-t}$ and define its density

$$\tilde{p}(t, \mathbf{x}) = p(T-t, \mathbf{x}), \quad (t, \mathbf{x}) \in [0, T] \times \mathbb{R}^d. \quad (\text{A.10})$$

A simple change of variables confirms that $\tilde{p}(t, \cdot)$ is indeed the probability density of $\mathbf{Y}_t$. Differentiating with respect to t and using (A.9) evaluated at $(T-t, \mathbf{x})$ yields

$$\frac{\partial \tilde{p}}{\partial t}(t, \mathbf{x}) = -\frac{\partial p}{\partial t}(T-t, \mathbf{x}) = \nabla \cdot (\mathbf{f}(T-t, \cdot) p(T-t, \cdot)) - \frac{1}{2} \nabla \cdot (\nabla \cdot (a(T-t, \cdot) p(T-t, \cdot))). \quad (\text{A.11})$$

Replacing $p(T - t, \mathbf{x})$ with $\tilde{p}(t, \mathbf{x})$ (and understanding that $\mathbf{f}$ and a are evaluated at $(T - t, \mathbf{x})$), we obtain

$$\frac{\partial \tilde{p}}{\partial t} = \nabla \cdot (\mathbf{f}(T - t, \cdot) \tilde{p}) - \frac{1}{2} \nabla \cdot (\nabla \cdot (a(T - t, \cdot) \tilde{p})). \quad (\text{A.12})$$

Now suppose that $\mathbf{Y}_t$ itself admits a representation as a diffusion process with drift $\tilde{\mathbf{f}}(t, \mathbf{x})$ and diffusion matrix $\tilde{a}(t, \mathbf{x}) = \tilde{G}(t, \mathbf{x}) \tilde{G}(t, \mathbf{x})^\top$. Then its density must satisfy the forward Kolmogorov equation associated with this reversed diffusion:

$$\frac{\partial \tilde{p}}{\partial t} = -\nabla \cdot (\tilde{\mathbf{f}} \tilde{p}) + \frac{1}{2} \nabla \cdot (\nabla \cdot (\tilde{a} \tilde{p})). \quad (\text{A.13})$$

Since both (A.12) and (A.13) describe the same function $\tilde{p}$, their right-hand sides must coincide for all $(t, \mathbf{x})$:

$$\begin{aligned} \nabla \cdot (\mathbf{f}(T - t, \cdot) \tilde{p}) - \frac{1}{2} \nabla \cdot (\nabla \cdot (a(T - t, \cdot) \tilde{p})) = \\ -\nabla \cdot (\tilde{\mathbf{f}} \tilde{p}) + \frac{1}{2} \nabla \cdot (\nabla \cdot (\tilde{a} \tilde{p})), \end{aligned} \quad (\text{A.14})$$

Rearranging terms gives

$$\nabla \cdot ((\mathbf{f}(T - t, \cdot) + \tilde{\mathbf{f}}) \tilde{p}) = \frac{1}{2} \nabla \cdot (\nabla \cdot (a(T - t, \cdot) \tilde{p} + \tilde{a} \tilde{p})). \quad (\text{A.15})$$

A natural identification that satisfies (A.15) for all sufficiently smooth $\tilde{p}$ is to take

$$\tilde{a}(t, \mathbf{x}) = a(T - t, \mathbf{x}). \quad (\text{A.16})$$

This choice is justified by the fact that the quadratic variation of $\mathbf{Y}_t$ satisfies $[\mathbf{Y}]_t = [\mathbf{X}]_{T-t}$ in law, and under our regularity assumptions, the diffusion matrix is uniquely determined by the quadratic variation. Substituting (A.16) into (A.15) simplifies it to

$$\nabla \cdot ((\mathbf{f}(T - t, \cdot) + \tilde{\mathbf{f}}) \tilde{p}) = \nabla \cdot (\nabla \cdot (a(T - t, \cdot) \tilde{p})). \quad (\text{A.17})$$

Equation (A.17) holds for every $(t, \mathbf{x})$. Integrating against a smooth compactly supported test function $\phi(\mathbf{x})$ and applying the divergence theorem yields

$$\int_{\mathbb{R}^d} \nabla \phi(\mathbf{x}) \cdot ((\mathbf{f}(T - t, \mathbf{x}) + \tilde{\mathbf{f}}(t, \mathbf{x})) \tilde{p}(t, \mathbf{x})) d\mathbf{x} = \int_{\mathbb{R}^d} \nabla \phi(\mathbf{x}) \cdot (\nabla \cdot (a(T - t, \mathbf{x}) \tilde{p}(t, \mathbf{x}))) d\mathbf{x},$$

valid for all $\phi \in C_c^\infty(\mathbb{R}^d)$. This implies equality of the two vector fields in the sense of distributions. By the smoothness of $\tilde{p}$, $\mathbf{f}$, and a , they are actually equal pointwise:

$$(\mathbf{f}(T-t, \mathbf{x}) + \tilde{\mathbf{f}}(t, \mathbf{x}))\tilde{p}(t, \mathbf{x}) = \nabla \cdot (a(T-t, \mathbf{x})\tilde{p}(t, \mathbf{x})), \quad \forall(t, \mathbf{x}). \quad (\text{A.18})$$

Solving for $\tilde{\mathbf{f}}$ and using the strict positivity of $\tilde{p}$ (a consequence of the hypoellipticity of the Kolmogorov equations under our non-degeneracy assumptions), we obtain

$$\tilde{\mathbf{f}}(t, \mathbf{x}) = -\mathbf{f}(T-t, \mathbf{x}) + \frac{\nabla_{\mathbf{x}} \cdot (a(T-t, \mathbf{x})\tilde{p}(t, \mathbf{x}))}{\tilde{p}(t, \mathbf{x})}. \quad (\text{A.19})$$

Recalling that $\tilde{p}(t, \mathbf{x}) = p(T-t, \mathbf{x})$ gives the desired formula

$$\tilde{\mathbf{f}}(t, \mathbf{x}) = -\mathbf{f}(T-t, \mathbf{x}) + \frac{\nabla_{\mathbf{x}} \cdot (p(T-t, \mathbf{x})a(T-t, \mathbf{x}))}{p(T-t, \mathbf{x})}. \quad (\text{A.20})$$

It remains to show that with these coefficients, the SDE (A.6) indeed admits a weak solution whose marginal distributions coincide with those of $\mathbf{X}_{T-t}$. To this end, define the reverse-time filtration $\tilde{\mathcal{F}}_t = \mathcal{F}_{T-t}$, which is increasing in t . For any bounded measurable function ϕ , consider the process

$$\tilde{M}_t = \mathbb{E}[\phi(\mathbf{X}_T) \mid \tilde{\mathcal{F}}_t]. \quad (\text{A.21})$$

One verifies that $\tilde{M}_t$ is a martingale with respect to $\tilde{\mathcal{F}}_t$. Using the smoothness of p and the backward Kolmogorov equation (A.4), an application of Itô's formula allows us to express $\tilde{M}_t$ in terms of $\mathbf{Y}_t$ and identify its drift and diffusion coefficients, which are precisely those given by (A.16) and (A.20). The martingale representation theorem then guarantees the existence of a Brownian motion $\tilde{\mathbf{W}}_t$ (possibly on an enlarged filtration) such that (A.6) holds. This completes the proof. $\square$

Remark on uniqueness and the McKean–Vlasov representation. Under the assumptions of the theorem, the density $p(t, \mathbf{x})$ is the unique classical solution of the forward Kolmogorov equation (A.3) and, consequently, of the corresponding backward equation with terminal condition $p(T, \mathbf{x}) = p_T(\mathbf{x})$. [10] study precisely this backward Fokker–Planck equation and provide sufficient conditions for its uniqueness. Under the specific assumptions studied in that work (e.g., Ornstein–Uhlenbeck dynamics, bounded non-degenerate coefficients, or time-homogeneous Hölder-continuous coefficients), the mapping from the initial distribution p_0 to the terminal distribution p_T is injective. This guarantees that, in those settings, the reverse-time SDE (A.6) is

the unique diffusion interpolating between the prior p_T and the data distribution p_0 . Furthermore, when the score term is expressed in terms of the density, (A.6) becomes a McKean–Vlasov SDE of the type analysed in that work. This connection provides a rigorous foundation for using a learned score network as a proxy for the true score in score-based generative models.

Appendix B

The Fokker-Planck-Kolmogorov Equation

This appendix presents a rigorous derivation of the Fokker-Planck-Kolmogorov (FPK) equation, its relationship to transition densities, and its application to the families of stochastic differential equations introduced in Chapter 3. The notation is fully aligned with that chapter: in particular, $\mathbf{D} = G G^\top$ denotes the diffusion matrix.

B.1 Preliminaries and Notation

We consider a stochastic differential equation (SDE) of the general form

$$d\mathbf{X}_t = \mathbf{f}(\mathbf{X}_t, t) dt + G(\mathbf{X}_t, t) d\mathbf{W}_t, \quad (\text{B.1})$$

where $\mathbf{W}_t$ is a standard D -dimensional Wiener process (i.e. $\mathbb{E}[dW_i dW_j] = \delta_{ij} dt$). The initial condition $\mathbf{X}_{t_0}$ is independent of the Brownian motion and has density $p(\mathbf{X}_{t_0})$. The probability density of the process at time t is denoted by $p(\mathbf{x}, t)$ or simply $p(\mathbf{x}(t))$.

To simplify notation we define the **diffusion matrix** of the process as

$$\mathbf{D}(\mathbf{x}, t) = G(\mathbf{x}, t) G(\mathbf{x}, t)^\top \in \mathbb{R}^{D \times D}. \quad (\text{B.2})$$

This is exactly the matrix $\mathbf{D}$ introduced in Chapter 3.

B.2 The Infinitesimal Generator

To derive the FPK equation, we first introduce the concept of the infinitesimal generator of an Itô process.

Definition B.2.1 (Infinitesimal Generator). *Let $\mathbf{X}_t$ be the solution of the SDE (B.1). The infinitesimal generator $\mathcal{A}_t$ of the process is the differential operator defined by*

$$\mathcal{A}_t \phi(\mathbf{x}, t) = \frac{\partial \phi(\mathbf{x}, t)}{\partial t} + \sum_{i=1}^D \frac{\partial \phi}{\partial x_i} f_i(\mathbf{x}, t) + \frac{1}{2} \sum_{i,j=1}^D \frac{\partial^2 \phi}{\partial x_i \partial x_j} D_{ij}(\mathbf{x}, t),$$

where $\phi(\mathbf{x}, t)$ is any twice differentiable function and $D_{ij}(\mathbf{x}, t)$ are the elements of the diffusion matrix (B.2).

The generator is derived directly from Itô's formula and characterizes the expected evolution of functions of the stochastic process. Its importance lies in the fact that the expectation of any function of the process satisfies

$$\frac{d}{dt} \mathbb{E}[\phi(\mathbf{X}_t, t)] = \mathbb{E}[\mathcal{A}_t \phi(\mathbf{X}_t, t)].$$

B.3 Derivation of the Fokker-Planck-Kolmogorov Equation

We now derive the equation governing the temporal evolution of the probability density $p(\mathbf{x}, t)$.

Theorem B.3.1 (Fokker-Planck-Kolmogorov Equation). *The probability density $p(\mathbf{x}, t)$ of the solution of the SDE (B.1) satisfies the partial differential equation*

$$\frac{\partial p(\mathbf{x}, t)}{\partial t} = - \sum_{i=1}^D \frac{\partial}{\partial x_i} [f_i(\mathbf{x}, t) p(\mathbf{x}, t)] + \frac{1}{2} \sum_{i,j=1}^D \frac{\partial^2}{\partial x_i \partial x_j} [D_{ij}(\mathbf{x}, t) p(\mathbf{x}, t)]. \quad (\text{B.3})$$

This equation is known as the Fokker-Planck-Kolmogorov equation or the forward Kolmogorov equation.

Proof. Let $\phi(\mathbf{x})$ be an arbitrary twice differentiable function with compact support. Applying Itô's formula to the process $\phi(\mathbf{X}_t)$, we obtain

$$d\phi = \sum_{i=1}^D \frac{\partial \phi}{\partial x_i} dX_i + \frac{1}{2} \sum_{i,j=1}^D \frac{\partial^2 \phi}{\partial x_i \partial x_j} dX_i dX_j.$$

Substituting dX_i from the SDE (B.1) and using the Itô multiplication rules ($dt dt = 0$, $dt dW_k = 0$, $dW_k dW_\ell = \delta_{k\ell} dt$), we obtain

$$\begin{aligned} d\phi &= \sum_{i=1}^D \frac{\partial \phi}{\partial x_i} f_i(\mathbf{x}, t) dt + \sum_{i=1}^D \frac{\partial \phi}{\partial x_i} (G(\mathbf{x}, t) d\mathbf{W})_i \\ &\quad + \frac{1}{2} \sum_{i,j=1}^D \frac{\partial^2 \phi}{\partial x_i \partial x_j} D_{ij}(\mathbf{x}, t) dt. \end{aligned}$$

Taking the expectation with respect to $\mathbf{x}$ on both sides and noting that $\mathbb{E}[(G d\mathbf{W})_i] = 0$, we obtain

$$\frac{d\mathbb{E}[\phi(\mathbf{X}_t)]}{dt} = \mathbb{E} \left[\sum_{i=1}^D \frac{\partial \phi}{\partial x_i} f_i(\mathbf{x}, t) \right] + \frac{1}{2} \mathbb{E} \left[\sum_{i,j=1}^D \frac{\partial^2 \phi}{\partial x_i \partial x_j} D_{ij}(\mathbf{x}, t) \right].$$

The left-hand side can be written as

$$\frac{d\mathbb{E}[\phi(\mathbf{X}_t)]}{dt} = \frac{d}{dt} \int_{\mathbb{R}^D} \phi(\mathbf{x}) p(\mathbf{x}, t) d\mathbf{x} = \int_{\mathbb{R}^D} \phi(\mathbf{x}) \frac{\partial p(\mathbf{x}, t)}{\partial t} d\mathbf{x}.$$

For the first term on the right-hand side, we apply integration by parts. Recalling the multidimensional formula

$$\int_{\Omega} \frac{\partial u}{\partial x_i} v d\mathbf{x} = \int_{\partial\Omega} u v n_i dS - \int_{\Omega} u \frac{\partial v}{\partial x_i} d\mathbf{x},$$

where $\mathbf{n}$ is the outward normal of the boundary $\partial\Omega$ and dS is its area element. Since the integration domain is all of $\mathbb{R}^D$ and ϕ has compact support, the boundary term vanishes, leaving

$$\int_{\mathbb{R}^D} \frac{\partial \phi}{\partial x_i} v d\mathbf{x} = - \int_{\mathbb{R}^D} \phi \frac{\partial v}{\partial x_i} d\mathbf{x}.$$

Applying this with $v(\mathbf{x}) = f_i(\mathbf{x}, t)p(\mathbf{x}, t)$, we obtain

$$\mathbb{E}\left[\frac{\partial\phi}{\partial x_i}f_i(\mathbf{x}, t)\right] = \int_{\mathbb{R}^D} \frac{\partial\phi}{\partial x_i}f_i(\mathbf{x}, t)p(\mathbf{x}, t) d\mathbf{x} = - \int_{\mathbb{R}^D} \phi(\mathbf{x}) \frac{\partial}{\partial x_i}[f_i(\mathbf{x}, t)p(\mathbf{x}, t)] d\mathbf{x}.$$

For the second term, we apply integration by parts twice. First with $u = \frac{\partial\phi}{\partial x_j}$ and $v = D_{ij}(\mathbf{x}, t)p(\mathbf{x}, t)$:

$$\begin{aligned} & \mathbb{E}\left[\frac{\partial^2\phi}{\partial x_i\partial x_j}D_{ij}(\mathbf{x}, t)\right] \\ &= \int_{\mathbb{R}^D} \frac{\partial^2\phi}{\partial x_i\partial x_j}D_{ij}(\mathbf{x}, t)p(\mathbf{x}, t) d\mathbf{x} \\ &= - \int_{\mathbb{R}^D} \frac{\partial\phi}{\partial x_j} \frac{\partial}{\partial x_i}[D_{ij}(\mathbf{x}, t)p(\mathbf{x}, t)] d\mathbf{x}. \end{aligned}$$

Applying integration by parts again with $u = \phi$ and $v = \frac{\partial}{\partial x_i}[D_{ij}(\mathbf{x}, t)p(\mathbf{x}, t)]$:

$$\mathbb{E}\left[\frac{\partial^2\phi}{\partial x_i\partial x_j}D_{ij}(\mathbf{x}, t)\right] = \int_{\mathbb{R}^D} \phi(\mathbf{x}) \frac{\partial^2}{\partial x_i\partial x_j}[D_{ij}(\mathbf{x}, t)p(\mathbf{x}, t)] d\mathbf{x}.$$

Substituting the expressions for the left-hand side and the two terms into the expectation equation, we obtain

$$\begin{aligned} \int_{\mathbb{R}^D} \phi(\mathbf{x}) \frac{\partial p(\mathbf{x}, t)}{\partial t} d\mathbf{x} &= - \sum_{i=1}^D \int_{\mathbb{R}^D} \phi(\mathbf{x}) \frac{\partial}{\partial x_i}[f_i(\mathbf{x}, t)p(\mathbf{x}, t)] d\mathbf{x} \\ &\quad + \frac{1}{2} \sum_{i,j=1}^D \int_{\mathbb{R}^D} \phi(\mathbf{x}) \frac{\partial^2}{\partial x_i\partial x_j}[D_{ij}(\mathbf{x}, t)p(\mathbf{x}, t)] d\mathbf{x}. \end{aligned}$$

Rearranging:

$$\int_{\mathbb{R}^D} \phi(\mathbf{x}) \left\{ \frac{\partial p}{\partial t} + \sum_{i=1}^D \frac{\partial}{\partial x_i}[f_i p] - \frac{1}{2} \sum_{i,j=1}^D \frac{\partial^2}{\partial x_i\partial x_j}[D_{ij} p] \right\} d\mathbf{x} = 0.$$

Since this equation must hold for every arbitrary test function ϕ , the integrand must vanish, yielding the FPK equation (B.3). $\square$

B.3.1 Operator Formulation

The FPK equation can be elegantly expressed using the adjoint operator of the infinitesimal generator. Defining the L^2 inner product

$$\langle \phi, \psi \rangle = \int_{\mathbb{R}^D} \phi(\mathbf{x}) \psi(\mathbf{x}) d\mathbf{x},$$

the expectation of $\phi(\mathbf{X}_t)$ can be written as

$$\mathbb{E}[\phi(\mathbf{X}_t)] = \langle \phi, p(\cdot, t) \rangle.$$

The **adjoint operator** $\mathcal{A}_t^*$ of the generator $\mathcal{A}_t$ is defined by

$$\langle \mathcal{A}_t \phi, \psi \rangle = \langle \phi, \mathcal{A}_t^* \psi \rangle,$$

for all appropriate functions ϕ and ψ . The preceding derivation shows that

$$\mathcal{A}_t^* p = - \sum_{i=1}^D \frac{\partial}{\partial x_i} [f_i p] + \frac{1}{2} \sum_{i,j=1}^D \frac{\partial^2}{\partial x_i \partial x_j} [D_{ij} p],$$

so that the FPK equation can be written compactly as

$$\frac{\partial p}{\partial t} = \mathcal{A}_t^* p.$$

B.4 Transition Densities and Markov Properties

A fundamental property of solutions to Itô SDEs is that they are Markov processes.

Definition B.4.1 (Markov Process). *A stochastic process $\mathbf{X}_t$ is a **Markov process** if its future is independent of its past given the present:*

$$p(\mathbf{X}_t | \mathcal{X}_s) = p(\mathbf{X}_t | \mathbf{X}_s), \quad \forall t \geq s,$$

where $\mathcal{X}_s = \{\mathbf{X}_\tau : \tau \leq s\}$ denotes the history of the process up to time s .

The **transition density** (or transition kernel) from time s to time t is denoted by

$$p(\mathbf{X}_t | \mathbf{X}_s) \equiv p(\mathbf{x}, t | \mathbf{y}, s),$$

where the right-hand side emphasizes the explicit dependence: $\mathbf{x}$ at time t given $\mathbf{y}$ at time s .

Theorem B.4.1 (Transition Density as Solution of the FPK). *The transition density $p(\mathbf{x}, t \mid \mathbf{y}, s)$ for $t \geq s$ is the solution of the FPK equation (B.3) with the degenerate initial condition*

$$p(\mathbf{x}, s \mid \mathbf{y}, s) = \delta(\mathbf{x} - \mathbf{y}),$$

where $\delta(\cdot)$ is the Dirac delta function.

Proof. The transition density characterizes the probability of finding the process at $\mathbf{x}$ at time t given that it started at $\mathbf{y}$ at time s . At time s , we know with certainty that the process is at $\mathbf{y}$, hence the initial condition above. The temporal evolution for $t > s$ is governed by the same FPK equation as the unconditional density, since the generator of the process does not depend on the initial condition. $\square$

B.4.1 Backward Kolmogorov Equation

In addition to the FPK (forward) equation, the transition density also satisfies the **backward Kolmogorov equation**.

Theorem B.4.2 (Backward Kolmogorov Equation). *The transition density $p(\mathbf{y}, t \mid \mathbf{x}, s)$ satisfies*

$$-\frac{\partial p(\mathbf{y}, t \mid \mathbf{x}, s)}{\partial s} = \mathcal{A}_s p(\mathbf{y}, t \mid \mathbf{x}, s),$$

with initial condition

$$p(\mathbf{y}, s \mid \mathbf{x}, s) = \delta(\mathbf{y} - \mathbf{x}),$$

where the generator $\mathcal{A}_s$ acts on the variable $\mathbf{x}$ at time s (the initial condition), not on $\mathbf{y}$ at time t .

Explicitly, the backward equation is:

$$-\frac{\partial p}{\partial s} = \sum_{i=1}^D f_i(\mathbf{x}, s) \frac{\partial p}{\partial x_i} + \frac{1}{2} \sum_{i,j=1}^D D_{ij}(\mathbf{x}, s) \frac{\partial^2 p}{\partial x_i \partial x_j}.$$

The negative sign in $\partial p / \partial s$ reflects that we are evolving backward in time from the initial condition.

B.4.2 Chapman-Kolmogorov Equation

The Markov property implies that transition densities satisfy the **Chapman Kolmogorov equation**.

Theorem B.4.3 (Chapman-Kolmogorov Equation). *For any three times $t_1 < t_2 < t_3$, the transition density satisfies*

$$p(\mathbf{x}_3, t_3 \mid \mathbf{x}_1, t_1) = \int_{\mathbb{R}^D} p(\mathbf{x}_3, t_3 \mid \mathbf{x}_2, t_2) p(\mathbf{x}_2, t_2 \mid \mathbf{x}_1, t_1) d\mathbf{x}_2.$$

This equation expresses the temporal consistency of transition densities: to go from $\mathbf{x}_1$ at t_1 to $\mathbf{x}_3$ at t_3 , the process must pass through some state $\mathbf{x}_2$ at the intermediate time t_2 , and the total probability is obtained by marginalizing over all intermediate possibilities.

B.5 Application to the SDE Families

We now apply the FPK theory to the four families of SDEs introduced in Chapter 3. For each family, we derive the explicit FPK equation and verify its consistency with the known transition kernels.

B.5.1 Variance Exploding (VE) SDE

The VE SDE is given by

$$d\mathbf{X}_t = \sqrt{\frac{d\sigma^2(t)}{dt}} d\mathbf{W}_t, \quad (\text{B.4})$$

which in the notation of (B.1) has

$$\mathbf{f}(\mathbf{x}, t) = \mathbf{0}, \quad G(\mathbf{x}, t) = \sqrt{\frac{d\sigma^2(t)}{dt}} \mathbf{I}.$$

The diffusion matrix is

$$\mathbf{D}(\mathbf{x}, t) = G(t)G(t)^\top = \frac{d\sigma^2(t)}{dt} \mathbf{I}.$$

Substituting into the FPK equation (B.3):

$$\frac{\partial p(\mathbf{x}, t)}{\partial t} = \frac{1}{2} \frac{d\sigma^2(t)}{dt} \nabla^2 p(\mathbf{x}, t), \quad (\text{B.5})$$

where $\nabla^2 = \sum_{i=1}^D \frac{\partial^2}{\partial x_i^2}$ is the Laplacian.

This is a pure **diffusion equation** with time-varying diffusion coefficient. Since the drift is zero, there is no deterministic transport, only stochastic dispersion.

Verification with the transition kernel. From Chapter 3, we know that the VE transition kernel is

$$p_{0,t}(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \mathbf{x}_0, \sigma^2(t)\mathbf{I}). \quad (\text{B.6})$$

To verify, we write the Gaussian explicitly:

$$p_{0,t}(\mathbf{x}_t | \mathbf{x}_0) = \frac{1}{(2\pi\sigma^2(t))^{D/2}} \exp\left(-\frac{\|\mathbf{x}_t - \mathbf{x}_0\|^2}{2\sigma^2(t)}\right).$$

Computing the required derivatives:

$$\begin{aligned} \frac{\partial p}{\partial t} &= p \left[-\frac{D}{2\sigma^2(t)} \frac{d\sigma^2(t)}{dt} + \frac{\|\mathbf{x}_t - \mathbf{x}_0\|^2}{2\sigma^4(t)} \frac{d\sigma^2(t)}{dt} \right], \\ \nabla^2 p &= p \left[\frac{D}{\sigma^2(t)} - \frac{\|\mathbf{x}_t - \mathbf{x}_0\|^2}{\sigma^4(t)} \right]. \end{aligned}$$

Substituting into (B.5):

$$\begin{aligned} \text{LHS} &= p \left[-\frac{D}{2\sigma^2} \dot{\sigma}^2 + \frac{\|\mathbf{x}_t - \mathbf{x}_0\|^2}{2\sigma^4} \dot{\sigma}^2 \right], \\ \text{RHS} &= \frac{1}{2} \dot{\sigma}^2 \cdot p \left[\frac{D}{\sigma^2} - \frac{\|\mathbf{x}_t - \mathbf{x}_0\|^2}{\sigma^4} \right] = p \left[\frac{D \dot{\sigma}^2}{2\sigma^2} - \frac{\|\mathbf{x}_t - \mathbf{x}_0\|^2 \dot{\sigma}^2}{2\sigma^4} \right], \end{aligned}$$

which confirms that both sides coincide, verifying the consistency.

B.5.2 Variance Preserving (VP) SDE

The VP SDE is given by

$$d\mathbf{X}_t = -\frac{1}{2}\beta(t)\mathbf{X}_t dt + \sqrt{\beta(t)} d\mathbf{W}_t, \quad (\text{B.7})$$

with coefficients

$$\mathbf{f}(\mathbf{x}, t) = -\frac{1}{2}\beta(t)\mathbf{x}, \quad G(\mathbf{x}, t) = \sqrt{\beta(t)}\mathbf{I}.$$

The diffusion matrix is

$$\mathbf{D}(\mathbf{x}, t) = \beta(t)\mathbf{I}.$$

The FPK equation becomes

$$\frac{\partial p}{\partial t} = \frac{\beta(t)}{2} \sum_{i=1}^D \frac{\partial}{\partial x_i} [x_i p] + \frac{\beta(t)}{2} \nabla^2 p. \quad (\text{B.8})$$

Using the identity $\frac{\partial}{\partial x_i} [x_i p] = p + x_i \frac{\partial p}{\partial x_i}$, this can be rewritten as

$$\frac{\partial p}{\partial t} = \frac{\beta(t)}{2} \left[Dp + \sum_{i=1}^D x_i \frac{\partial p}{\partial x_i} + \nabla^2 p \right] = \frac{\beta(t)}{2} [Dp + \mathbf{x} \cdot \nabla p + \nabla^2 p].$$

Verification with the transition kernel. The VP transition kernel is

$$p_{0,t}(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}\left(\mathbf{x}_t; e^{-\bar{\beta}(t)/2} \mathbf{x}_0, (1 - e^{-\bar{\beta}(t)})\mathbf{I}\right), \quad (\text{B.9})$$

where $\bar{\beta}(t) = \int_0^t \beta(s) ds$. Writing

$$\mu(t) = e^{-\bar{\beta}(t)/2}, \quad \sigma^2(t) = 1 - e^{-\bar{\beta}(t)},$$

we have

$$\frac{d\mu}{dt} = -\frac{\beta(t)}{2} \mu(t), \quad \frac{d\sigma^2}{dt} = \beta(t) e^{-\bar{\beta}(t)} = \beta(t) \mu^2(t).$$

The density is

$$p = \frac{1}{(2\pi\sigma^2)^{D/2}} \exp\left(-\frac{\|\mathbf{x}_t - \mu\mathbf{x}_0\|^2}{2\sigma^2}\right).$$

The required derivatives are:

$$\begin{aligned} \frac{\partial p}{\partial t} &= p \left[-\frac{D}{2\sigma^2} \dot{\sigma}^2 + \frac{\|\mathbf{x}_t - \mu\mathbf{x}_0\|^2}{2\sigma^4} \dot{\sigma}^2 + \frac{(\mathbf{x}_t - \mu\mathbf{x}_0) \cdot (-\dot{\mu}\mathbf{x}_0)}{\sigma^2} \right], \\ \frac{\partial p}{\partial x_i} &= -p \frac{x_i - \mu x_{0,i}}{\sigma^2}, \\ \frac{\partial^2 p}{\partial x_i^2} &= p \left[\frac{(x_i - \mu x_{0,i})^2}{\sigma^4} - \frac{1}{\sigma^2} \right]. \end{aligned}$$

Substituting the parameter derivatives and simplifying, one can verify that the FPK equation (B.8) is satisfied. The algebraic details are tedious but straightforward.

B.5.3 Linear Variance (LV) SDE

The LV SDE proposed in Chapter 3 is

$$d\mathbf{X}_t = -\theta\mathbf{X}_t dt + \sqrt{1 + 2\theta t} d\mathbf{W}_t, \quad (\text{B.10})$$

with coefficients

$$\mathbf{f}(\mathbf{x}, t) = -\theta\mathbf{x}, \quad G(\mathbf{x}, t) = \sqrt{1 + 2\theta t} \mathbf{I}.$$

The diffusion matrix is

$$\mathbf{D}(\mathbf{x}, t) = (1 + 2\theta t)\mathbf{I}.$$

The FPK equation is

$$\frac{\partial p}{\partial t} = \theta \sum_{i=1}^D \frac{\partial}{\partial x_i} [x_i p] + \frac{1 + 2\theta t}{2} \nabla^2 p. \quad (\text{B.11})$$

Expanding the first term:

$$\frac{\partial p}{\partial t} = \theta D p + \theta \mathbf{x} \cdot \nabla p + \frac{1 + 2\theta t}{2} \nabla^2 p. \quad (\text{B.12})$$

Verification with the transition kernel. The transition kernel of the LV SDE is

$$p_{0,t}(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; e^{-\theta t} \mathbf{x}_0, t\mathbf{I}), \quad (\text{B.13})$$

with mean $\mu(t) = e^{-\theta t}$ and variance $\sigma^2(t) = t$. The parameter derivatives are

$$\frac{d\mu}{dt} = -\theta e^{-\theta t} = -\theta \mu(t), \quad \frac{d\sigma^2}{dt} = 1.$$

The density is

$$p = \frac{1}{(2\pi t)^{D/2}} \exp\left(-\frac{\|\mathbf{x}_t - e^{-\theta t} \mathbf{x}_0\|^2}{2t}\right).$$

Computing the time derivative:

$$\begin{aligned}\frac{\partial p}{\partial t} &= p \left[-\frac{D}{2t} + \frac{\|\mathbf{x}_t - \mu \mathbf{x}_0\|^2}{2t^2} + \frac{(\mathbf{x}_t - \mu \mathbf{x}_0) \cdot (-\theta \mu \mathbf{x}_0)}{t} \right] \\ &= p \left[-\frac{D}{2t} + \frac{\|\mathbf{x}_t - \mu \mathbf{x}_0\|^2}{2t^2} - \frac{\theta \mu}{t} (\mathbf{x}_t - \mu \mathbf{x}_0) \cdot \mathbf{x}_0 \right].\end{aligned}$$

On the other hand:

$$\begin{aligned}\nabla^2 p &= p \left[\frac{D}{t} - \frac{\|\mathbf{x}_t - \mu \mathbf{x}_0\|^2}{t^2} \right], \\ \mathbf{x} \cdot \nabla p &= -p \frac{\mathbf{x}_t \cdot (\mathbf{x}_t - \mu \mathbf{x}_0)}{t}.\end{aligned}$$

Substituting into the right-hand side of (B.12):

$$\begin{aligned}\text{RHS} &= \theta D p + \theta p \left[-\frac{\mathbf{x}_t \cdot (\mathbf{x}_t - \mu \mathbf{x}_0)}{t} \right] + \frac{1 + 2\theta t}{2} p \left[\frac{D}{t} - \frac{\|\mathbf{x}_t - \mu \mathbf{x}_0\|^2}{t^2} \right] \\ &= p \left[\theta D - \frac{\theta}{t} \mathbf{x}_t \cdot (\mathbf{x}_t - \mu \mathbf{x}_0) + \frac{(1 + 2\theta t)D}{2t} - \frac{(1 + 2\theta t)\|\mathbf{x}_t - \mu \mathbf{x}_0\|^2}{2t^2} \right].\end{aligned}$$

Simplifying and rearranging terms (noting that $\mathbf{x}_t \cdot (\mathbf{x}_t - \mu \mathbf{x}_0) = \|\mathbf{x}_t\|^2 - \mu \mathbf{x}_t \cdot \mathbf{x}_0$), one can verify algebraically that both sides coincide. The key is to use $\mu = e^{-\theta t}$ and the relation $\|\mathbf{x}_t - \mu \mathbf{x}_0\|^2 = \|\mathbf{x}_t\|^2 - 2\mu \mathbf{x}_t \cdot \mathbf{x}_0 + \mu^2 \|\mathbf{x}_0\|^2$.

The verification confirms that the Gaussian transition kernel derived in Chapter 3 is indeed the solution of the corresponding FPK equation.

B.6 Equations for Moments and Covariances

Although the FPK equation provides the complete probabilistic description of the process, often only the moments (mean and covariance) are required. We can derive ordinary differential equations for these moments directly from the generator.

B.6.1 General Derivation

Consider the general SDE (B.1). From the generator expectation equation, we have

$$\frac{d}{dt}\mathbb{E}[\phi(\mathbf{X}_t, t)] = \mathbb{E}\left[\frac{\partial\phi}{\partial t}\right] + \mathbb{E}\left[\sum_{i=1}^D \frac{\partial\phi}{\partial x_i} f_i(\mathbf{x}, t)\right] + \frac{1}{2}\mathbb{E}\left[\sum_{i,j=1}^D \frac{\partial^2\phi}{\partial x_i \partial x_j} D_{ij}(\mathbf{x}, t)\right].$$

Equation for the mean. Choosing $\phi(\mathbf{x}, t) = x_u$ (the u -th component of $\mathbf{x}$), we obtain

$$\frac{d\mathbb{E}[X_u]}{dt} = \mathbb{E}[f_u(\mathbf{X}_t, t)].$$

Denoting $\mathbf{m}(t) = \mathbb{E}[\mathbf{X}_t]$, this is written vectorially as

$$\frac{d\mathbf{m}}{dt} = \mathbb{E}[\mathbf{f}(\mathbf{X}_t, t)].$$

Equation for the covariance. Choosing $\phi(\mathbf{x}, t) = (x_u - m_u(t))(x_v - m_v(t))$, where $m_u(t)$ and $m_v(t)$ are the components of the mean, and using the moment equation, we obtain (after calculations):

$$\frac{d}{dt}\mathbb{E}[(X_u - m_u)(X_v - m_v)] = \mathbb{E}[(X_v - m_v)f_u(\mathbf{X}_t, t)] + \mathbb{E}[(X_u - m_u)f_v(\mathbf{X}_t, t)] + \mathbb{E}[D_{uv}(\mathbf{X}_t, t)].$$

Denoting $\mathbf{P}(t) = \mathbb{E}[(\mathbf{X}_t - \mathbf{m}(t))(\mathbf{X}_t - \mathbf{m}(t))^T]$ as the covariance matrix, this is written as

$$\frac{d\mathbf{P}}{dt} = \mathbb{E}[\mathbf{f}(\mathbf{X}_t, t)(\mathbf{X}_t - \mathbf{m})^T] + \mathbb{E}[(\mathbf{X}_t - \mathbf{m})\mathbf{f}^T(\mathbf{X}_t, t)] + \mathbb{E}[\mathbf{D}(\mathbf{X}_t, t)].$$

Important note. The equations for mean and covariance are not closed in general, since the expectations on the right-hand side depend on the complete density $p(\mathbf{x}, t)$, which would be obtained by solving the FPK equation. However, these equations are exact and closed for linear (affine) SDEs, as demonstrated below.

B.6.2 Linear (Affine) Case

For an affine SDE of the form

$$d\mathbf{x} = (\mathbf{F}(t)\mathbf{x} + \mathbf{c}(t)) dt + G(t) d\mathbf{W}_t, \quad (\text{B.14})$$

where $\mathbf{F}(t)$, $\mathbf{c}(t)$, and $G(t)$ are deterministic, the moment equations close.

The drift is $\mathbf{f}(\mathbf{x}, t) = \mathbf{F}(t)\mathbf{x} + \mathbf{c}(t)$ and the diffusion matrix is $\mathbf{D}(t) = \mathbf{G}(t)\mathbf{G}(t)^\top$ (independent of $\mathbf{x}$).

Mean. From the mean equation:

$$\frac{d\mathbf{m}}{dt} = \mathbb{E}[\mathbf{F}(t)\mathbf{X}_t + \mathbf{c}(t)] = \mathbf{F}(t)\mathbb{E}[\mathbf{X}_t] + \mathbf{c}(t) = \mathbf{F}(t)\mathbf{m} + \mathbf{c}(t).$$

This is a linear ODE for $\mathbf{m}(t)$.

Covariance. From the covariance equation:

$$\begin{aligned} \frac{d\mathbf{P}}{dt} &= \mathbb{E}[(\mathbf{F}\mathbf{X}_t + \mathbf{c})(\mathbf{X}_t - \mathbf{m})^T] + \mathbb{E}[(\mathbf{X}_t - \mathbf{m})(\mathbf{F}\mathbf{X}_t + \mathbf{c})^T] + \mathbf{D}(t) \\ &= \mathbf{F}\mathbb{E}[\mathbf{X}_t(\mathbf{X}_t - \mathbf{m})^T] + \mathbb{E}[(\mathbf{X}_t - \mathbf{m})\mathbf{X}_t^T]\mathbf{F}^T + \mathbf{D}(t) \\ &= \mathbf{F}\mathbf{P} + \mathbf{P}\mathbf{F}^T + \mathbf{D}(t), \end{aligned}$$

where we have used $\mathbb{E}[\mathbf{c}(\mathbf{X}_t - \mathbf{m})^T] = \mathbf{0}$ and $\mathbb{E}[\mathbf{X}_t(\mathbf{X}_t - \mathbf{m})^T] = \mathbb{E}[(\mathbf{X}_t - \mathbf{m} + \mathbf{m})(\mathbf{X}_t - \mathbf{m})^T] = \mathbf{P}$.

The covariance equation is the **Lyapunov differential equation** for the covariance, and can be solved explicitly using the fundamental matrix of the system.

B.6.3 Application to the LV SDE

For the LV SDE (B.10), we have

$$\mathbf{F}(t) = -\theta\mathbf{I}, \quad \mathbf{c}(t) = \mathbf{0}, \quad \mathbf{D}(t) = (1 + 2\theta t)\mathbf{I}.$$

Mean. From the mean ODE:

$$\frac{d\mathbf{m}}{dt} = -\theta\mathbf{m}, \quad \mathbf{m}(0) = \mathbf{x}_0,$$

whose solution is

$$\mathbf{m}(t) = e^{-\theta t}\mathbf{x}_0,$$

consistent with the transition kernel (B.13).

Covariance. From the covariance equation:

$$\frac{d\mathbf{P}}{dt} = -\theta\mathbf{P} - \mathbf{P}\theta + (1 + 2\theta t)\mathbf{I} = -2\theta\mathbf{P} + (1 + 2\theta t)\mathbf{I}, \quad \mathbf{P}(0) = \mathbf{0}.$$

This is a scalar ODE (since $\mathbf{P}(t) = p(t)\mathbf{I}$ by symmetry):

$$\frac{dp}{dt} = -2\theta p + (1 + 2\theta t), \quad p(0) = 0.$$

This is a first-order linear ODE. Using the integrating factor $\mu(t) = e^{2\theta t}$:

$$\frac{d}{dt}[e^{2\theta t} p] = e^{2\theta t}(1 + 2\theta t).$$

Integrating:

$$e^{2\theta t} p(t) = \int_0^t e^{2\theta s}(1 + 2\theta s) ds.$$

Computing the integral (using integration by parts or recognizing that $\frac{d}{ds}[se^{2\theta s}] = e^{2\theta s}(1 + 2\theta s)$):

$$\int_0^t e^{2\theta s}(1 + 2\theta s) ds = [se^{2\theta s}]_0^t = te^{2\theta t}.$$

Therefore:

$$p(t) = e^{-2\theta t} \cdot te^{2\theta t} = t,$$

and thus

$$\mathbf{P}(t) = t\mathbf{I},$$

again consistent with the transition kernel (B.13).

This calculation illustrates that the moment equations provide an alternative route to deriving the mean and covariance of the transition kernel, without solving the complete FPK equation. For linear SDEs, this approach is often more efficient.